

[REDACTED]

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Exhibit 97

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

EXHIBIT 97

Domestic Industry Claim Chart for Claims of U.S. Patent No. 12,675,407 (the “’407 Patent”) Based on Practicing ’407 Products

As shown in the chart below, the Practicing ’407 Products, which include Netlist DDR5 VLP RDIMMs, Netlist Lightning DDR5 RDIMMs, Netlist DDR5 LP MRDIMMs, [REDACTED], and SK hynix RDIMMs and MRDIMMs, practice at least claim 1 of the ’407 patent. It should be understood that Netlist’s contentions are not limited to these products but instead encompass any and all reasonably similar products and components that practice the claims in the same or similar manner as discussed below. Netlist may supplement, modify or amend its contentions under the doctrine of equivalents based on any claim constructions, additional information obtained through formal discovery or other means, and/or based on other circumstances including any assertions by Respondents that the Practicing ’407 Products do not practice the ’407 patent as to any one or more of the limitations in the claim(s) below.

To the extent Respondents contend that any differences are alleged to exist between the asserted claims below and the Practicing ’407 Products, such differences are insubstantial and the Practicing ’407 Products perform substantially the same function, in substantially the same way, to yield substantially the same result, and therefore the Practicing ’407 Products practice the claims below under the doctrine of equivalents. Netlist may supplement, modify or amend its contentions under the doctrine of equivalents based on any claim constructions, additional information obtained through formal discovery or other means, and/or based on other circumstances including any assertions that the Practicing ’407 Products do not satisfy the technical domestic industry by Respondents as to any one or more of the limitations in the claims below.

Claim 1	Evidence of Use
[1.pre] A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines,	To the extent the preamble is limiting, the Practicing ’407 Products ¹ include a memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and data signal lines. For example: The Practicing ’407 Products comprise a “memory module.” <i>See, e.g.,</i>

¹ Analysis for certain limitations below uses RDIMM format as an example, but based on currently available information MRDIMMs would operate in materially the same way in all relevant parts. *See* JEDEC Press Release (planned features for MRDIMMs include “Utilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor & Pinout, SPD, PMIC, and TS for ease of adoption”).

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control/address (C/A)
signal lines and data
signal lines, the memory
module comprising:

In the server DRAM section, SK hynix presented a lineup of DDR5-based memory modules for servers. The spotlight was on the DDR5 RDIMM* and MRDIMM* products built using the 1c node,* the sixth generation of the 10 nm process technology. The company also showcased various DRAM solutions designed to maximize AI server performance and reduce power consumption, including its 96 GB DDR5 RDIMM, 256 GB 3DS* DDR5 RDIMM, and DDR5 MRDIMM in both 96 GB and 256 GB capacities.

* Registered Dual In-line Memory Module (RDIMM): A server memory module product in which multiple DRAM chips are mounted on a substrate.

* Multiplexed Rank Dual In-line Memory Module (MRDIMM): A server memory module product in which multiple DRAM chips are mounted on a substrate. Speed is enhanced by operating two ranks—the basic operating units of the module—simultaneously.

* 10 nm DRAM process technology has progressed through six generations: 1x, 1y, 1z, 1a, 1b, and 1c. In August 2024, SK hynix became the first in the world to successfully develop 1c process technology.

* 3D Stacked Memory (3DS): A high-bandwidth memory product in which two or more DRAM chips are packaged together and interconnected using TSV technology.

Ex. CC-63 (SK Hynix Blog Post) at 5.



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Ex. CC-33 (Netlist DDR5 DIMMs).



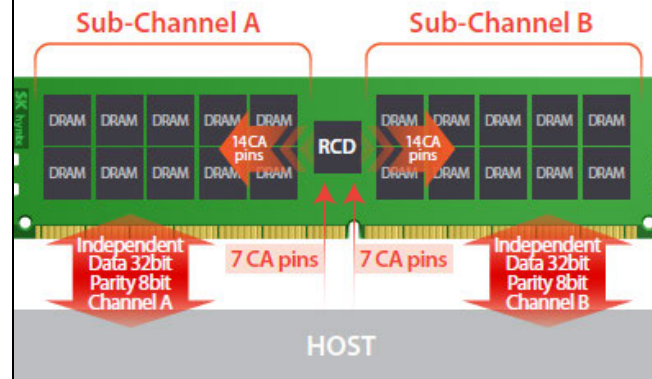
Netlist DDR5 MRDIMM products optimize AI training, inference, and HPC workloads with low latency and high bandwidth.

Netlist's MRDIMM series offers exceptional performance and unmatched speed, enabling users to fully optimize their high performance computing systems without the need for any adjustments.

Ex. CC-71 (Netlist DDR5 MRDIMM Product Brief) at 1.

The latest DDR5 DRAM standard turbo-boosts RDIMM module performance and technology for the era of AI and Big Data. SK hynix expects its DDR5 RDIMMs to reach over 8,000Mbps in data rates with research into continued speed enhancements.

Features



| New Sub-channel DRAM Architecture

The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

Ex. CC-40 (SK Hynix A New Standard) at 5.

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Seoul, December 8, 2022

SK hynix Inc. (or “the company”, www.skhynix.com) announced today that it has developed working samples of DDR5* Multiplexer Combined Ranks* (MCR) Dual In-line Memory Module, the world’s fastest server DRAM product. The new product has been confirmed to operate at the data rate of minimum 8Gbps, and at least 80% faster than 4.8Gbps of the existing DDR5 products.

MCR DIMM is an achievement coming from out-of-the-box thinking with an aim to improve the operation speed of DDR5. Challenging the prevailing concept that the operation speed of DDR5 relies on that of DRAM chip itself, engineers sought to find a way to improve the speed of modules instead of chips for development of the latest product.

SK hynix designed the product in a way that enables simultaneous operation of two ranks by utilizing the data buffer* installed onto the MCR DIMM based on Intel’s MCR technology.

**Double Data Rate (DDR), a DRAM standard mainly used for servers and client applications, has been developed up to the fifth generation. The MCR DIMM is a module product with multiple DRAM chips attached to the board and improved speed as a result of two ranks operating simultaneously.*

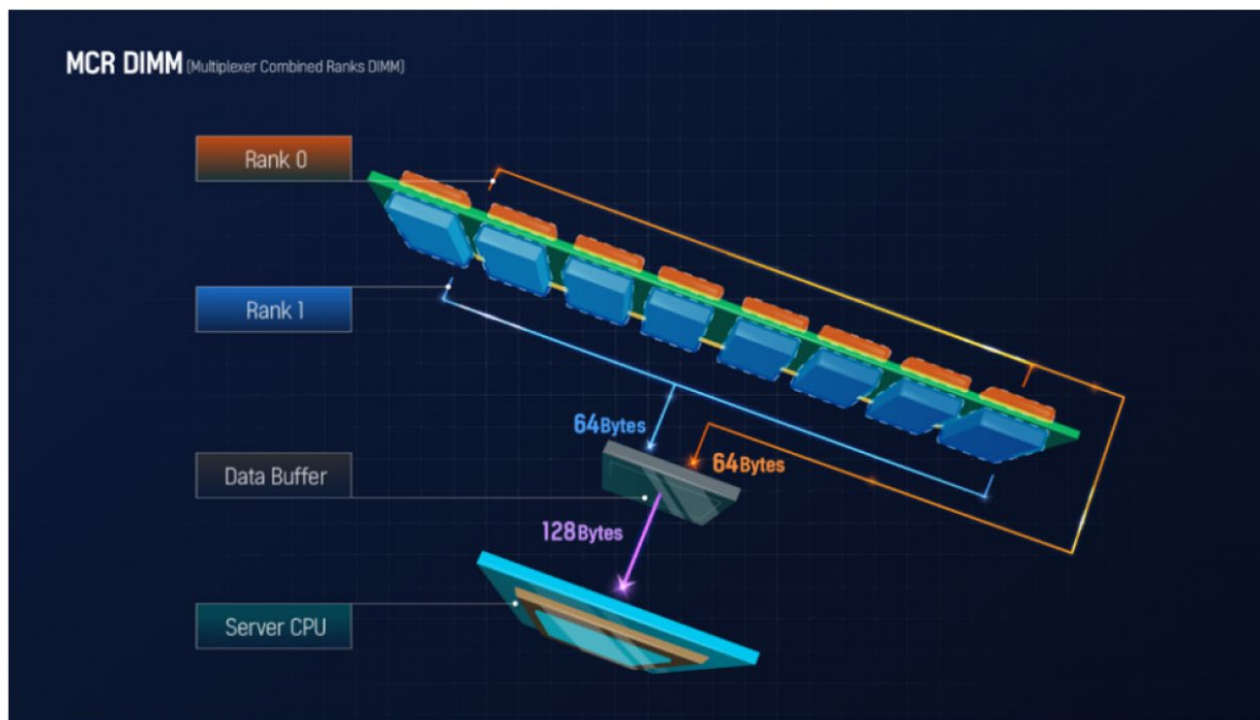
**Rank: A collection of basic transfer units of the data sent to CPU from DRAM module. A rank typically refers to 64 bytes of data to be transferred to central processing unit as a bundle.*

**Buffer: A component that optimizes signal transmission performance between DRAM and CPU. Mainly installed onto modules for servers requiring high performance and reliability*

Ex. CC-65 (SK Hynix Develops MCR DIMM) at 1.

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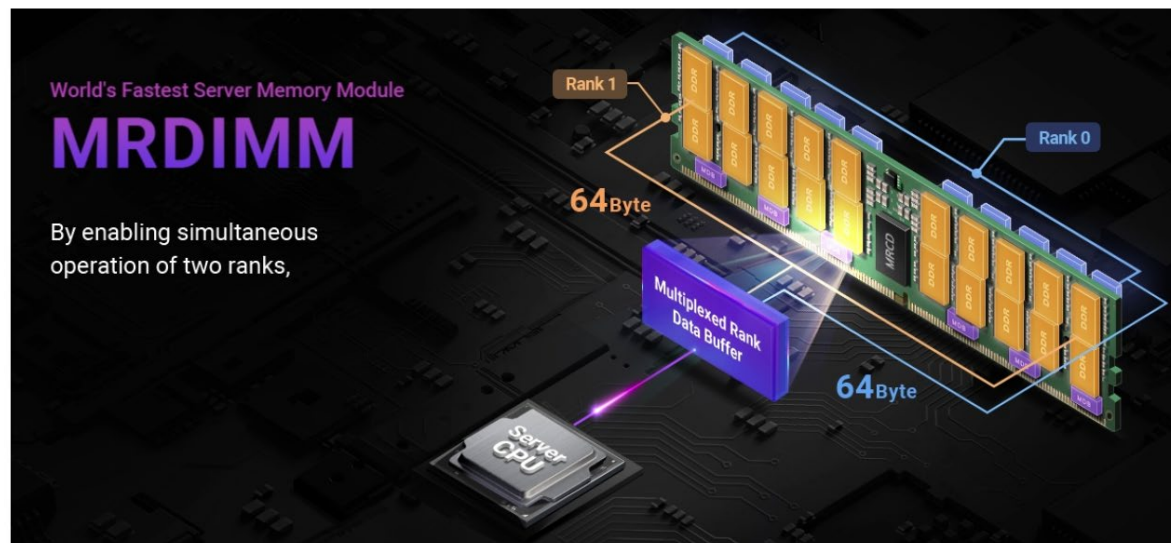
By enabling simultaneous operation of two ranks, MCR DIMM allows transmission of 128 bytes of data to CPU at once, compared with 64 bytes fetched generally in conventional DRAM module. An increase in the amount of data sent to the CPU each time supports the data transfer rate of minimum 8Gbps, twice as fast as a single DRAM.



Ex. CC-65 (SK Hynix Develops MCR DIMM) at 1.

How MRDIMM Works

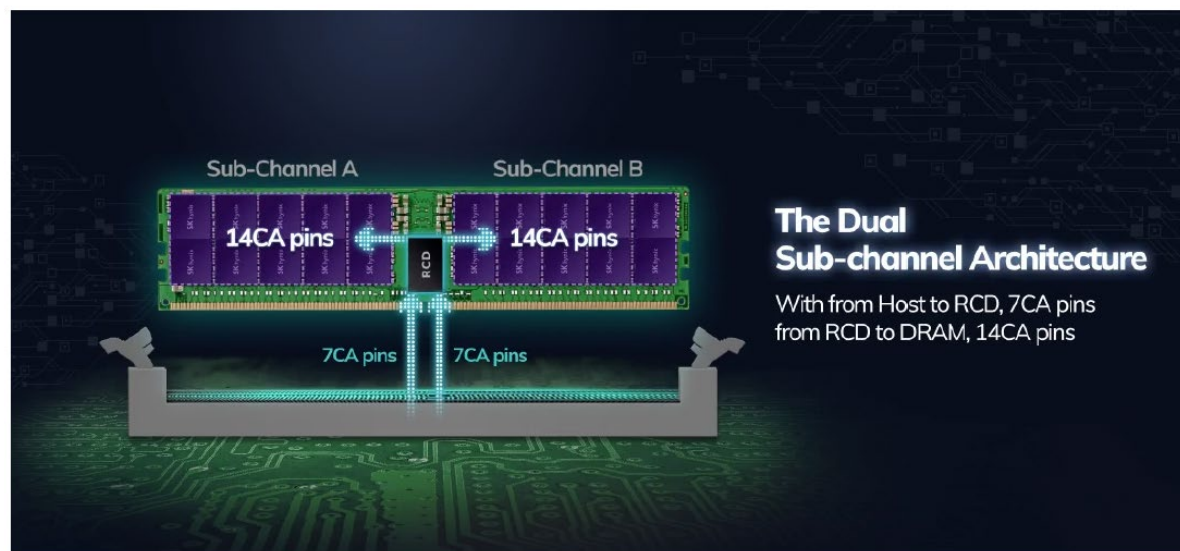
By facilitating simultaneous operation of two ranks, MRDIMM enables the transmission of 128 bytes of data to the CPU at once, compared to the 64 bytes typically fetched by conventional DRAM modules. This increase in the amount of data sent to the CPU each time supports twice the speed of a single DRAM.



Ex. CC-42 (SK Hynix MRDIMM) at 2.

Double Sub-channel DRAM Architecture

With a dual sub-channel structure, DDR5 RDIMM increases the number of CA (Command-Address) pins to seven double-data-rate CA buses each from host to RCD and 14 single-data-rate buses each on RCD to DRAM. Having two independent 40-bit channels thus enables 2x bandwidth and higher bus (signal) efficiency compared to a single-channel DDR4 DIMM.



Ex. CC-41 (SK hynix RDIMM) at 2.

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SERVER DDR5 RDIMM

Ex. CC-34 (SK hynix Products).

RDIMM

[VIEW MORE](#)



DDR5
Generation

16-256GB
Module Density

4800-6400Mbps and higher
Speed

DDR4
Generation

16-256GB
Module Density

2666-3200Mbps
Speed

Application



Server



Networking

- DIMM with a register (Register Clock Driver, RCD) for command/address signals between the DRAM and system memory controller
- Supports x4, x8 DRAM and x4 3DS stacks

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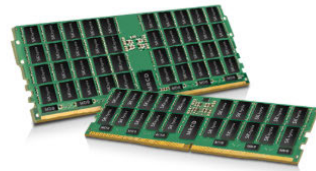
MRDIMM

[VIEW MORE](#)

DDR5
Generation

32-128GB
Module Density

up to 8800Mbps
Speed



Application



- DIMM with a register (Register Clock Driver, Data Buffer) for enabling simultaneous operation of two ranks, which increases bandwidth.
- Supports x4, x8 DRAM

Ex. CC-35 (SK hynix Module).



Ex. CC-36 at IMG_5114 (showing an SK hynix DDR5 RDIMM Product).

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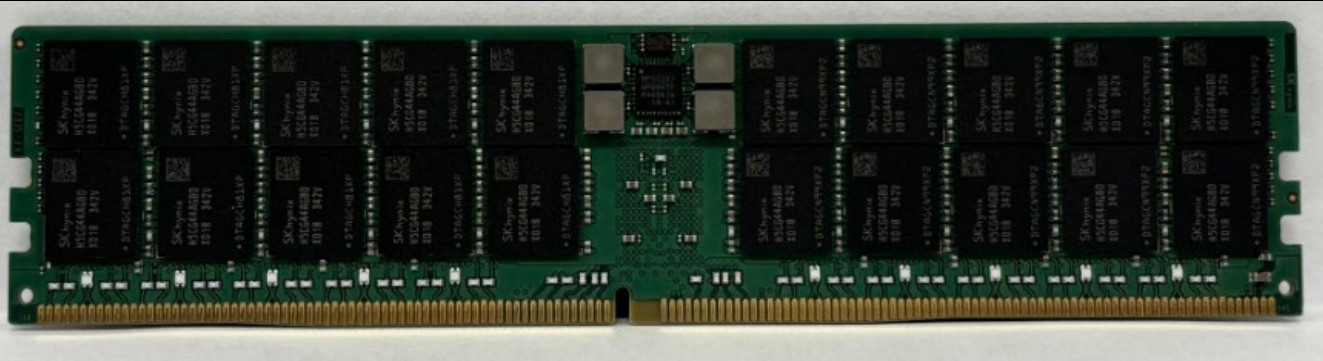


Ex. CC-36 at IMG_5117 (showing an SK hynix DDR5 RDIMM Product).



Ex. CC-36 at IMG_5112 (showing a Netlist DDR5 RDIMM Overclock Product).

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Ex. CC-36 at IMG_5109 (showing Netlist DDR5 RDIMM Overclock Product).



Ex. CC-36 at IMG_5126 (showing Netlist DDR5 MRDIMM Product).

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Ex. CC-36 at IMG_5122 (showing Netlist DDR5 MRDIMM Product).



Ex. CC-36 at IMG_5124 (showing Netlist DDR5 MRDIMM Product).

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Ex. CC-36 at IMG_5125 (showing Netlist DDR5 MRDIMM Product).

“DIMM” in RDIMM and MRDIMM stands for dual inline *memory module*. For example, Netlist’s DDR5 DIMMs web page is in the “Products / *Memory Modules*” section, advertising that “DDR5 DIMMs are the *physical modules that contain DDR5 memory chips*” and listing RDIMMs, VLP RDIMMs, and MRDIMMs as examples of the memory modules offered by Netlist. Ex. CC-33 (Netlist DDR5 DIMMs). Additionally, Netlist’s Lightning DDR5 DIMM web page advertises “*memory Lightning Modules*,” specifically listing DDR5 RDIMM Overclock Products and DDR5 RDIMM Ultra Low Latency RDIMM Products. Ex. CC-37 (Netlist Lightning DDR5 DIMM).

As another example,

Datasheet Version 1.1) at 1.

As yet another example,

Ex. CC-38C (Netlist DDR5 VLP RDIMM

Ex.

[REDACTED]

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	CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 1. The memory module of the Practicing '407 Products are further operable in a computer system having a memory controller (such as a CPU) and a system bus. For example: “DDR5 DIMMs are the physical modules that contain <i>DDR5 memory chips and are used to expand the memory capacity of a computer system.</i> DDR5 DIMMs have larger burst lengths than DDR4, with a burst length of 16 vs. 8, which allows for a greater amount of data to be transferred <i>between the memory and the CPU.</i>” Ex. CC-33 (Netlist DDR5 DIMMs). As another example: “<i>The Netlist DDR5 memory is designed for the latest AMD and Intel processors supporting the DDR5 standard.</i> With Plug and Play functionality profiles, memory automatically operates to the highest published DDR5 JEDEC data rates up to 6800 Mbps, with no changes to the BIOS, CPU Core, and voltage. DDR5 memory is ideal for AI Inference, Video Editing, Simulation, Gaming Development, and Database workloads giving your system a boost in performance without having to overclock the processor or memory. System performance is boosted by up to 12% depending on workload.” Ex. CC-56 (Netlist Low Latency DDR5 RDIMM Product Brief) at 1. [REDACTED] [REDACTED] Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 1.
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MRDIMM PERFORMANCE

Netlist MRDIMM are designed to enhance data center performance, particularly for AI, high-performance computing (HPC), and data analytics applications. It offers up to a 38% increase in bandwidth compared to traditional DDR5 RDIMMs, making it a high-bandwidth, low-latency solution. Available in capacities ranging from 32GB to 128GB, MRDIMM supports high-density memory configurations while improving performance per watt for energy efficiency. It is compatible with the next generation of Intel® Xeon® processors, codenamed Granite Rapids, and comes in both low-profile and tall form factor (TFF) modules, providing flexibility for various server configurations.

More Bandwidth on Granite Rapids AP (37.5%) & SP (12.5%) platforms

Rank Configuration	64GB 16Gb 1U 4R x 8	64GB 16Gb 1U 2R x 4	128GB 16Gb 2U 56mm Height	128GB 32Gb 1U 31mm Height
Consistent Performance	a 37.5% increase in bandwidth per socket for MRDIMM at 8800 MT/s versus DDR5 at 6400 MT/s with no more than 5ns latency penalty with no restriction on R/W ratios		a 37.5% increase in bandwidth per socket for MRDIMM at 8800 MT/s versus DDR5 at 6400 MT/s with no more than 5ns latency penalty with no restriction on R/W ratios	
Matched System Configuration	1 x Socket, & 64GB 1DPC configuration		1 x Socket, & 128GB 1DPC configuration	
Same Capacity	768 GB data Processing visual information & e-commerce, social media, entertainment		1.5 TB data Computer vision & Natural Language Processing Model (NLP) DB	
Save More	0.90	1.0	0.75	1.0
	CPU 1 x Intel Xeon processor (GNR) Memory 128GB x 16 DDR5 DIMMs for 1.5TB		CPU 1 x Intel Xeon processor (GNR) Memory 128GB x 16 DDR5 DIMMs for 1.5TB	

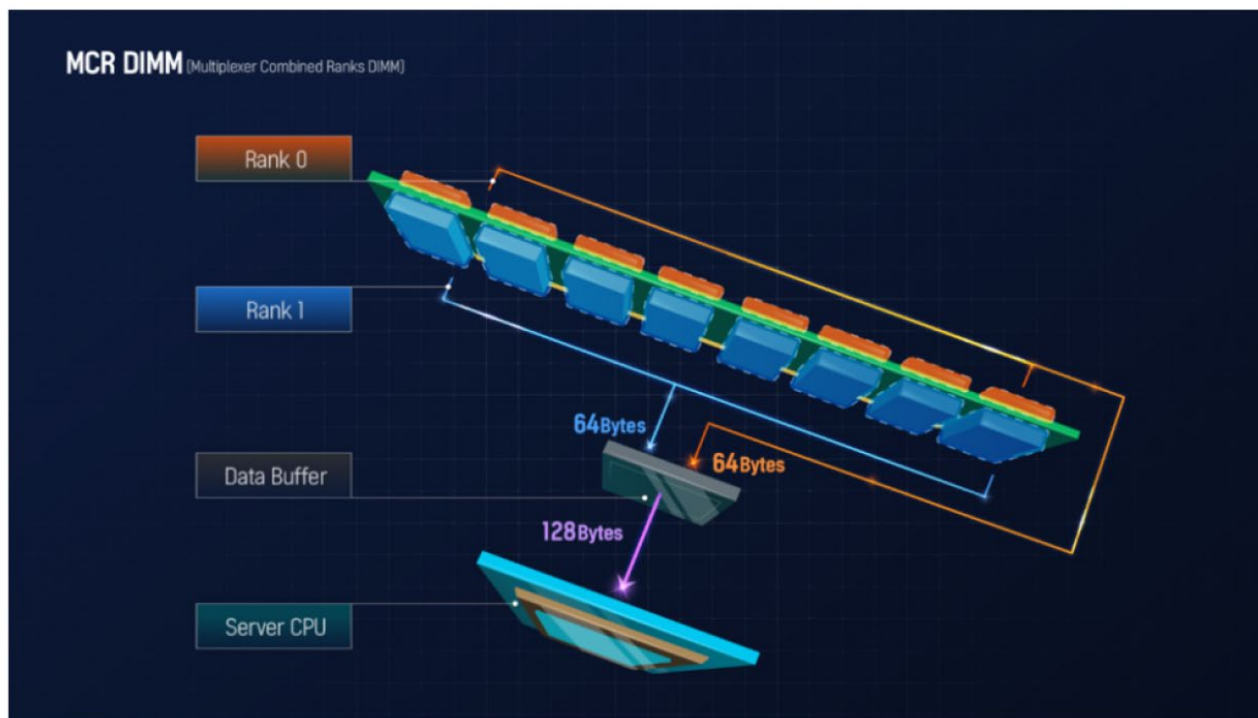
Ex. CC-71 (Netlist DDR5 MRDIMM Product Brief) at 1.

Compatibility	Optimized for Intel Xeon 6 processors. They maintain the same connector and physical layout as RDIMMs, allowing easy integration into existing systems without modifications
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Ex. CC-71 (Netlist DDR5 MRDIMM Product Brief) at 2.

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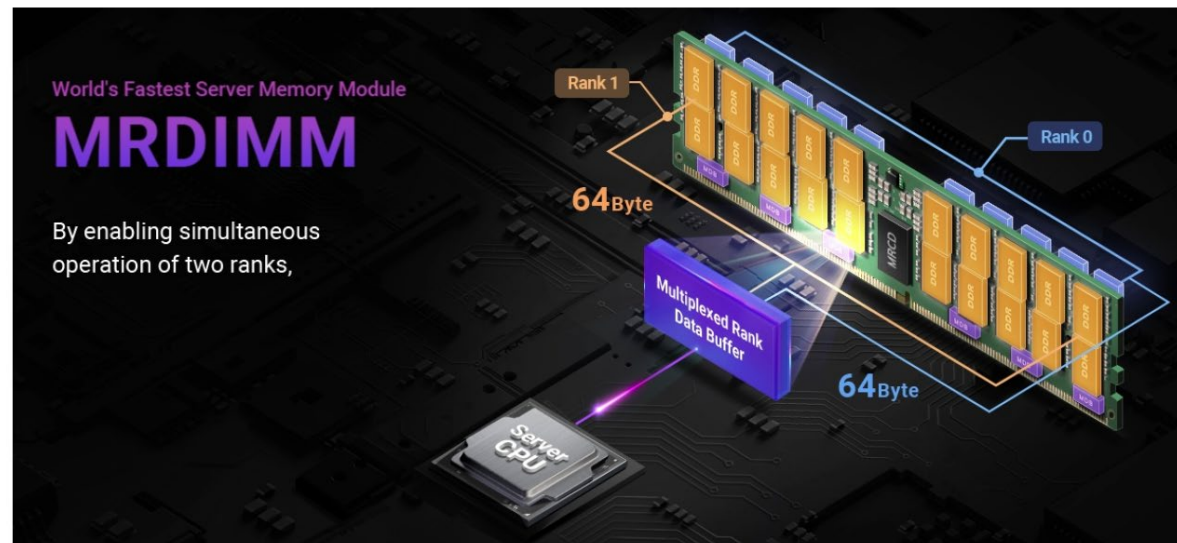
By enabling simultaneous operation of two ranks, MCR DIMM allows transmission of 128 bytes of data to CPU at once, compared with 64 bytes fetched generally in conventional DRAM module. An increase in the amount of data sent to the CPU each time supports the data transfer rate of minimum 8Gbps, twice as fast as a single DRAM.



Ex. CC-65 (SK Hynix Develops MCR DIMM) at 1 (depicting the 128 Byte system bus used to operatively connect the memory module to the memory controller (“Server CPU”)).

How MRDIMM Works

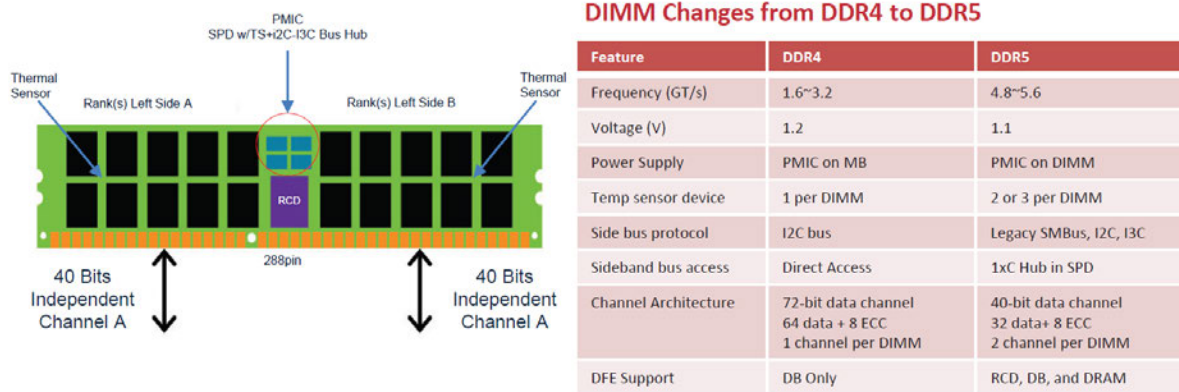
By facilitating simultaneous operation of two ranks, MRDIMM enables the transmission of 128 bytes of data to the CPU at once, compared to the 64 bytes typically fetched by conventional DRAM modules. This increase in the amount of data sent to the CPU each time supports twice the speed of a single DRAM.



Ex. CC-42 (SK Hynix MRDIMM) at 2 (depicting the dark purple system bus used to operatively connect the memory module to the memory controller (“Server CPU”).).

Additionally, the Practicing ’407 Products are JEDEC compliant, further demonstrating they are memory modules operable in computer systems having memory controllers and system busses. *See, e.g.:*

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Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1.

Netlist DDR5 Memory Module

Emerging Workloads Driving New Memory Requirements

FEATURES

- Data rates up to 5600 Mbps
- Burst Length (BL16)
- On-Die Error Correction Coding (ECC)
- Post-Package Repair
- Refresh Management
- CRC (write & read)
- Decision Feedback Equalization (DFE)

BENEFITS

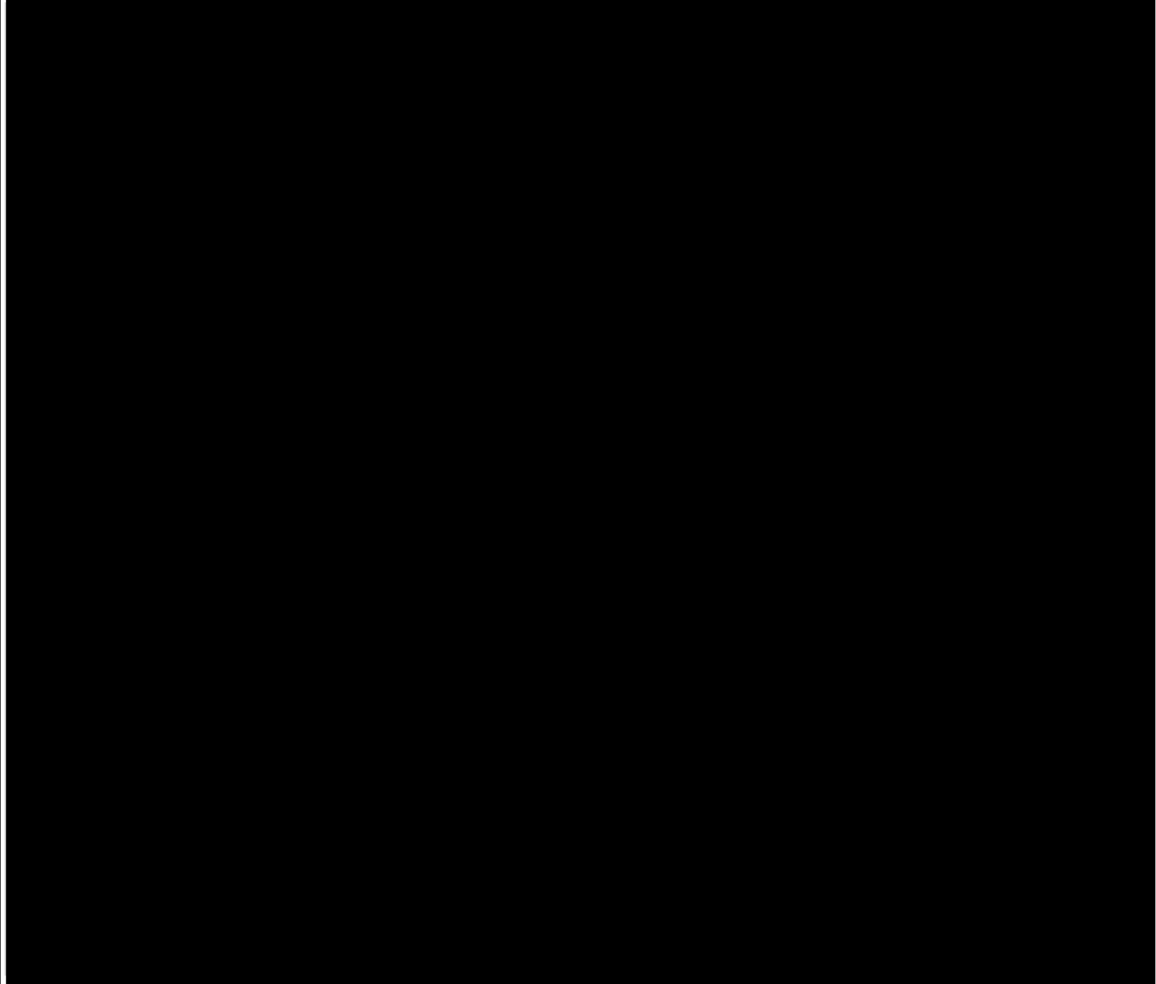
- Locked BOM Support
- JEDEC Compliant
- 100% System level testing
- Technical support
- Customization

Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 2.

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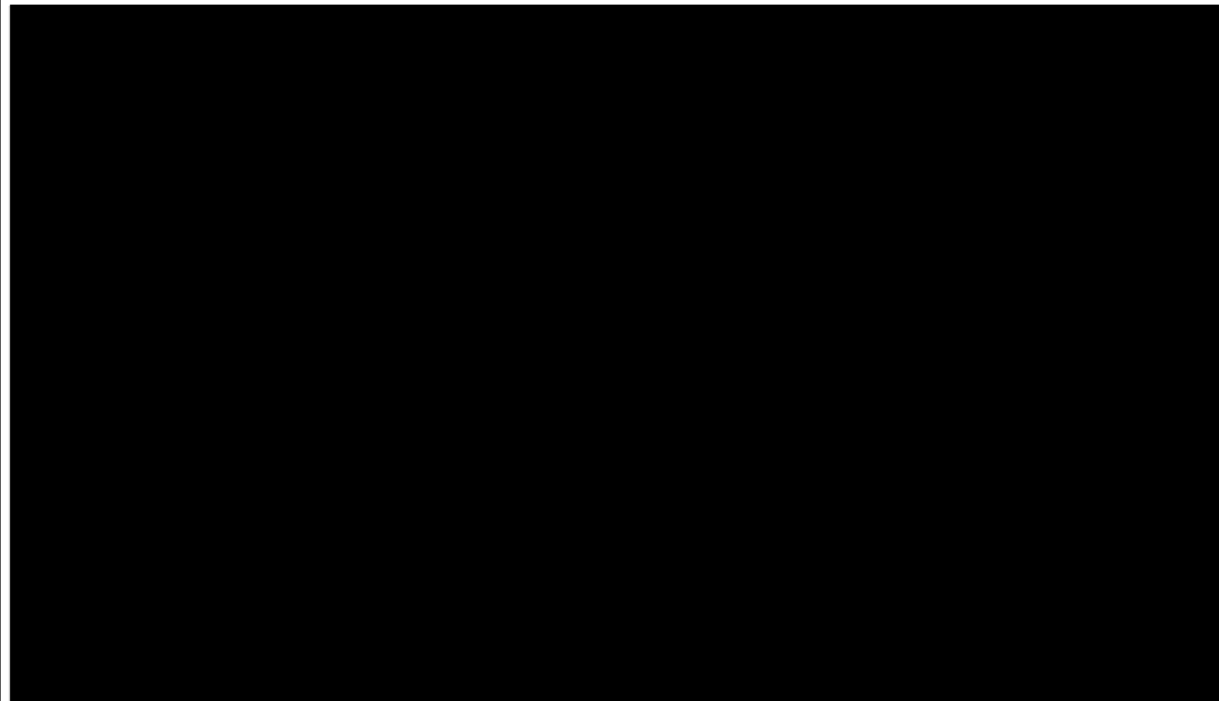
	DDR5 DIMMs FEATURES • Data rates up to 6400 Mbps • Burst Length (BL16) • On-Die Error Correction Coding (ECC) • Post-Package Repair • Refresh Management • CRC (write & read) • Decision Feedback Equalization (DFE) BENEFITS • Locked BOM Support • JEDEC Compliant • 100% System level testing • Technical support • Customization Ex. CC-33 (Netlist DDR5 DIMMs) at 2.
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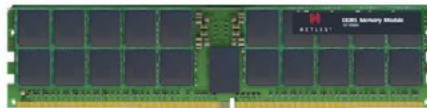
		
	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 1.	

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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 1.



Netlist's Ultra Low Latency DDR5 RDIMM products are the ultimate solution for Data Center Workloads like AI Inference, Video Editing, Gaming Development, and Database

Netlist's Ultra Low Latency RDIMM family delivers extreme performance and unrivaled speed, empowering users to maximize their computing systems' potential without any modifications.

Netlist is a leading manufacturer of high-performance memory and the Netlist Ultra Low Latency (ULL) JEDEC DDR5 RDIMMs offer extremely low latency and high speed on a JEDEC RDIMM. DDR5 is the latest generation, cutting-edge memory technology for Servers and Workstations and Netlist ULL JEDEC DDR5 RDIMMs integrate all the enhanced features of DDR5 into a standard JEDEC form factor. Netlist ULL memory delivers superior speed, stability, and efficiency.

Ex. CC-56 (Netlist Low Latency DDR5 RDIMM Product Brief) at 1-2.

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- **Lightning Ultra Low Latency:** Engineered for minimal delay, our memory solutions deliver faster access times improved performance. With JEDEC compatibility, our DDR5 RDIMMs ensure seamless integration with a wide range of systems, providing a plug-and-play experience. This compatibility, combined with advanced Low Latency capabilities, can boost performance by up to 12% in certain workloads, making it ideal for demanding applications.

Ex. CC-37 (Netlist Lightning DDR5 DIMM) at 1.

- **Compatibility & Stability:** Our DDR5 Lightning Modules are designed for system plug-and-play, increasing native speeds from 4800 Mbps to 72000 Mbps. JEDEC compliance ensures that the Lightning Modules default to higher speeds, utilizing JEDEC-standard PCB, DRAM, RCD, PMIC, and clock components. Built with server-grade DRAM and memory logic, these Lightning Modules offer cost-optimized OEM solutions, available as individual DIMMs in trays rather than kits. They undergo rigorous system-level testing to ensure OEM durability, with locked and optimized BOMs for consistent performance. Additionally, we offer memory customization to meet specific needs.

Ex. CC-37 (Netlist Lightning DDR5 DIMM) at 2.

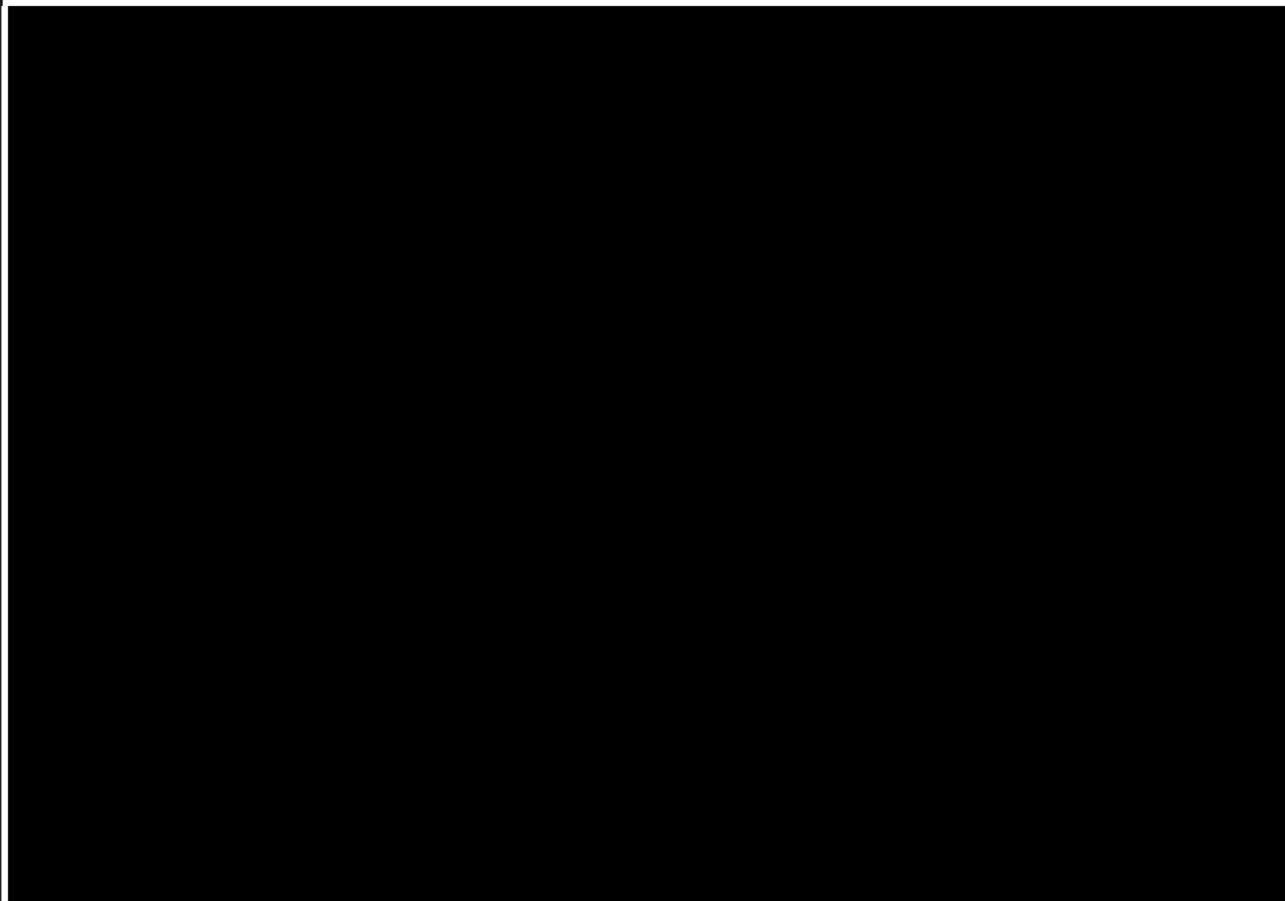
This standard defines the electrical and mechanical requirements for 288-position, 1.1 Volt (VDD and VDDQ), DDR5 Registered (RDIMM), Double Data Rate (DDR), Synchronous DRAM Dual In-Line Memory Modules (DIMM). These Registered DDR5 SDRAM DIMMs are intended for use in server, workstation, and database environments.

Ex. CC-88 DDR5 Registered Dual Inline Memory Module (RDIMM) Common Standard, JESD305B.01 v. 3.00, JEDEC, (Feb. 2026) (“JEDEC 305B”) at 1.

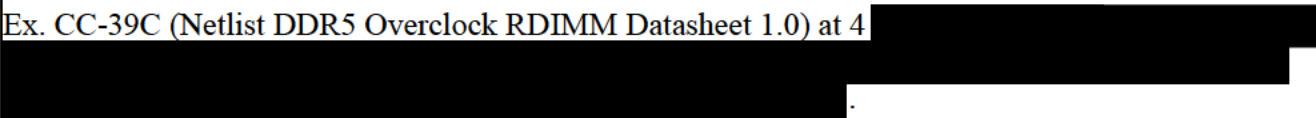
[REDACTED]

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Furthermore, the system bus of the computer systems that the Practicing '407 Products are operable in includes one or more clock signal lines, control/address (C/A) signal lines, and data signal lines. *See, e.g.,*

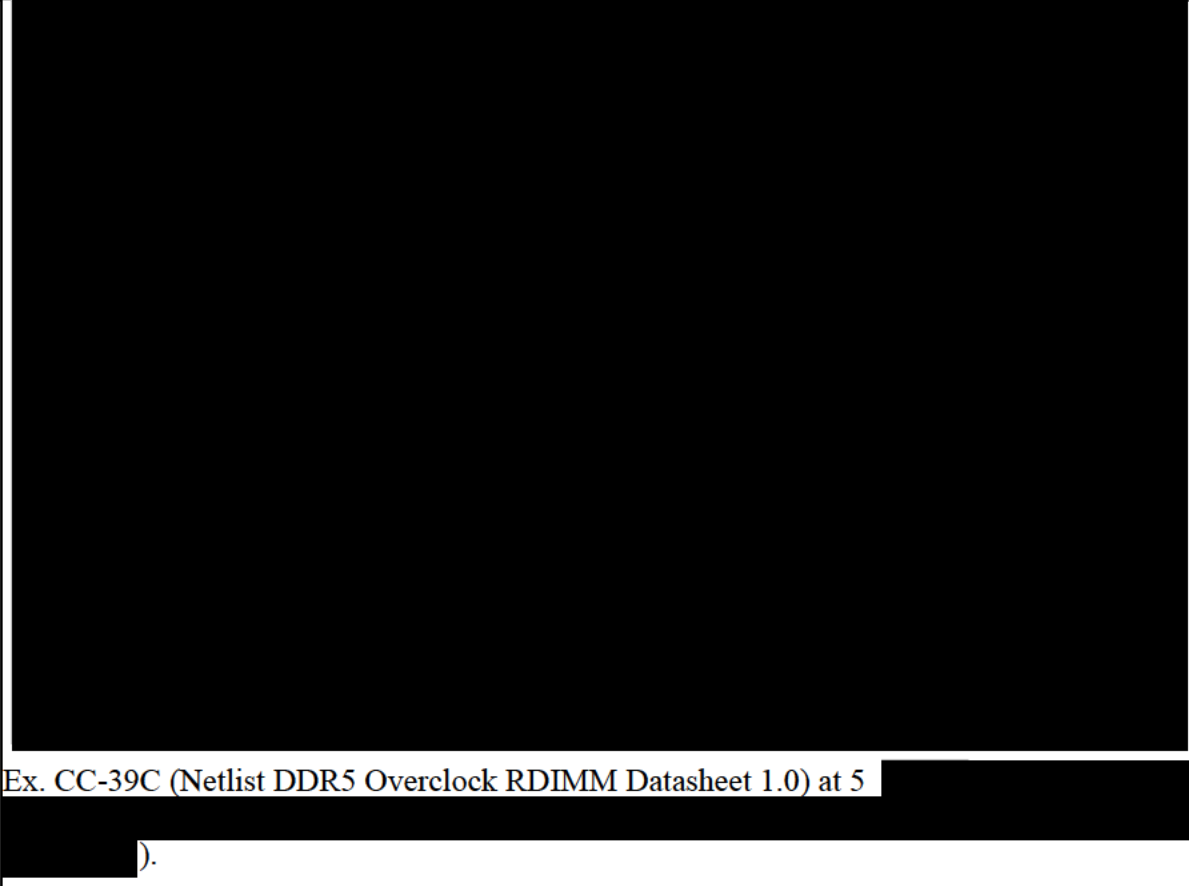


Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 4



[REDACTED]

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	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 5	
	[REDACTED]	
	[REDACTED]).	

[REDACTED]

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[REDACTED]

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 5 [REDACTED]
[REDACTED].

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Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 6.

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	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 7 [REDACTED] [REDACTED])
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	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 8	[REDACTED]
	[REDACTED]	)
	[REDACTED]	
	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 9	[REDACTED]

[REDACTED]

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Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 10

[REDACTED]

[REDACTED]

[REDACTED]

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	[REDACTED]	
	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 4	[REDACTED]
	[REDACTED]	[REDACTED]

[REDACTED]

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	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 5 [REDACTED] [REDACTED]
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[REDACTED]

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[REDACTED]

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	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 5 [REDACTED] [REDACTED] [REDACTED]).
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[illegible]

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	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 6 [REDACTED] [REDACTED]
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

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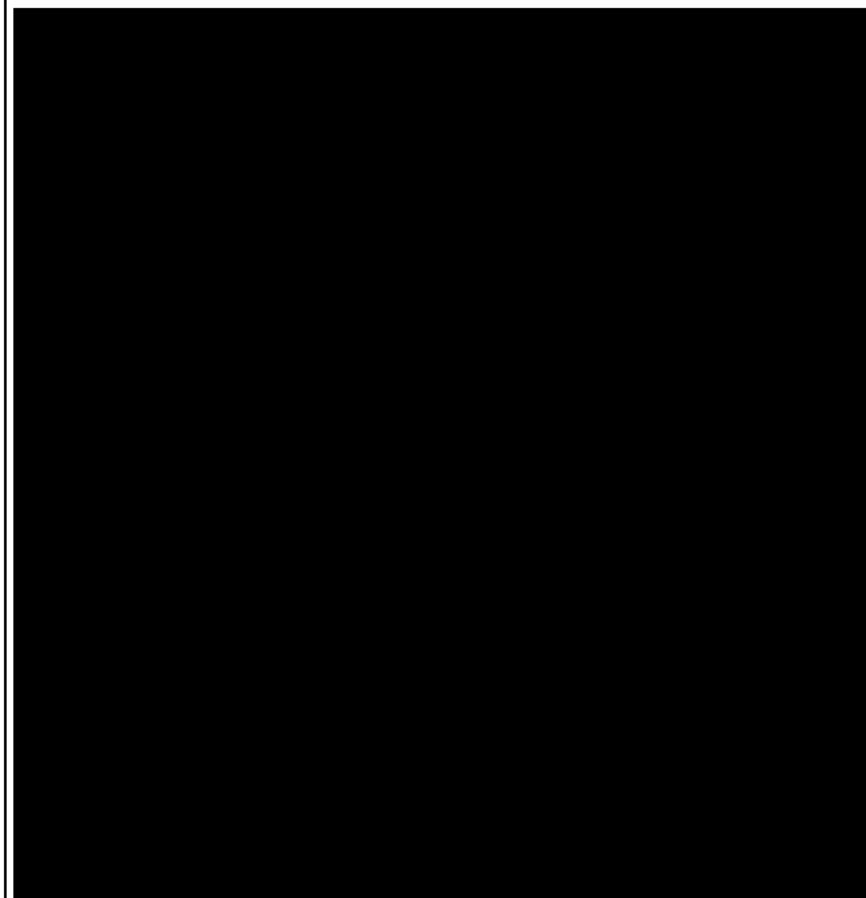
[REDACTED]

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	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 7 [REDACTED]). Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 8.
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[REDACTED]

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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 9

[REDACTED]
[REDACTED]).

As explained *supra*, the Practicing '407 Products are JEDEC compliant, including the use of a JEDEC compliant 288-pin edge connector used to connect to the system bus. The JEDEC DDR5 RDIMM standard

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connector pinout and signal description further indicates the system bus of the computer system includes one or more clock signal lines, control/address (C/A) signal lines, and data signal lines. *See, e.g.,*

Table 3 — Pin Definitions

Pin Name	Description	Pin Name	Description
CA[6:0]_A CA[6:0]_B	Command and Address (CA) Bus	DQ[31:0]_A DQ[31:0]_B	DIMM memory Data bus channel A and B
CS[1:0]_A_n CS[1:0]_B_n	Control (Chip Select) (CS_n)	CB[7:0]_A CB[7:0]_B	DIMM ECC Checkbits (CB) channel A and B
PAR_A PAR_B	Parity input	DQS[9:0]_A_t DQS[9:0]_B_t	Data Strobes (positive line of differential pair)
CK_t	Clock (true/positive)	DQS[9:0]_A_c DQS[9:0]_B_c	Data Strobes (negative line of differential pair)
CK_c	Clock (complement/negative)	TDQS[9:5]_A_t TDQS[9:5]_B_t	Not valid for x4 operation. Enabled via Mode Register.

Ex. CC-88 (JEDEC 305B) at 3 (illustrating clock signal lines (*e.g.*, “CK_t” and “CK_c”), control/address (C/A) signal lines (*e.g.*, “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”), and data signal lines (*e.g.*, “DQ[]_A” and “DQ[]_B”)).

Table 4 — Input/Output Functional Description

Symbol	Type	I/O Levels	Function
CK_t, CK_c,	Input	VDD	Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.
CA[6:0]_A CA[6:0]_B	Input	VDD	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during Mode Register Set commands.

Ex. CC-88 (JEDEC 305B) at 4 (illustrating clock signal lines (*e.g.*, “CK_t” and “CK_c”) and control/address (C/A) signal lines (*e.g.*, “CA[]_A” and “CA[]_B”)).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

DQ[31:0]_A DQ[31:0]_B	Input/ Output	VDD	Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.
CB[7:0]_A CB[7:0]_B	Input/ Output	VDD	ECC Checkbits Input/ Output: Bi-directional data bus - for 8-bit ECC (EC8) all 8 bits are used - for 4-bit ECC (EC4) [3:0] bits are used; [7:4] bits are floating

Ex. CC-88 (JEDEC 305B) at 4 (illustrating data signal lines (e.g., “DQ[]_A” and “DQ[]_B”)).

Table 4 — Input/Output Functional Description (cont'd)

Symbol	Type	I/O Levels	Function
DQS[9:0]_A_t DQS[9:0]_B_t	Input/ Output	VDD	Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. The Data Strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential Data Strobe only and does not support single-ended.
DQS[9:0]_A_c DQS[9:0]_B_c			
TDQS[9:5]_A_t TDQS[9:5]_B_t	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.
TDQS[9:5]_A_c TDQS[9:5]_B_c	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.

Ex. CC-88 (JEDEC 305B) at 5.

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Front Side Pin Label	Pin	Pin	Back side Pin Label	Front Side Pin Label	Pin	Pin	Back side Pin Label
VIN_BULK	1	145	VIN_BULK	PAR_A	74	218	CK_c
RFU	2	146	VIN_BULK	VSS	75	219	VSS
VIN_MGMT	3	147	PCAMP				Key
HSCL	4	148	HSA				
HSDA	5	149	RFU	CA0_B	76	220	RFU / No Pin Present
VSS	6	150	RFU	VSS	77	221	CA1_B
DQ0_A	7	151	VSS	CA2_B	78	222	VSS
VSS	8	152	DQ2_A	VSS	79	223	CA3_B
DQ1_A	9	153	VSS	CA4_B	80	224	VSS
VSS	10	154	DQ3_A	VSS	81	225	CA5_B
DQS0_A_t	11	155	VSS	CA6_B	82	226	VSS
DQS0_A_c	12	156	DQS5_A_c, TDQS5_A_c	VSS	83	227	PAR_B
VSS	13	157	DQS5_A_t, TDQS5_A_t	CS0_B_n	84	228	VSS
DQ4_A	14	158	VSS	VSS	85	229	CS1_B_n
VSS	15	159	DQ6_A	DLBDQ	86	230	VSS
DQ5_A	16	160	VSS	DLBDQS	87	231	RFU
VSS	17	161	DQ7_A	VSS	88	232	RFU
DQ8_A	18	162	VSS	CB4_B	89	233	VSS
VSS	19	163	DQ10_A	VSS	90	234	CB6_B
DQ9_A	20	164	VSS	CB5_B	91	235	VSS
VSS	21	165	DQ11_A	VSS	92	236	CB7_B
DQS1_A_t	22	166	VSS	DQS9_B_t, TDQS9_B_t	93	237	VSS
DQS1_A_c	23	167	DQS6_A_c, TDQS6_A_c	DQS9_B_c, TDQS9_B_c	94	238	DQS4_B_c
VSS	24	168	DQS6_A_t, TDQS6_A_t	VSS	95	239	DQS4_B_t
DQ12_A	25	169	VSS	CB0_B	96	240	VSS
VSS	26	170	DQ14_A	VSS	97	241	CB2_B
DQ13_A	27	171	VSS	CB1_B	98	242	VSS
VSS	28	172	DQ15_A	VSS	99	243	CB3_B
DQ16_A	29	173	VSS	DQ0_B	100	244	VSS
VSS	30	174	DQ18_A	VSS	101	245	DQ2_B
DQ17_A	31	175	VSS	DQ1_B	102	246	VSS
VSS	32	176	DQ19_A	VSS	103	247	DQ3_B
DQS2_A_t	33	177	VSS	DQS0_B_t	104	248	VSS
DQS2_A_c	34	178	DQS7_A_c, TDQS7_A_c	DQS0_B_c	105	249	DQS5_B_c, TDQS5_B_c
VSS	35	179	DQS7_A_t, TDQS7_A_t	VSS	106	250	DQS5_B_t, TDQS5_B_t
DQ20_A	36	180	VSS	DQ4_B	107	251	VSS
VSS	37	181	DQ22_A	VSS	108	252	DQ6_B
DQ21_A	38	182	VSS	DQ5_B	109	253	VSS

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

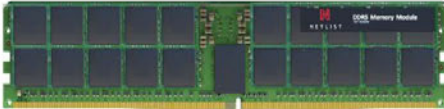
	Ex. CC-88 (JEDEC 305B) at 6 (illustrating clock signal lines (<i>e.g.</i> , “CK_t” and “CK_c”), control/address (C/A) signal lines (<i>e.g.</i> , “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”), and data signal lines (<i>e.g.</i> , “DQ[]_A” and “DQ[]_B”).
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Front Side Pin Label	Pin	Pin	Back side Pin Label	Front Side Pin Label	Pin	Pin	Back side Pin Label
VSS	39	183	DQ23_A	VSS	110	254	DQ7_B
DQ24_A	40	184	VSS	DQ8_B	111	255	VSS
VSS	41	185	DQ26_A	VSS	112	256	DQ10_B
DQ25_A	42	186	VSS	DQ9_B	113	257	VSS
VSS	43	187	DQ27_A	VSS	114	258	DQ11_B
DQS3_A_t	44	188	VSS	DQS1_B_t	115	259	VSS
DQS3_A_c	45	189	DQS8_A_c, TDQS8_A_c	DQS1_B_c	116	260	DQS6_B_c, TDQS6_B_c
VSS	46	190	DQS8_A_t, TDQS8_A_t	VSS	117	261	DQS6_B_t, TDQS6_B_t
DQ28_A	47	191	VSS	DQ12_B	118	262	VSS
VSS	48	192	DQ30_A	VSS	119	263	DQ14_B
DQ29_A	49	193	VSS	DQ13_B	120	264	VSS
VSS	50	194	DQ31_A	VSS	121	265	DQ15_B
CB0_A	51	195	VSS	DQ16_B	122	266	VSS
VSS	52	196	CB2_A	VSS	123	267	DQ18_B
CB1_A	53	197	VSS	DQ17_B	124	268	VSS
VSS	54	198	CB3_A	VSS	125	269	DQ19_B
DQS4_A_t	55	199	VSS	DQS2_B_t	126	270	VSS
DQS4_A_c	56	200	DQS9_A_c, TDQS9_A_c	DQS2_B_c	127	271	DQS7_B_c, TDQS7_B_c
VSS	57	201	DQS9_A_t, TDQS9_A_t	VSS	128	272	DQS7_B_t, TDQS7_B_t
CB4_A	58	202	VSS	DQ20_B	129	273	VSS
VSS	59	203	CB6_A	VSS	130	274	DQ22_B
CB5_A	60	204	VSS	DQ21_B	131	275	VSS
VSS	61	205	CB7_A	VSS	132	276	DQ23_B
ALERT_n	62	206	VSS	DQ24_B	133	277	VSS
VSS	63	207	RESET_n	VSS	134	278	DQ26_B
CS0_A_n	64	208	VSS	DQ25_B	135	279	VSS
VSS	65	209	CS1_A_n	VSS	136	280	DQ27_B
CA0_A	66	210	VSS	DQS3_B_t	137	281	VSS
VSS	67	211	CA1_A	DQS3_B_c	138	282	DQS8_B_c, TDQS8_B_c
CA2_A	68	212	VSS	VSS	139	283	DQS8_B_t, TDQS8_B_t
VSS	69	213	CA3_A	DQ28_B	140	284	VSS
CA4_A	70	214	VSS	VSS	141	285	DQ30_B
VSS	71	215	CA5_A	DQ29_B	142	286	VSS
CA6_A	72	216	VSS	VSS	143	287	DQ31_B
VSS	73	217	CK_t	RFU	144	288	VSS

NOTE 1 Individual Annex standards for RDIMM define pin use where multiple functions are possible.
NOTE 2 Pin 220 for 8000 Mbps and faster RDIMM raw cards do not have the edge pin present.

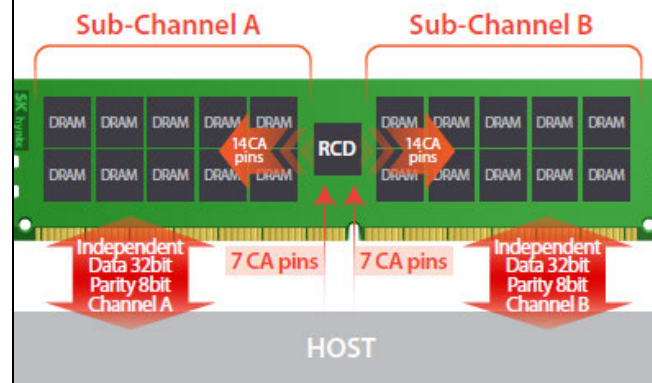
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Ex. CC-88 (JEDEC 305B) at 7 (illustrating clock signal lines (<i>e.g.</i> , “CK_t” and “CK_c”), control/address (C/A) signal lines (<i>e.g.</i> , “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”), and data signal lines (<i>e.g.</i> , “DQ[]_A” and “DQ[]_B”).
[1.a] a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus;	The Practicing '407 Products comprise a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus. For example: The Practicing '407 Products include a printed circuit board (PCB) having connectors configured to provide electrical connections. For example, the Practicing '407 Products include a printed circuit board shown in green below, with connectors shown in gold/yellow below that are configured to provide electrical connections. <i>See, e.g.,</i>:  VLP RDIMM Ex. CC-33 (Netlist DDR5 DIMMs).  Netlist DDR5 MRDIMM products optimize AI training, inference, and HPC workloads with low latency and high bandwidth. Netlist's MRDIMM series offers exceptional performance and unmatched speed, enabling users to fully optimize their high performance computing systems without the need for any adjustments. Ex. CC-71 (Netlist DDR5 MRDIMM Product Brief) at 1.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

The latest DDR5 DRAM standard turbo-boosts RDIMM module performance and technology for the era of AI and Big Data. SK hynix expects its DDR5 RDIMMs to reach over 8,000Mbps in data rates with research into continued speed enhancements.

Features



| New Sub-channel DRAM Architecture

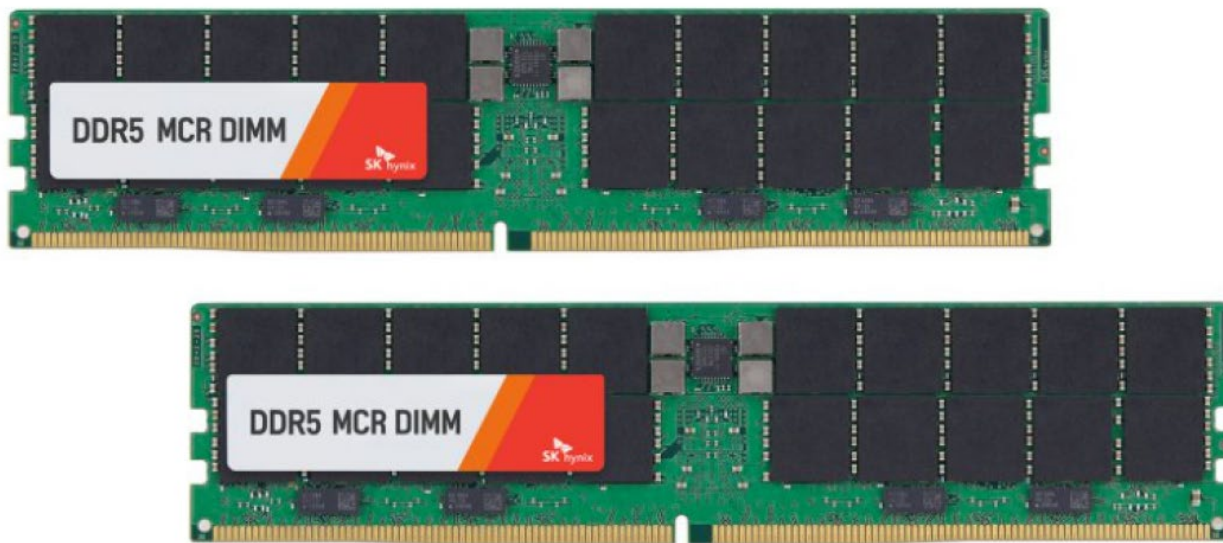
The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

Ex. CC-40 (SK Hynix A New Standard) at 5.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	In the server DRAM section, SK hynix presented a lineup of DDR5-based memory modules for servers. The spotlight was on the DDR5 RDIMM* and MRDIMM* products built using the 1c node,* the sixth generation of the 10 nm process technology. The company also showcased various DRAM solutions designed to maximize AI server performance and reduce power consumption, including its 96 GB DDR5 RDIMM, 256 GB 3DS* DDR5 RDIMM, and DDR5 MRDIMM in both 96 GB and 256 GB capacities. * Registered Dual In-line Memory Module (RDIMM): A server memory module product in which multiple DRAM chips are mounted on a substrate. * Multiplexed Rank Dual In-line Memory Module (MRDIMM): A server memory module product in which multiple DRAM chips are mounted on a substrate. Speed is enhanced by operating two ranks—the basic operating units of the module—simultaneously. * 10 nm DRAM process technology has progressed through six generations: 1x, 1y, 1z, 1a, 1b, and 1c. In August 2024, SK hynix became the first in the world to successfully develop 1c process technology. * 3D Stacked Memory (3DS): A high-bandwidth memory product in which two or more DRAM chips are packaged together and interconnected using TSV technology. Ex. CC-63 (SK Hynix Blog Post) at 5.
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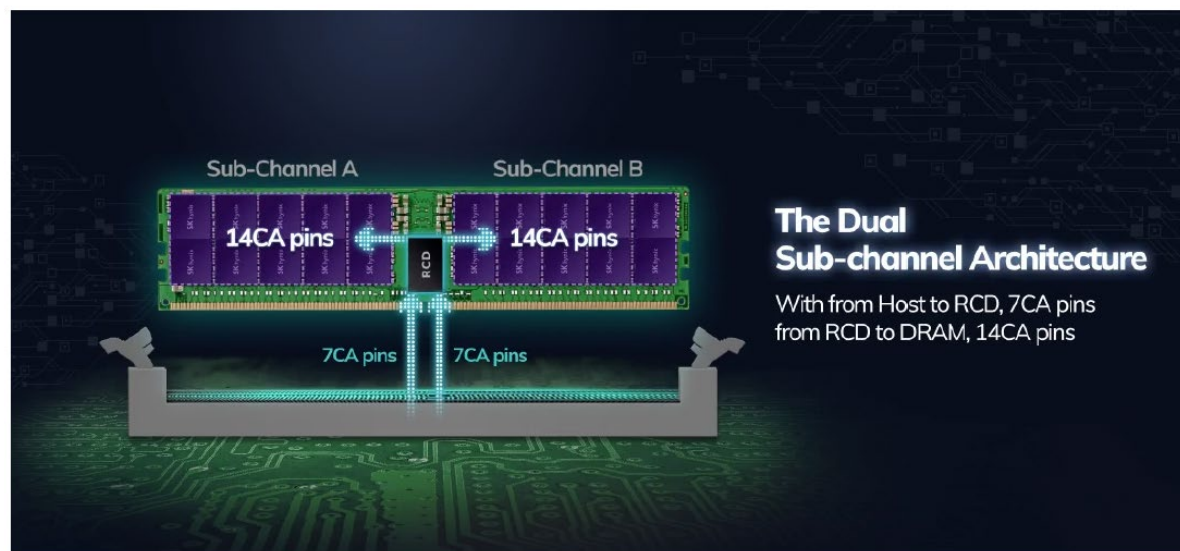
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



Ex. CC-65 (SK Hynix Develops MCR DIMM) at 2.

Double Sub-channel DRAM Architecture

With a dual sub-channel structure, DDR5 RDIMM increases the number of CA (Command-Address) pins to seven double-data-rate CA buses each from host to RCD and 14 single-data-rate buses each on RCD to DRAM. Having two independent 40-bit channels thus enables 2x bandwidth and higher bus (signal) efficiency compared to a single-channel DDR4 DIMM.



Ex. CC-41 (SK hynix RDIMM) at 2.



U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

SERVER DDR5 RDIMM

Ex. CC-34 (SK hynix Products).

RDIMM

VIEW MORE



DDR5
Generation

16-256GB
Module Density

4800-6400Mbps and higher
Speed

DDR4
Generation

16-256GB
Module Density

2666-3200Mbps
Speed

Application



- DIMM with a register (Register Clock Driver, RCD) for command/address signals between the DRAM and system memory controller
- Supports x4, x8 DRAM and x4 3DS stacks

MRDIMM

VIEW MORE



DDR5
Generation

32-128GB
Module Density

up to 8800Mbps
Speed

Application



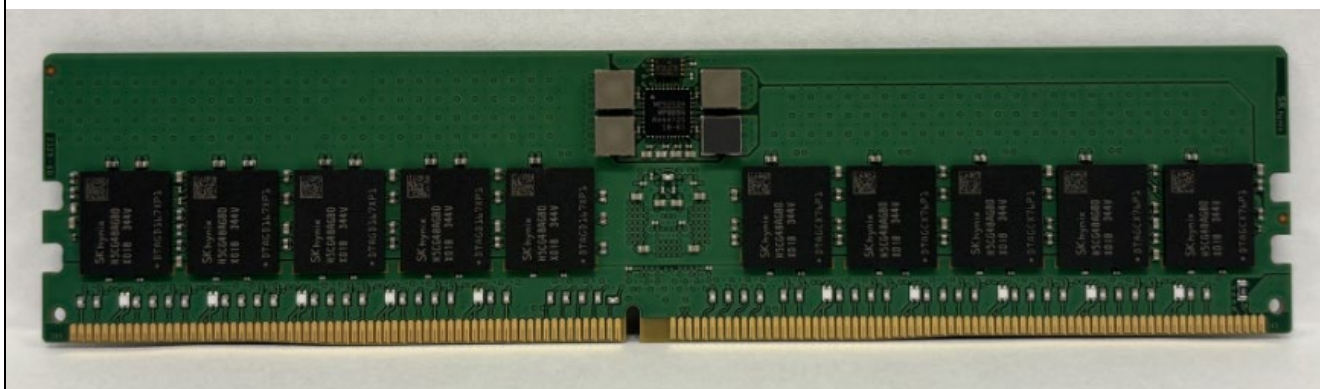
- DIMM with a register (Register Clock Driver, Data Buffer) for enabling simultaneous operation of two ranks, which increases bandwidth.
- Supports x4, x8 DRAM

Ex. CC-35 (SK hynix Module).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



Ex. CC-36 at IMG_5114 (showing SK hynix DDR5 RDIMM Product).

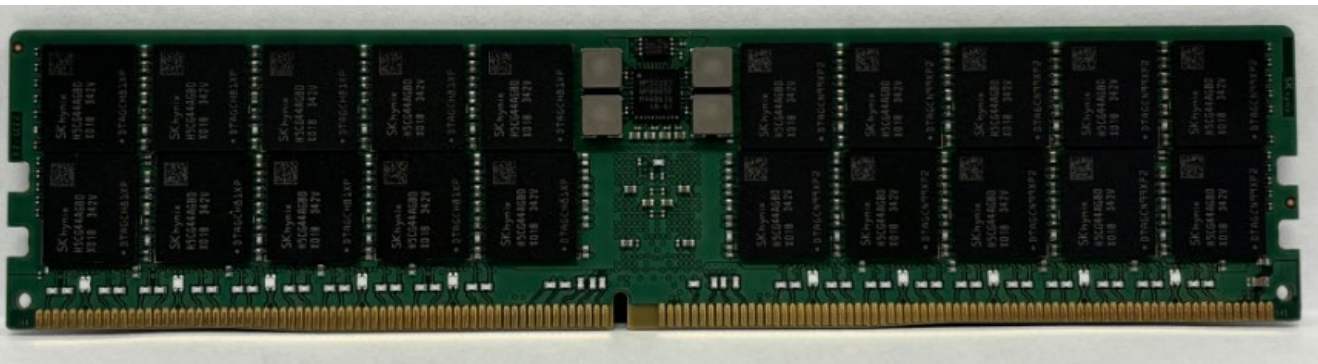


Ex. CC-36 at IMG_5117 (showing SK hynix DDR5 RDIMM Product).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



Ex. CC-36 at IMG_5112 (showing Netlist DDR5 RDIMM Overclock Product).



Ex. CC-36 at IMG_5109 (showing Netlist DDR5 RDIMM Overclock Product).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

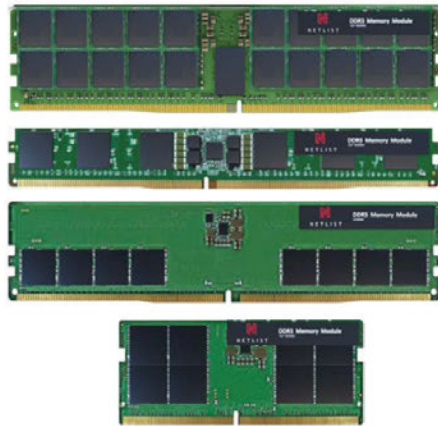


Ex. CC-36 at IMG_5126 (showing Netlist DDR5 MRDIMM Product).



Ex. CC-36 at IMG_5122 (showing Netlist DDR5 MRDIMM Product).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

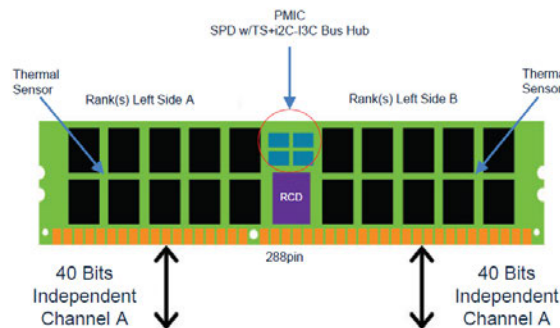


Netlist's DDR5 DIMM products are designed to address a broad range of applications including Enterprise and Cloud

DDR5 DIMM is a new architecture family with several different form factors to support the needs of the latest processors and emerging applications.

Core counts of new processors continue to accelerate year over year. DDR5 feature and performance capability allow CPU cores to scale due to an increase of 2X bandwidth on a DDR5 DIMM. DDR5 memory bandwidth and capacity will scale and be cost-effective with reliability to meet the ever-growing demands of future applications. Diversification of memory solutions and technology is needed to improve and support this scale-out effect. Enterprise, Cloud Hosting, Edge, Telco, HPC, AI/ML, and Data Centers are just a few examples. Netlist DDR5 family provides a long-term DIMM roadmap on Registered DIMM, Unbuffered DIMM ECC and non ECC, and SODIMM form factor to support these multiple applications.

Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1.



DIMM Changes from DDR4 to DDR5

Feature	DDR4	DDR5
Frequency (GT/s)	1.6~3.2	4.8~5.6
Voltage (V)	1.2	1.1
Power Supply	PMIC on MB	PMIC on DIMM
Temp sensor device	1 per DIMM	2 or 3 per DIMM
Side bus protocol	I2C bus	Legacy SMBus, I2C, I3C
Sideband bus access	Direct Access	1xC Hub in SPD
Channel Architecture	72-bit data channel 64 data + 8 ECC 1 channel per DIMM	40-bit data channel 32 data + 8 ECC 2 channel per DIMM
DFE Support	DB Only	RCD, DB, and DRAM

Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Netlist's Lightning Overclocked DDR5 RDIMM products just made performance upgrades easier in next generation workstations and servers  Netlist's Lightning RDIMM family delivers exceptional performance and unrivaled speed, empowering users to unleash the full potential of their computing systems. Netlist overclocked DDR5 RDIMM products represent an exciting advancement in memory technology, pushing the boundaries of performance and responsiveness in computing systems. With the transition to DDR5, Netlist's overclocking capabilities enable users to achieve higher memory speeds, enhanced data transfer rates, and increased bandwidth, resulting in improved overall system performance. By leveraging advanced engineering techniques and innovative circuitry, Netlist's overclocked DDR5 RDIMMs provide users with the flexibility to fine-tune their memory settings and unlock the full potential of their systems. These cutting-edge products offer a compelling solution for Workstations, and Servers seeking optimal performance and a seamless computing experience in the ever-evolving landscape of modern technology. Each DIMM is tested to run on x86 platforms at the programmed XMP profile latency timing. Ex. CC-57 (Netlist Lightning DDR5 High Perform. Module Product Brief) at 1. Furthermore, the electrical connections provided by the Practicing '407 Products' connectors are between the memory module and the system bus. For example, the Practicing '407 Products (comprising memory modules) include edge connectors with pin assignments for clock signal lines, control/address (C/A) signal lines, and data signal lines included in the system bus. <i>See, e.g.,</i>
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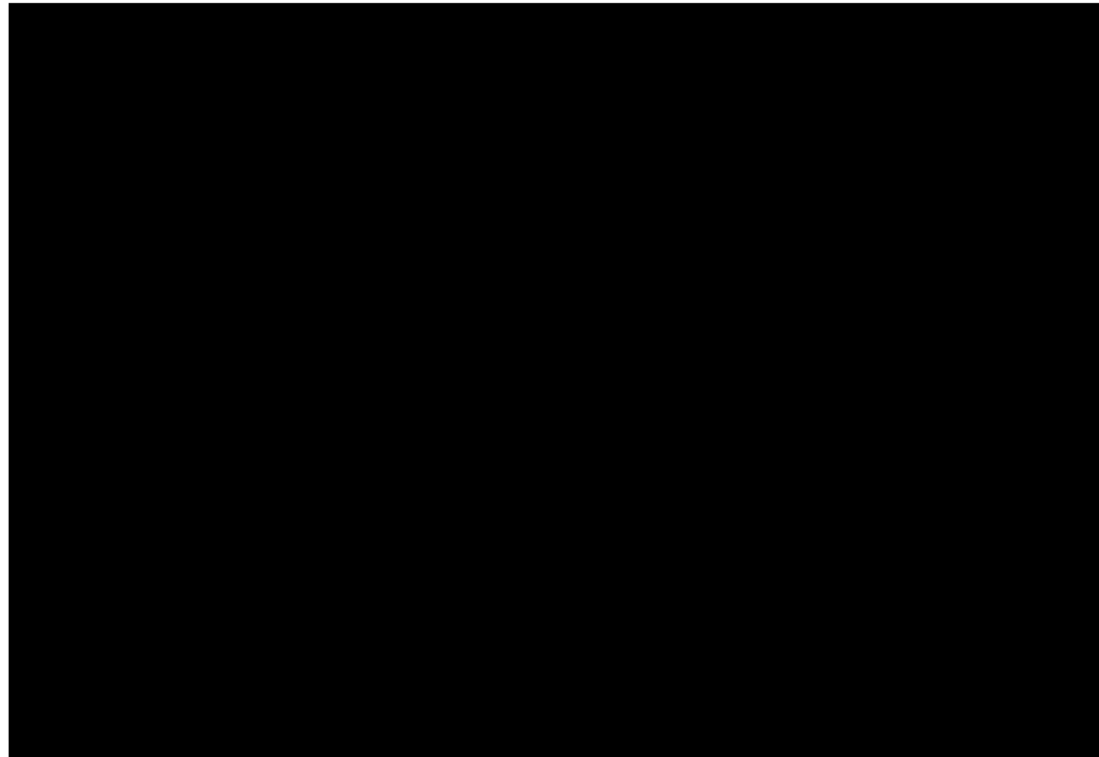
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 1.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 4.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 5.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 5.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 6.

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 7.
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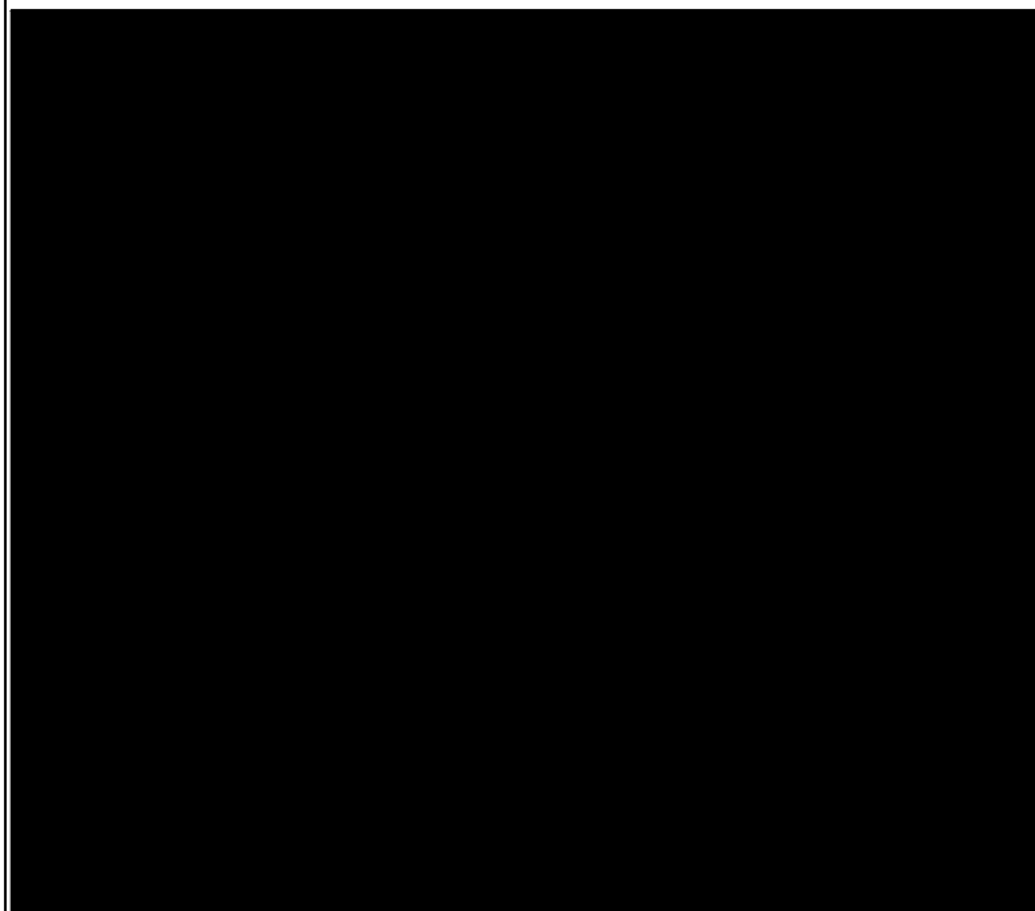
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 8.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	A large black rectangular redaction box covering the majority of the table cell content. Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 9.
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 10.

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 1

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 4.

Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 5.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

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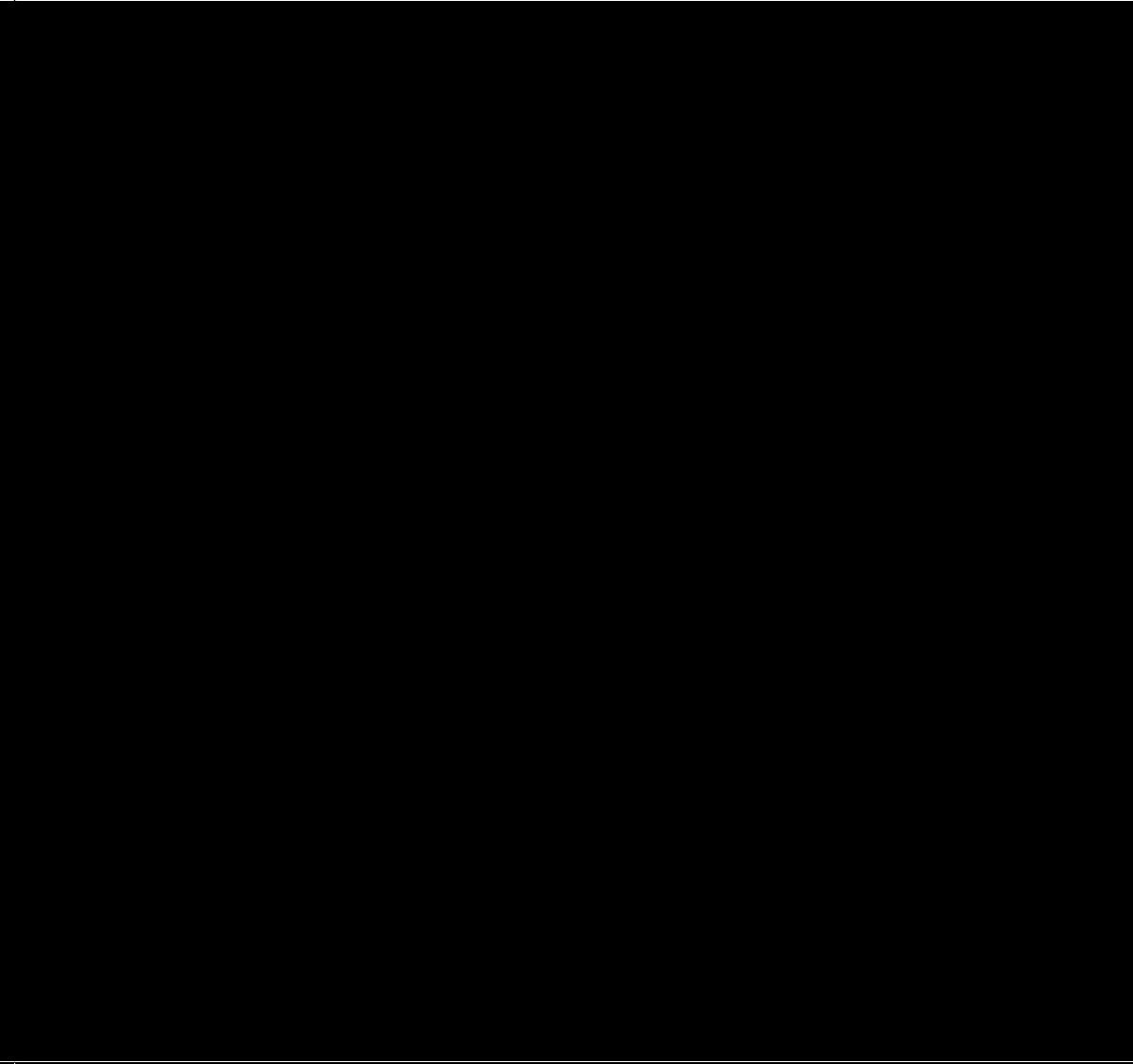
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 5.
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Date	Description	Debit	Credit	Balance
	Opening Balance			
	Cash Sales			
	Sales Tax			
	Accounts Receivable			
	Inventory			
	Prepaid Insurance			
	Accounts Payable			
	Wages Payable			
	Interest Payable			
	Income Tax Payable			
	Retained Earnings			
	Total			

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 6.

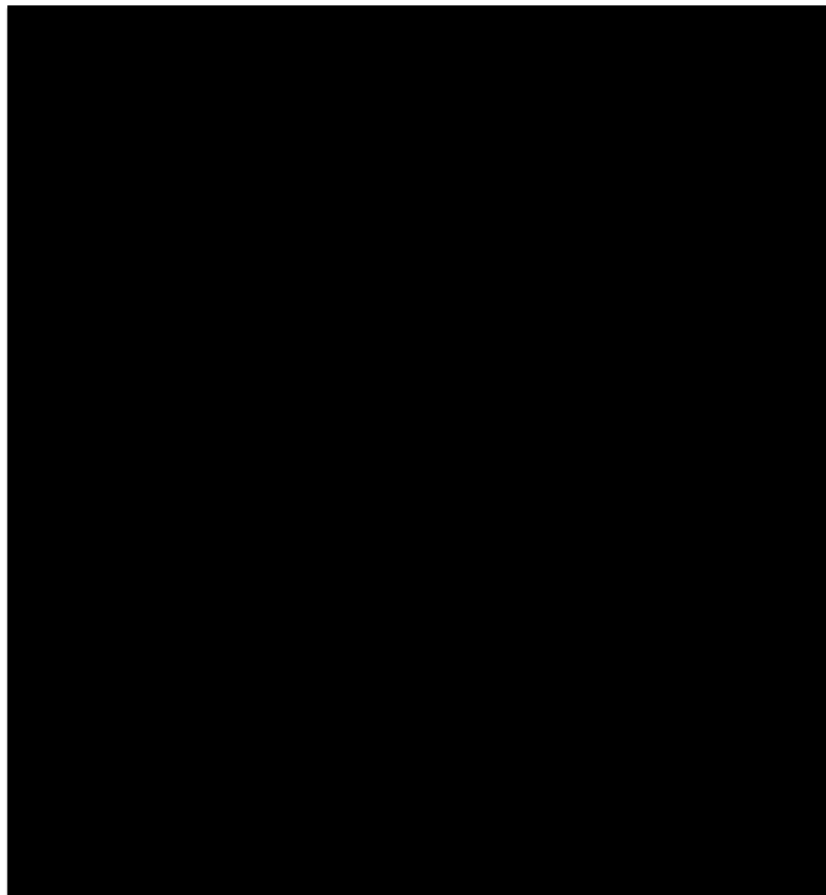


U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 7  Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 8.
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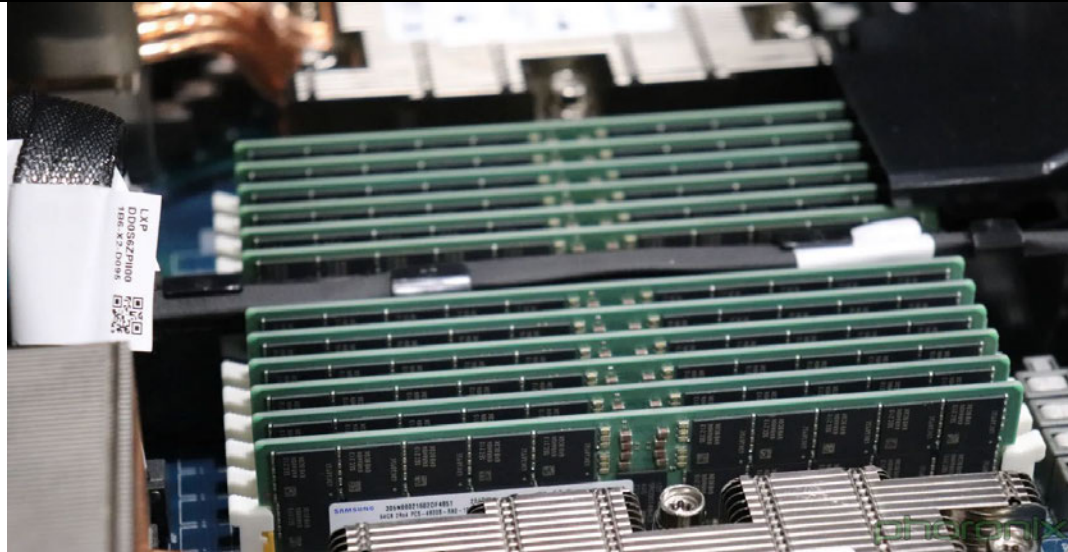
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 9.

As explained *supra*, the Practicing '407 Products are JEDEC standard compliant and are therefore configured to fit into a JEDEC-compliant memory slot, as illustrated below:

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



Ex. CC-70 (DDR5 Memory Channel Scaling Performance).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

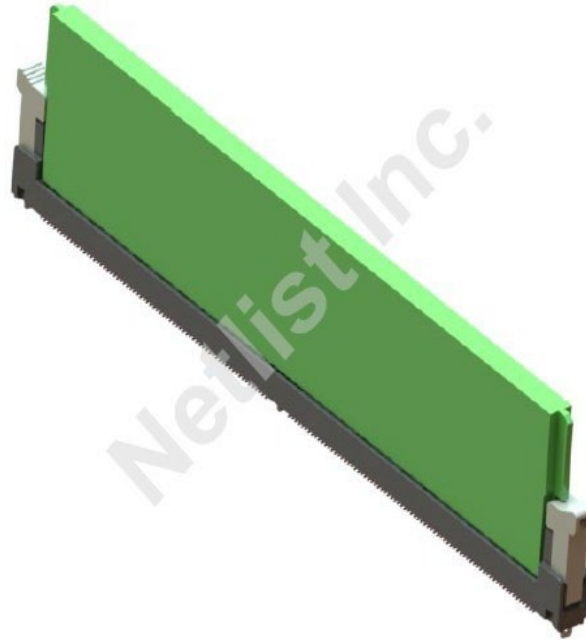


Figure 1 DDR5 socket and module (SMT)

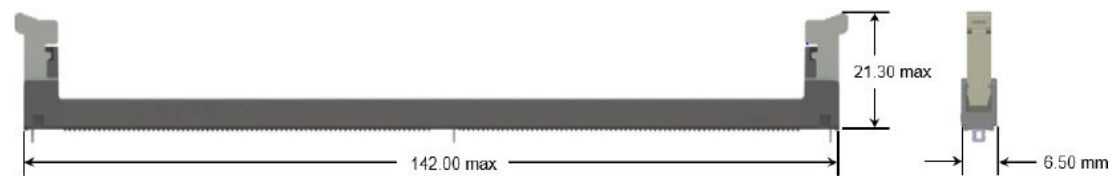
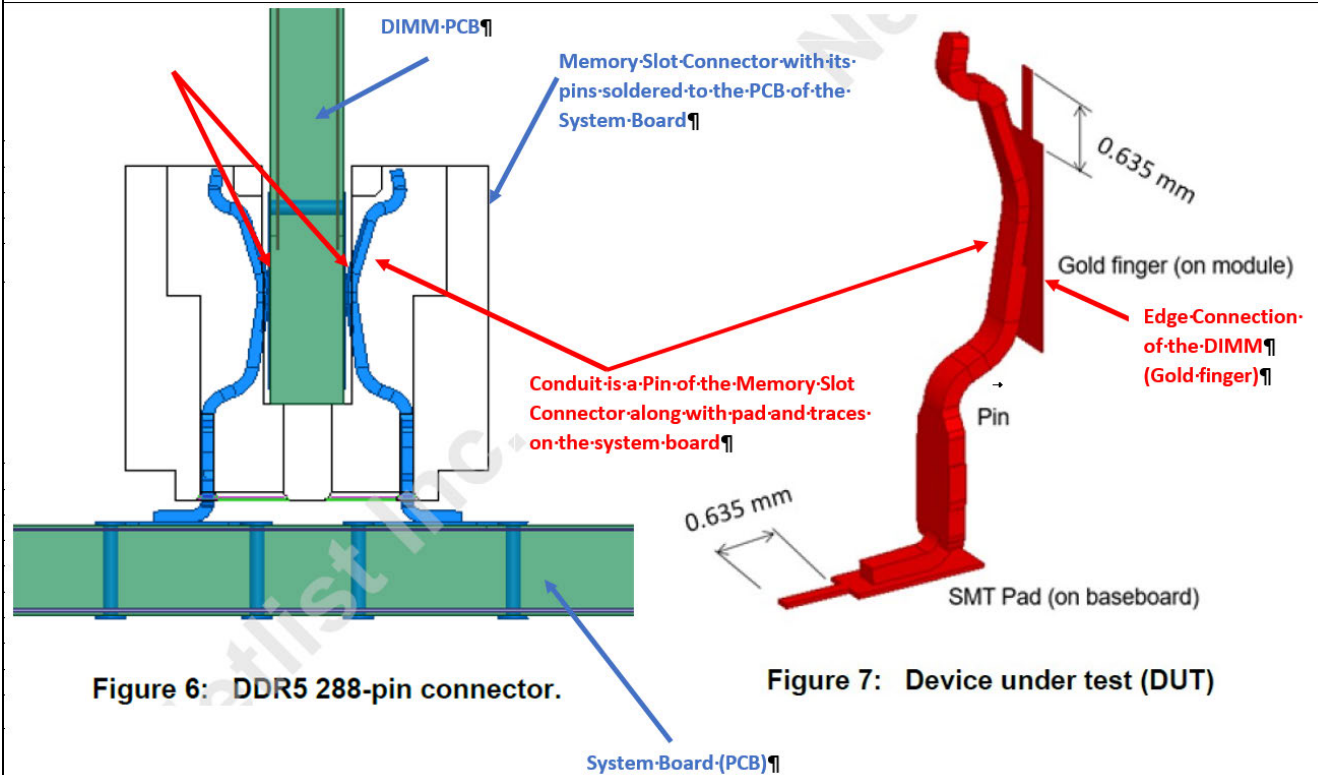


Figure 2 DDR5 U/R/LR DIMM Surface Mount (SMT) Connector Socket Outline

Ex. CC-73 (JEDEC PS-005B) at 6.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Contacting the RDIMM's edge connectors with the corresponding pins in the memory slot forms electrical connections between the RDIMM and the system bus:



Ex. CC-73 (JEDEC PS-005B) at 13, Figs. 6–7 (annotated) (illustrating an edge connection on a DIMM being electrically connected with a corresponding pin on the memory slot).

As explained *supra*, the Practicing '407 Products use a JEDEC compliant 288-pin edge connector. The JEDEC DDR5 RDIMM standard further demonstrates the Practicing '407 Products' edge connectors provide electrical connections between the memory module and the system bus because it includes connector pin assignments for clock signal lines, control/address (C/A) signal lines, and data signal lines included in the system bus. *See, e.g.,*

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Table 3 — Pin Definitions

Pin Name	Description	Pin Name	Description
CA[6:0]_A CA[6:0]_B	Command and Address (CA) Bus	DQ[31:0]_A DQ[31:0]_B	DIMM memory Data bus channel A and B
CS[1:0]_A_n CS[1:0]_B_n	Control (Chip Select) (CS_n)	CB[7:0]_A CB[7:0]_B	DIMM ECC Checkbits (CB) channel A and B
PAR_A PAR_B	Parity input	DQS[9:0]_A_t DQS[9:0]_B_t	Data Strobes (positive line of differential pair)
CK_t	Clock (true/positive)	DQS[9:0]_A_c DQS[9:0]_B_c	Data Strobes (negative line of differential pair)
CK_c	Clock (complement/negative)	TDQS[9:5]_A_t TDQS[9:5]_B_t	Not valid for x4 operation. Enabled via Mode Register.

Ex. CC-88 (JEDEC 305B) at 3.

Table 4 — Input/Output Functional Description

Symbol	Type	I/O Levels	Function
CK_t, CK_c,	Input	VDD	Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.
CA[6:0]_A CA[6:0]_B	Input	VDD	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during Mode Register Set commands.

Ex. CC-88 (JEDEC 305B) at 4.

DQ[31:0]_A DQ[31:0]_B	Input/ Output	VDD	Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.
CB[7:0]_A CB[7:0]_B	Input/ Output	VDD	ECC Checkbits Input/ Output: Bi-directional data bus - for 8-bit ECC (EC8) all 8 bits are used - for 4-bit ECC (EC4) [3:0] bits are used; [7:4] bits are floating

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-88 (JEDEC 305B) at 4.

Table 4 — Input/Output Functional Description (cont'd)

Symbol	Type	I/O Levels	Function
DQS[9:0]_A_t DQS[9:0]_B_t	Input/ Output	VDD	Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. The Data Strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential Data Strobe only and does not support single-ended.
DQS[9:0]_A_c DQS[9:0]_B_c			
TDQS[9:5]_A_t TDQS[9:5]_B_t	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.
TDQS[9:5]_A_c TDQS[9:5]_B_c	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.

Ex. CC-88 (JEDEC 305B) at 5.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Front Side Pin Label	Pin	Pin	Back side Pin Label	Front Side Pin Label	Pin	Pin	Back side Pin Label
VIN_BULK	1	145	VIN_BULK	PAR_A	74	218	CK_c
RFU	2	146	VIN_BULK	VSS	75	219	VSS
VIN_MGMT	3	147	PCAMP				Key
HSCL	4	148	HSA				
HSDA	5	149	RFU	CA0_B	76	220	RFU / No Pin Present
VSS	6	150	RFU	VSS	77	221	CA1_B
DQ0_A	7	151	VSS	CA2_B	78	222	VSS
VSS	8	152	DQ2_A	VSS	79	223	CA3_B
DQ1_A	9	153	VSS	CA4_B	80	224	VSS
VSS	10	154	DQ3_A	VSS	81	225	CA5_B
DQS0_A_t	11	155	VSS	CA6_B	82	226	VSS
DQS0_A_c	12	156	DQS5_A_c, TDQS5_A_c	VSS	83	227	PAR_B
VSS	13	157	DQS5_A_t, TDQS5_A_t	CS0_B_n	84	228	VSS
DQ4_A	14	158	VSS	VSS	85	229	CS1_B_n
VSS	15	159	DQ6_A	DLBDQ	86	230	VSS
DQ5_A	16	160	VSS	DLBDQS	87	231	RFU
VSS	17	161	DQ7_A	VSS	88	232	RFU
DQ8_A	18	162	VSS	CB4_B	89	233	VSS
VSS	19	163	DQ10_A	VSS	90	234	CB6_B
DQ9_A	20	164	VSS	CB5_B	91	235	VSS
VSS	21	165	DQ11_A	VSS	92	236	CB7_B
DQS1_A_t	22	166	VSS	DQS9_B_t, TDQS9_B_t	93	237	VSS
DQS1_A_c	23	167	DQS6_A_c, TDQS6_A_c	DQS9_B_c, TDQS9_B_c	94	238	DQS4_B_c
VSS	24	168	DQS6_A_t, TDQS6_A_t	VSS	95	239	DQS4_B_t
DQ12_A	25	169	VSS	CB0_B	96	240	VSS
VSS	26	170	DQ14_A	VSS	97	241	CB2_B
DQ13_A	27	171	VSS	CB1_B	98	242	VSS
VSS	28	172	DQ15_A	VSS	99	243	CB3_B
DQ16_A	29	173	VSS	DQ0_B	100	244	VSS
VSS	30	174	DQ18_A	VSS	101	245	DQ2_B
DQ17_A	31	175	VSS	DQ1_B	102	246	VSS
VSS	32	176	DQ19_A	VSS	103	247	DQ3_B
DQS2_A_t	33	177	VSS	DQS0_B_t	104	248	VSS
DQS2_A_c	34	178	DQS7_A_c, TDQS7_A_c	DQS0_B_c	105	249	DQS5_B_c, TDQS5_B_c
VSS	35	179	DQS7_A_t, TDQS7_A_t	VSS	106	250	DQS5_B_t, TDQS5_B_t
DQ20_A	36	180	VSS	DQ4_B	107	251	VSS
VSS	37	181	DQ22_A	VSS	108	252	DQ6_B
DQ21_A	38	182	VSS	DQ5_B	109	253	VSS

Ex. CC-88 (JEDEC 305B) at 6.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Front Side Pin Label	Pin	Pin	Back side Pin Label	Front Side Pin Label	Pin	Pin	Back side Pin Label
VSS	39	183	DQ23_A	VSS	110	254	DQ7_B
DQ24_A	40	184	VSS	DQ8_B	111	255	VSS
VSS	41	185	DQ26_A	VSS	112	256	DQ10_B
DQ25_A	42	186	VSS	DQ9_B	113	257	VSS
VSS	43	187	DQ27_A	VSS	114	258	DQ11_B
DQS3_A_t	44	188	VSS	DQS1_B_t	115	259	VSS
DQS3_A_c	45	189	DQS8_A_c, TDQS8_A_c	DQS1_B_c	116	260	DQS6_B_c, TDQS6_B_c
VSS	46	190	DQS8_A_t, TDQS8_A_t	VSS	117	261	DQS6_B_t, TDQS6_B_t
DQ28_A	47	191	VSS	DQ12_B	118	262	VSS
VSS	48	192	DQ30_A	VSS	119	263	DQ14_B
DQ29_A	49	193	VSS	DQ13_B	120	264	VSS
VSS	50	194	DQ31_A	VSS	121	265	DQ15_B
CB0_A	51	195	VSS	DQ16_B	122	266	VSS
VSS	52	196	CB2_A	VSS	123	267	DQ18_B
CB1_A	53	197	VSS	DQ17_B	124	268	VSS
VSS	54	198	CB3_A	VSS	125	269	DQ19_B
DQS4_A_t	55	199	VSS	DQS2_B_t	126	270	VSS
DQS4_A_c	56	200	DQS9_A_c, TDQS9_A_c	DQS2_B_c	127	271	DQS7_B_c, TDQS7_B_c
VSS	57	201	DQS9_A_t, TDQS9_A_t	VSS	128	272	DQS7_B_t, TDQS7_B_t
CB4_A	58	202	VSS	DQ20_B	129	273	VSS
VSS	59	203	CB6_A	VSS	130	274	DQ22_B
CB5_A	60	204	VSS	DQ21_B	131	275	VSS
VSS	61	205	CB7_A	VSS	132	276	DQ23_B
ALERT_n	62	206	VSS	DQ24_B	133	277	VSS
VSS	63	207	RESET_n	VSS	134	278	DQ26_B
CS0_A_n	64	208	VSS	DQ25_B	135	279	VSS
VSS	65	209	CS1_A_n	VSS	136	280	DQ27_B
CA0_A	66	210	VSS	DQS3_B_t	137	281	VSS
VSS	67	211	CA1_A	DQS3_B_c	138	282	DQS8_B_c, TDQS8_B_c
CA2_A	68	212	VSS	VSS	139	283	DQS8_B_t, TDQS8_B_t
VSS	69	213	CA3_A	DQ28_B	140	284	VSS
CA4_A	70	214	VSS	VSS	141	285	DQ30_B
VSS	71	215	CA5_A	DQ29_B	142	286	VSS
CA6_A	72	216	VSS	VSS	143	287	DQ31_B
VSS	73	217	CK_t	RFU	144	288	VSS

NOTE 1 Individual Annex standards for RDIMM define pin use where multiple functions are possible.
NOTE 2 Pin 220 for 8000 Mbps and faster RDIMM raw cards do not have the edge pin present.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Ex. CC-88 (JEDEC 305B) at 7.
[1.b] memory devices mounted on the PCB and organized in a plurality of groups, wherein a respective group of the memory devices is configurable to communicate data with the memory controller via a corresponding subset of the data signal lines;	The Practicing '407 Products comprise memory devices mounted on the PCB and organized in a plurality of groups, wherein a respective group of the memory devices is configurable to communicate data with the memory controller via a corresponding subset of the data signal lines. For example, The Practicing '407 Products have memory devices (<i>e.g.</i>, SDRAM and DRAM devices) mounted on the PCB and organized in a plurality of groups. <i>See, e.g.</i>,: Ex. CC-33 (Netlist DDR5 DIMMs) (black memory devices mounted in rows on each side of the PCB). Netlist DDR5 MRDIMM products optimize AI training, inference, and HPC workloads with low latency and high bandwidth. Netlist's MRDIMM series offers exceptional performance and unmatched speed, enabling users to fully optimize their high performance computing systems without the need for any adjustments. Ex. CC-71 (Netlist DDR5 MRDIMM Product Brief) at 1 (black memory devices mounted in rows on each side of the PCB).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

In the server DRAM section, SK hynix presented a lineup of DDR5-based memory modules for servers. The spotlight was on the DDR5 RDIMM* and MRDIMM* products built using the 1c node,* the sixth generation of the 10 nm process technology. The company also showcased various DRAM solutions designed to maximize AI server performance and reduce power consumption, including its 96 GB DDR5 RDIMM, 256 GB 3DS* DDR5 RDIMM, and DDR5 MRDIMM in both 96 GB and 256 GB capacities.

* Registered Dual In-line Memory Module (RDIMM): A server memory module product in which multiple DRAM chips are mounted on a substrate.

* Multiplexed Rank Dual In-line Memory Module (MRDIMM): A server memory module product in which multiple DRAM chips are mounted on a substrate. Speed is enhanced by operating two ranks—the basic operating units of the module—simultaneously.

* 10 nm DRAM process technology has progressed through six generations: 1x, 1y, 1z, 1a, 1b, and 1c. In August 2024, SK hynix became the first in the world to successfully develop 1c process technology.

* 3D Stacked Memory (3DS): A high-bandwidth memory product in which two or more DRAM chips are packaged together and interconnected using TSV technology.

Ex. CC-63 (SK Hynix Blog Post) at 5.



SERVER DDR5 RDIMM

Ex. CC-34 (SK hynix Products) (black memory devices mounted in rows on each side of the PCB).

RDIMM

[VIEW MORE](#)



DDR5
Generation

16-256GB
Module Density

4800-6400Mbps and higher
Speed

DDR4
Generation

16-256GB
Module Density

2666-3200Mbps
Speed

Application



Server

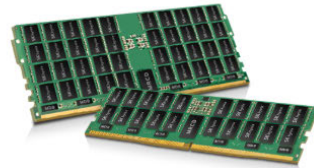


Networking

- DIMM with a register (Register Clock Driver, RCD) for command/address signals between the DRAM and system memory controller
- Supports x4, x8 DRAM and x4 3DS stacks

MRDIMM

[VIEW MORE](#)



DDR5
Generation

32-128GB
Module Density

up to 8800Mbps
Speed

Application



Server

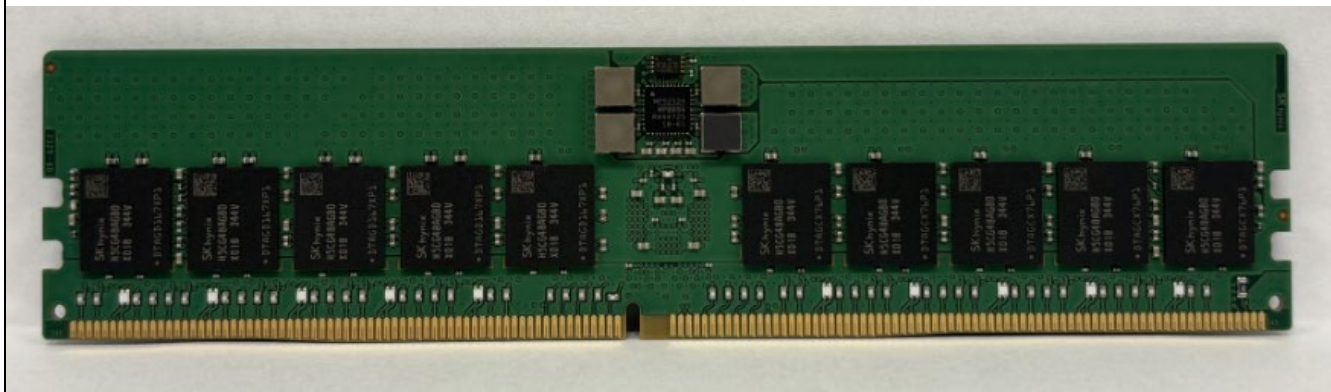
- DIMM with a register (Register Clock Driver, Data Buffer) for enabling simultaneous operation of two ranks, which increases bandwidth.
- Supports x4, x8 DRAM

Ex. CC-35 (SK hynix Module) (black memory devices mounted in rows on each side of the PCB).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



Ex. CC-36 at IMG_5114 (showing SK hynix DDR5 RDIMM Product) (black memory devices mounted in rows on each side of the PCB).

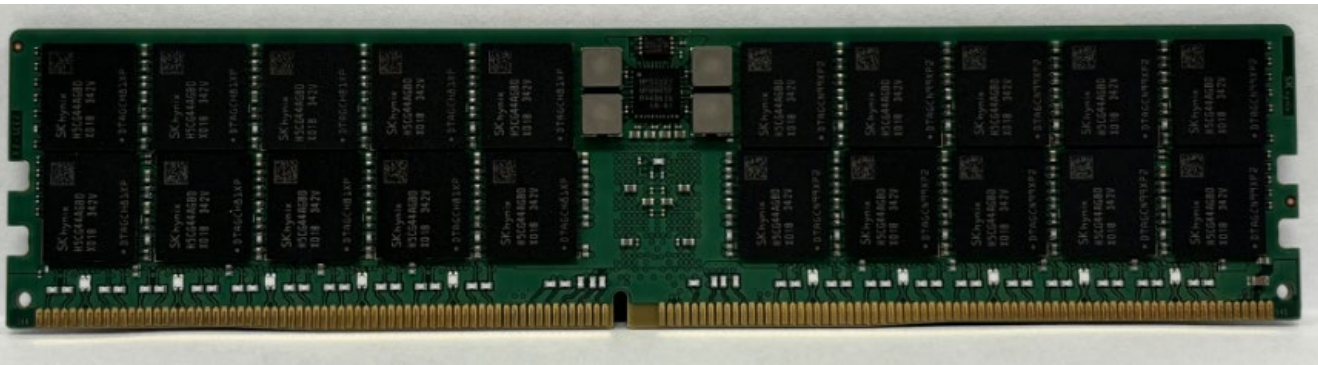


Ex. CC-36 at IMG_5117 (showing SK hynix DDR5 RDIMM Product) (black memory devices mounted in rows on each side of the PCB).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

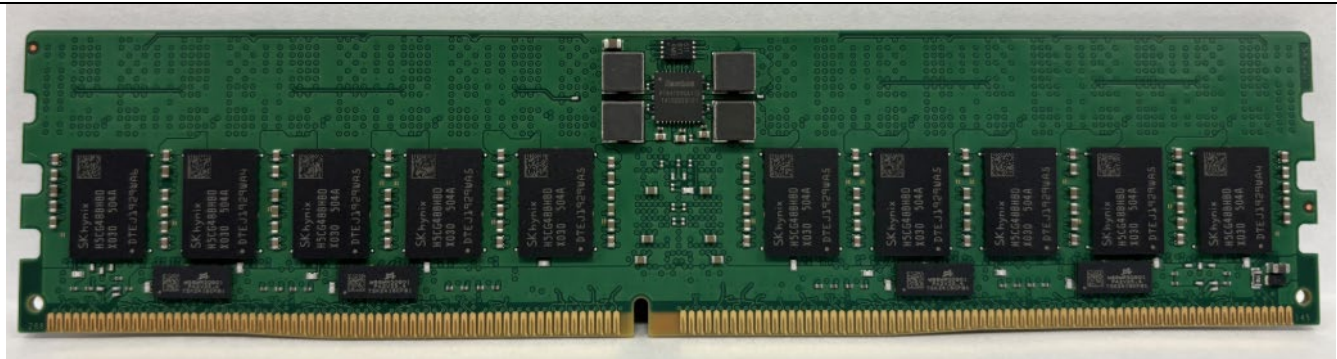


Ex. CC-36 at IMG_5112 (showing Netlist DDR5 RDIMM Overclock Product) (black memory devices mounted in rows on each side of the PCB).



Ex. CC-36 at IMG_5109 (showing Netlist DDR5 RDIMM Overclock Product) (black memory devices mounted in rows on each side of the PCB).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

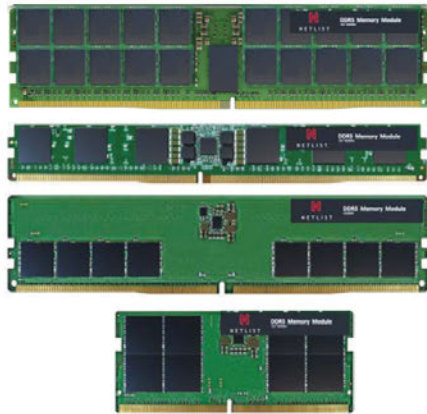


Ex. CC-36 at IMG_5126 (showing Netlist DDR5 MRDIMM Product) (black memory devices mounted in rows on each side of the PCB).



Ex. CC-36 at IMG_5122 (showing Netlist DDR5 MRDIMM Product) (black memory devices mounted in rows on each side of the PCB).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

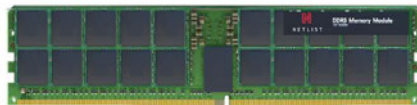


Netlist's DDR5 DIMM products are designed to address a broad range of applications including Enterprise and Cloud

DDR5 DIMM is a new architecture family with several different form factors to support the needs of the latest processors and emerging applications.

Core counts of new processors continue to accelerate year over year. DDR5 feature and performance capability allow CPU cores to scale due to an increase of 2X bandwidth on a DDR5 DIMM. DDR5 memory bandwidth and capacity will scale and be cost-effective with reliability to meet the ever-growing demands of future applications. Diversification of memory solutions and technology is needed to improve and support this scale-out effect. Enterprise, Cloud Hosting, Edge, Telco, HPC, AI/ML, and Data Centers are just a few examples. Netlist DDR5 family provides a long-term DIMM roadmap on Registered DIMM, Unbuffered DIMM ECC and non ECC, and SODIMM form factor to support these multiple applications.

Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1 (black memory devices mounted in rows on each side of the PCB).



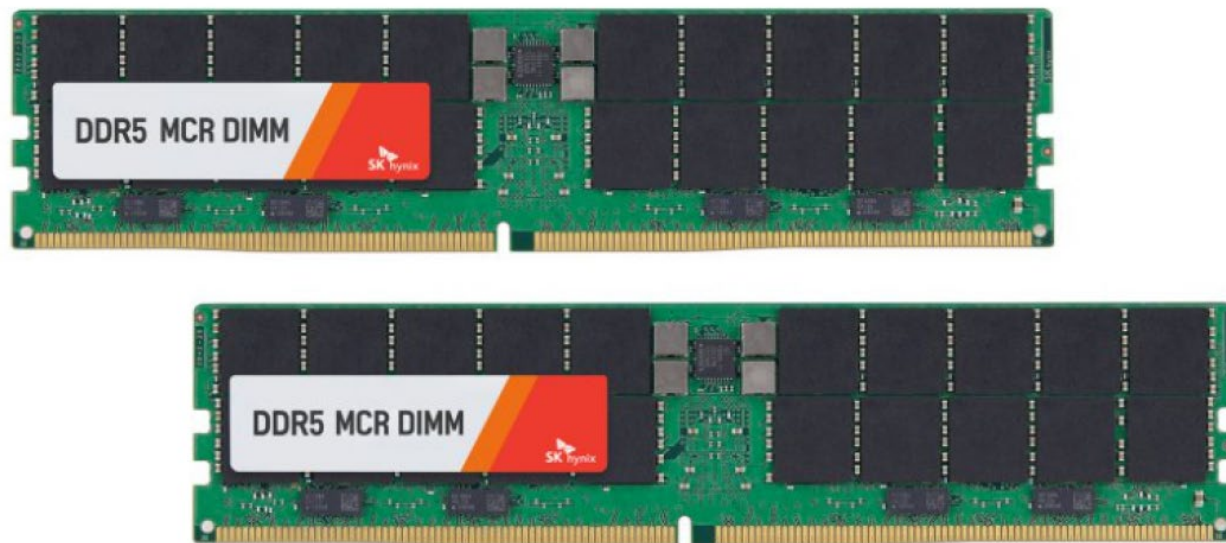
Netlist's Lightning Overclocked DDR5 RDIMM products just made performance upgrades easier in next generation workstations and servers

Netlist's Lightning RDIMM family delivers exceptional performance and unrivaled speed, empowering users to unleash the full potential of their computing systems.

Netlist overclocked DDR5 RDIMM products represent an exciting advancement in memory technology, pushing the boundaries of performance and responsiveness in computing systems. With the transition to DDR5, Netlist's overclocking capabilities enable users to achieve higher memory speeds, enhanced data transfer rates, and increased bandwidth, resulting in improved overall system performance. By leveraging advanced engineering techniques and innovative circuitry, Netlist's overclocked DDR5 RDIMMs provide users with the flexibility to fine-tune their memory settings and unlock the full potential of their systems. These cutting-edge products offer a compelling solution for Workstations, and Servers seeking optimal performance and a seamless computing experience in the ever-evolving landscape of modern technology. Each DIMM is tested to run on x86 platforms at the programmed XMP profile latency timing.

Ex. CC-57 (Netlist Lightning DDR5 High Perform. Module Product Brief) at 1 (black memory devices mounted in rows on each side of the PCB).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



Ex. CC-65 (SK Hynix Develops MCR DIMM) at 2 (black memory devices mounted in rows on each side of the PCB).

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 1 [REDACTED] [REDACTED]
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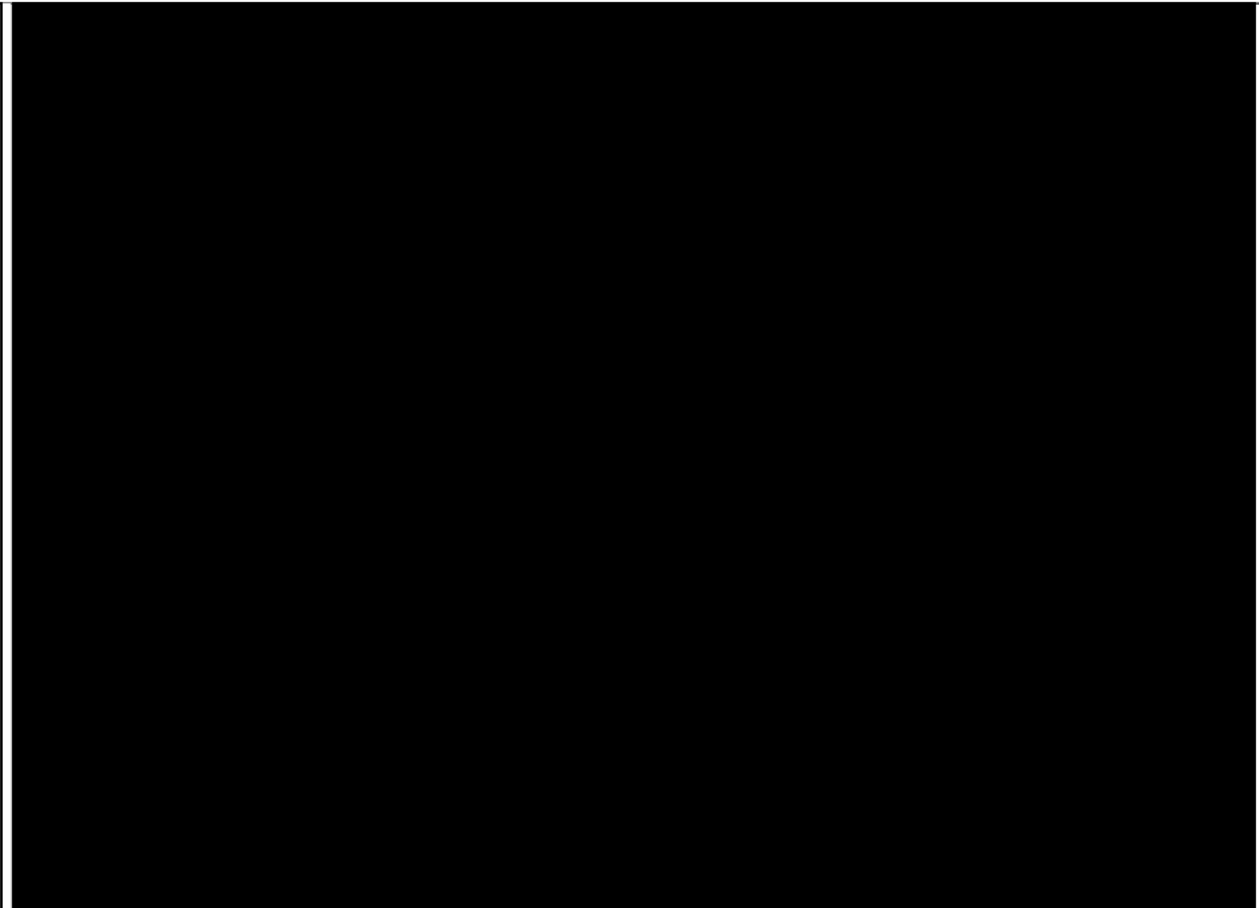
[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 10 [REDACTED] [REDACTED]
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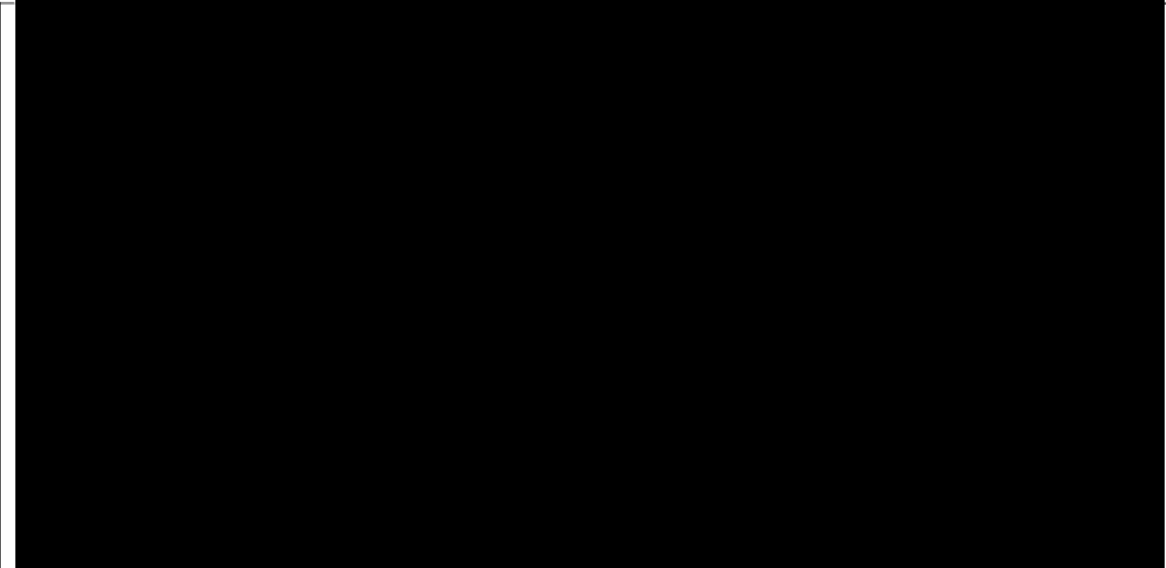
[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

		
	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 1 [REDACTED]	

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	 Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 9 [REDACTED] [REDACTED]
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

DDR5 is the new standard of double data rate (DDR) DRAM that improves on DDR4 in both speed and power. SK hynix presents the world's first DDR5 in a server solution that will enable the era of AI and big data.

See the Official Press Release

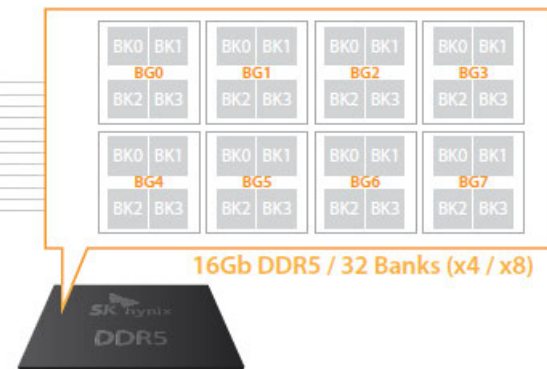
<https://news.skhynix.com/sk-hynix-launches-worlds-first-ddr5-dram/>



Key Benefits

Dynamic Scalability

2x memory access availability over DDR4 with 32 banks in 8 bank groups, both twice that of DDR4

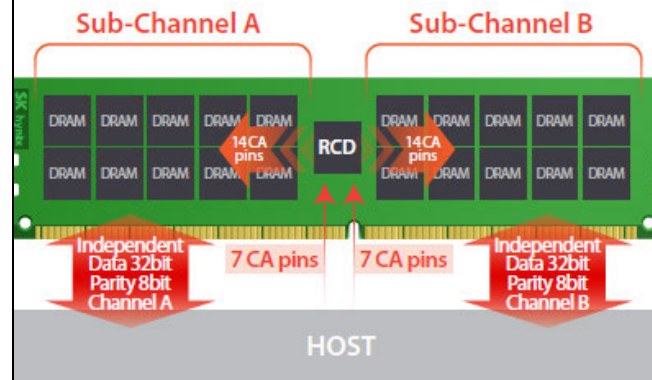


Ex. CC-40 (SK Hynix A New Standard) at 3.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

The latest DDR5 DRAM standard turbo-boosts RDIMM module performance and technology for the era of AI and Big Data. SK hynix expects its DDR5 RDIMMs to reach over 8,000Mbps in data rates with research into continued speed enhancements.

Features



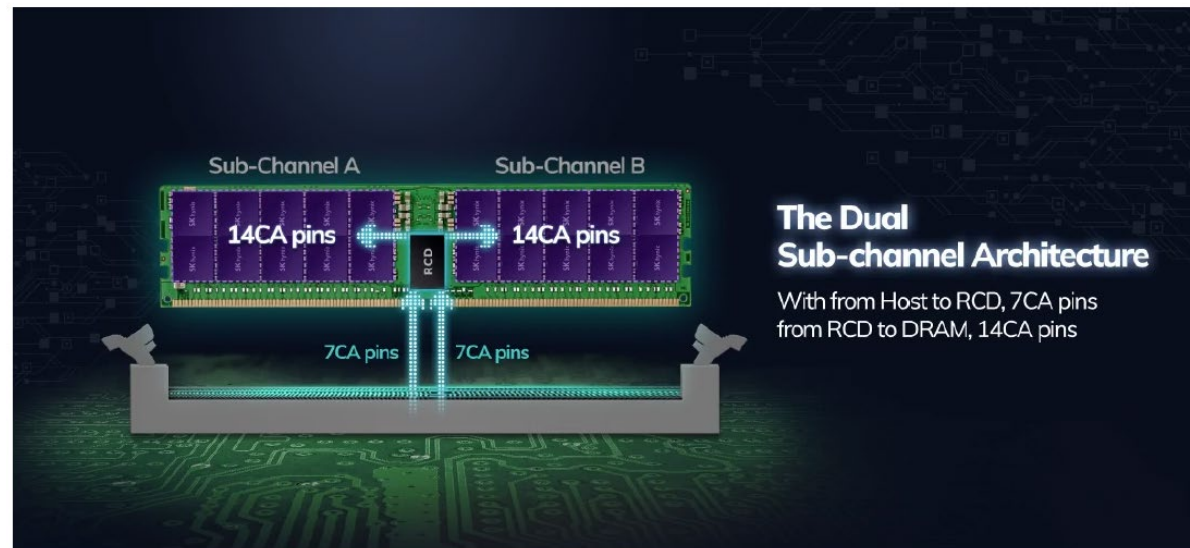
| New Sub-channel DRAM Architecture

The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

Ex. CC-40 (SK Hynix A New Standard) at 5.

Double Sub-channel DRAM Architecture

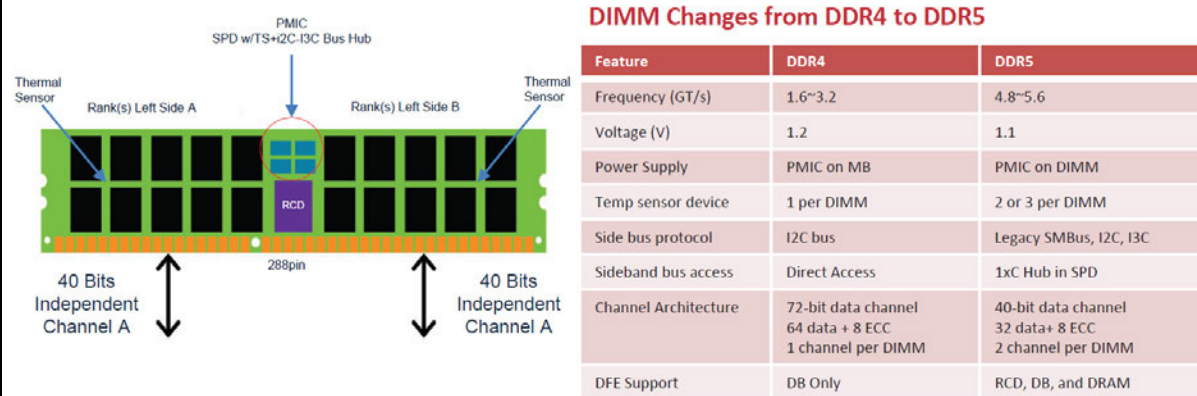
With a dual sub-channel structure, DDR5 RDIMM increases the number of CA (Command-Address) pins to seven double-data-rate CA buses each from host to RCD and 14 single-data-rate buses each on RCD to DRAM. Having two independent 40-bit channels thus enables 2x bandwidth and higher bus (signal) efficiency compared to a single-channel DDR4 DIMM.



Ex. CC-41 (SK hynix RDIMM) at 2.

Furthermore, the memory devices are organized in a plurality of groups, wherein a respective group of the memory devices is configured to communicate data with the memory controller via a corresponding subset of the data signal lines. For example, the Practicing '407 Products include memory devices that are organized into groups that communicate data with the memory controller via subsets of data signal lines within two independent channels. *See, e.g.,*:

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1.

DDR5 RDIMM Features

Features	Description
Frequency (GT/s)	4.8~6.4
Voltage (VDDQ)	1.1 (Native) 1.4V (XMP)
Power Supply	PMIC on DIMM
Temp sensor device	2 or 3 per DIMM
Side bus protocol	Legacy SMBus, I2C, I3C
Sideband bus access	I3C Hub in SPD
Channel Architecture	40-bit data channel, 32 data+ 8 ECC 2 channel per DIMM
DFE Support	RCD and DRAM
PCB Mechanical Outline	JEDEC (Height 31.25mm)
Environmental	0°C – 95°C, RoHS Compliant and Halogen-Free

DDR5 RDIMM Capacity & Latency Timing Profiles

Capacity	Speed	Latency	Comment
ALL	5600Mbps	46-45-45@1.1V	JEDEC (Default)
ALL	5600Mbps	40-40-40@1.1V	JEDEC
16GB 1Rx8 32GB 1Rx4 32GB 2Rx8	6000Mbps	28-36-36@1.4V	XMP Profile #1
16GB 1Rx8 32GB 1Rx4 32GB 2Rx8 64GB 2Rx4	6000Mbps	30-36-36@1.4V	XMP Profile #2
16GB 1Rx8 32GB 1Rx4 32GB 2Rx8 64GB 2Rx4	6000Mbps	34-36-36@1.15V	XMP Profile #3

* Q4'23 availability

Ex. CC-57 (Netlist Lightning DDR5 High Perform. Module Product Brief) at 1 (“2 channel per DIMM.”).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	<table><tr><th>Features</th><th>Description</th></tr><tr><td>Bandwidth</td><td>Up to 8.8K MT/s, providing a 39% bandwidth upgrade</td></tr><tr><td>Latency</td><td>Up to 40% lower latency compared to traditional DDR5</td></tr><tr><td>Capacity</td><td>Available in sizes from 32GB to 128GB</td></tr><tr><td>Parallel DRAM Activation</td><td>Enables pairs of DRAM ranks to be activated and accessed in parallel</td></tr></table>	Features	Description	Bandwidth	Up to 8.8K MT/s, providing a 39% bandwidth upgrade	Latency	Up to 40% lower latency compared to traditional DDR5	Capacity	Available in sizes from 32GB to 128GB	Parallel DRAM Activation	Enables pairs of DRAM ranks to be activated and accessed in parallel	
Features	Description											
Bandwidth	Up to 8.8K MT/s, providing a 39% bandwidth upgrade											
Latency	Up to 40% lower latency compared to traditional DDR5											
Capacity	Available in sizes from 32GB to 128GB											
Parallel DRAM Activation	Enables pairs of DRAM ranks to be activated and accessed in parallel											
	Ex. CC-71 (Netlist DDR5 MRDIMM Product Brief) at 1 (“Parallel DRAM Activation”).											

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 4 [REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 10.

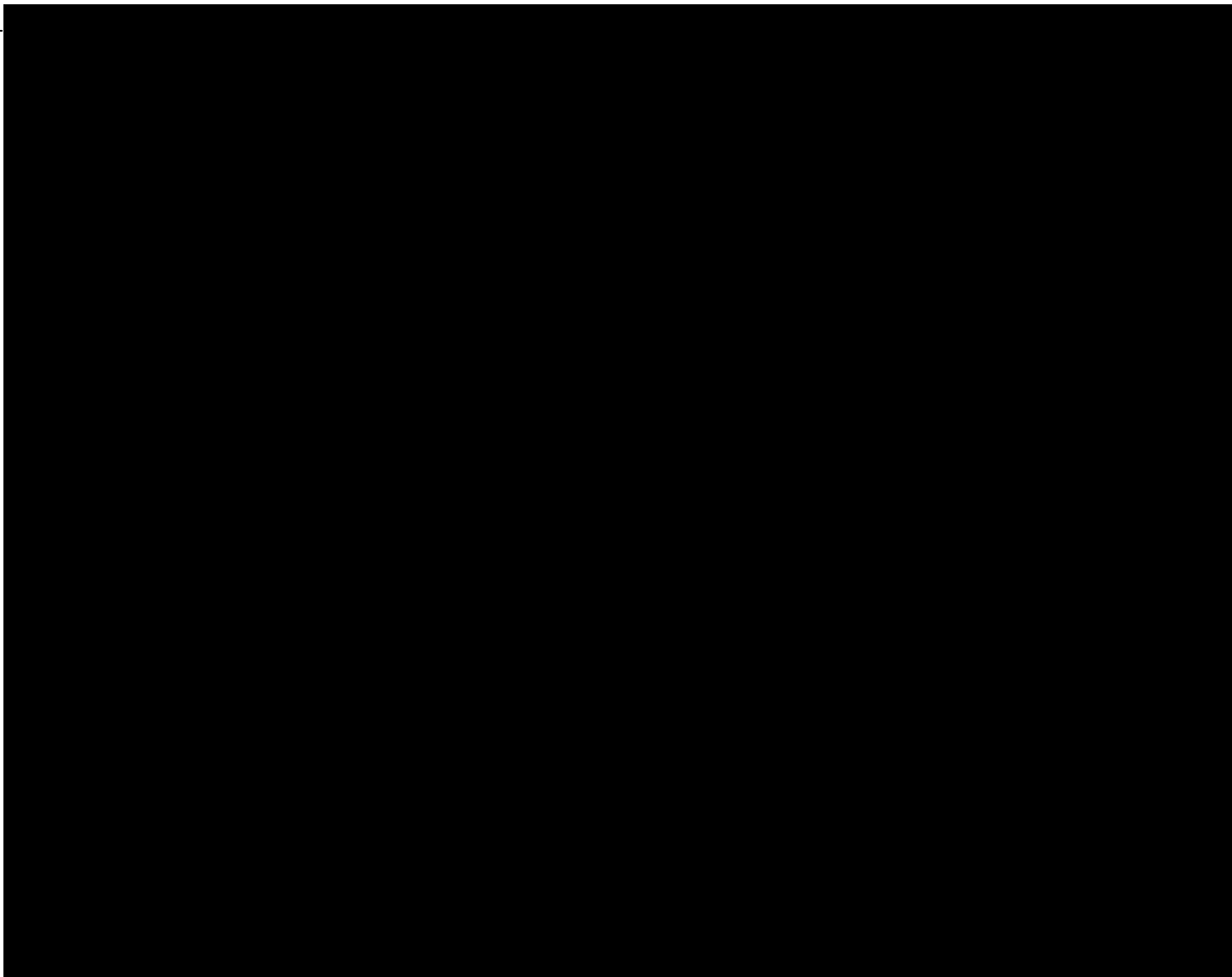
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 11.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

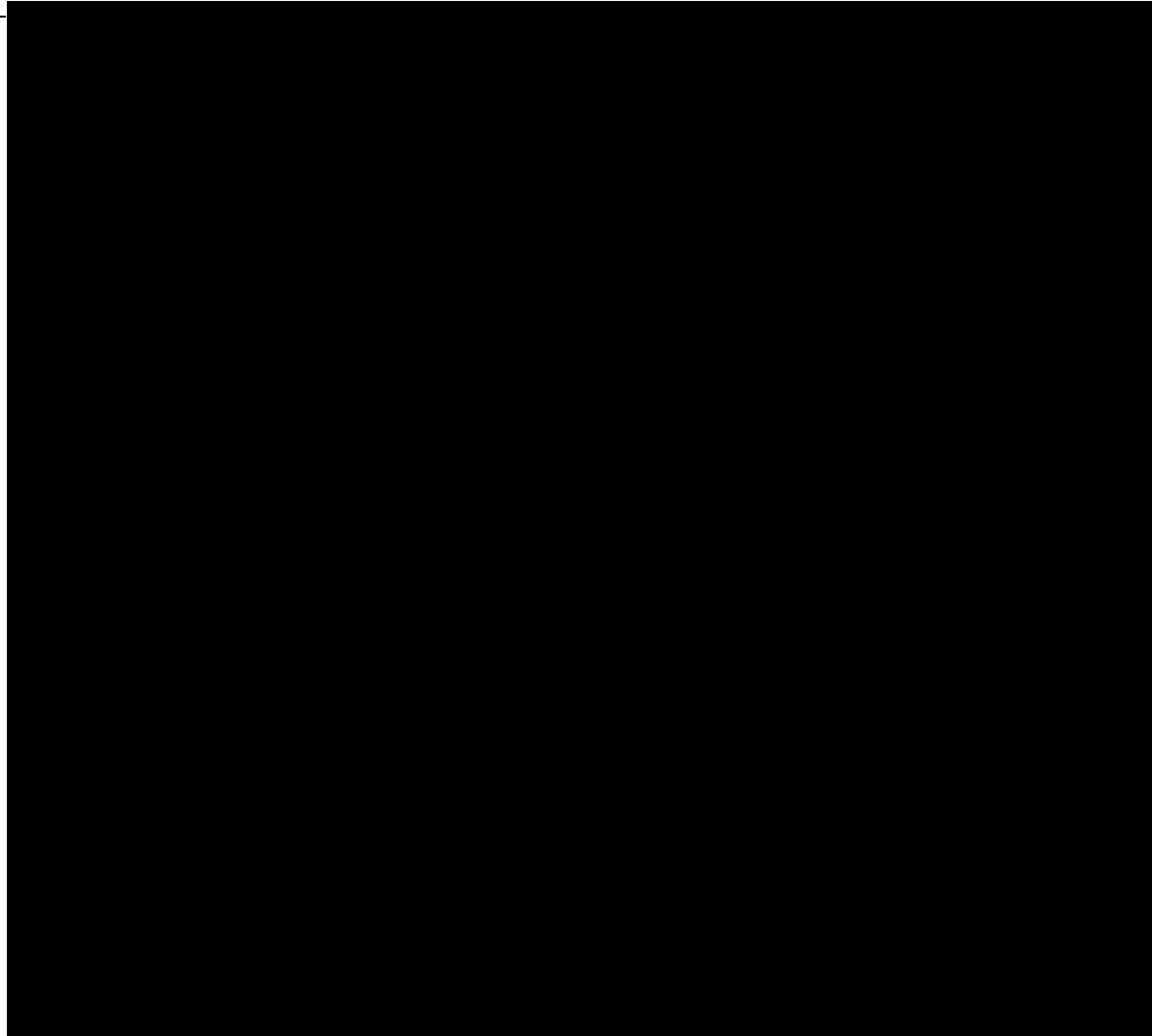
Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 12.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



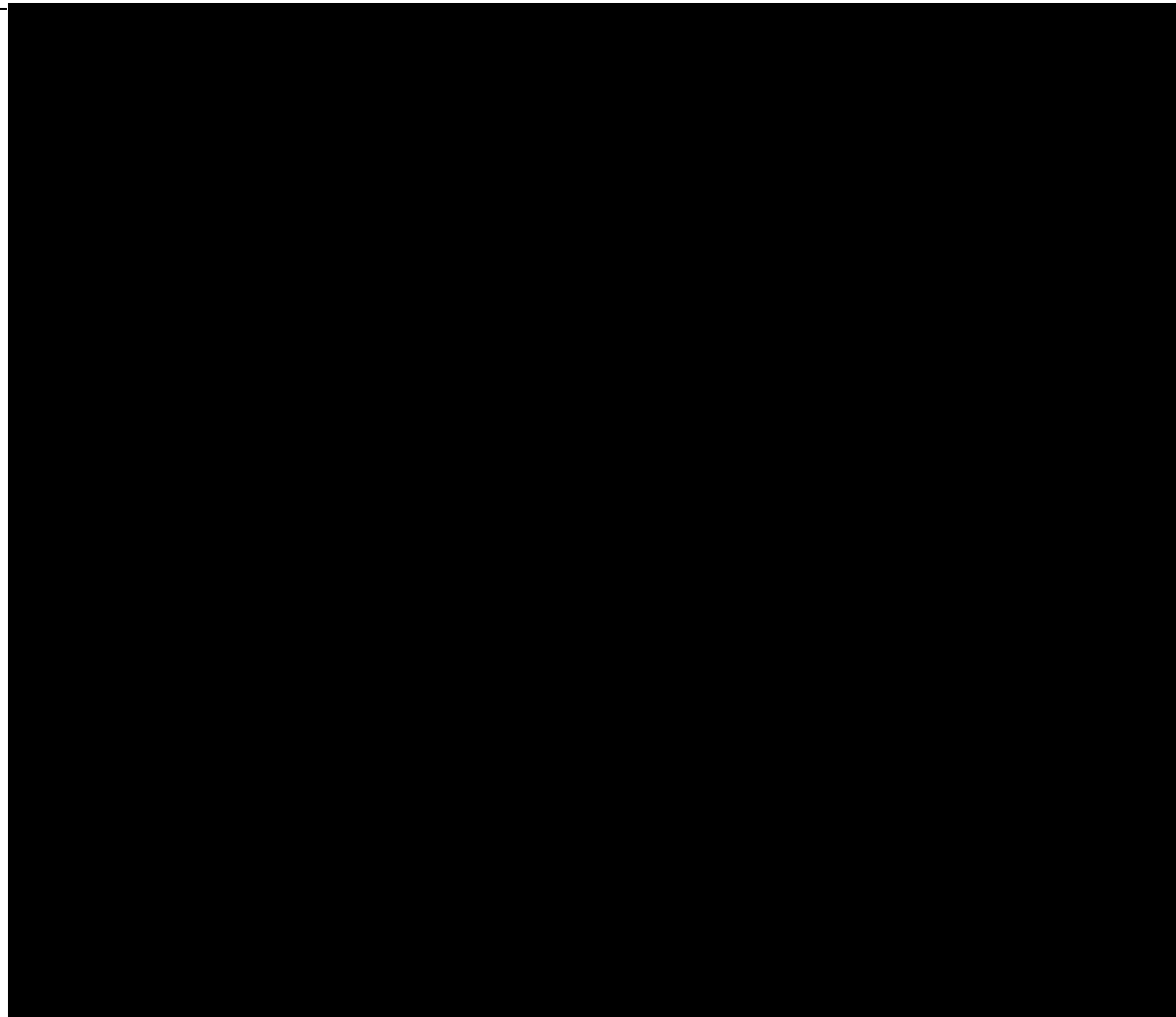
Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 10.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



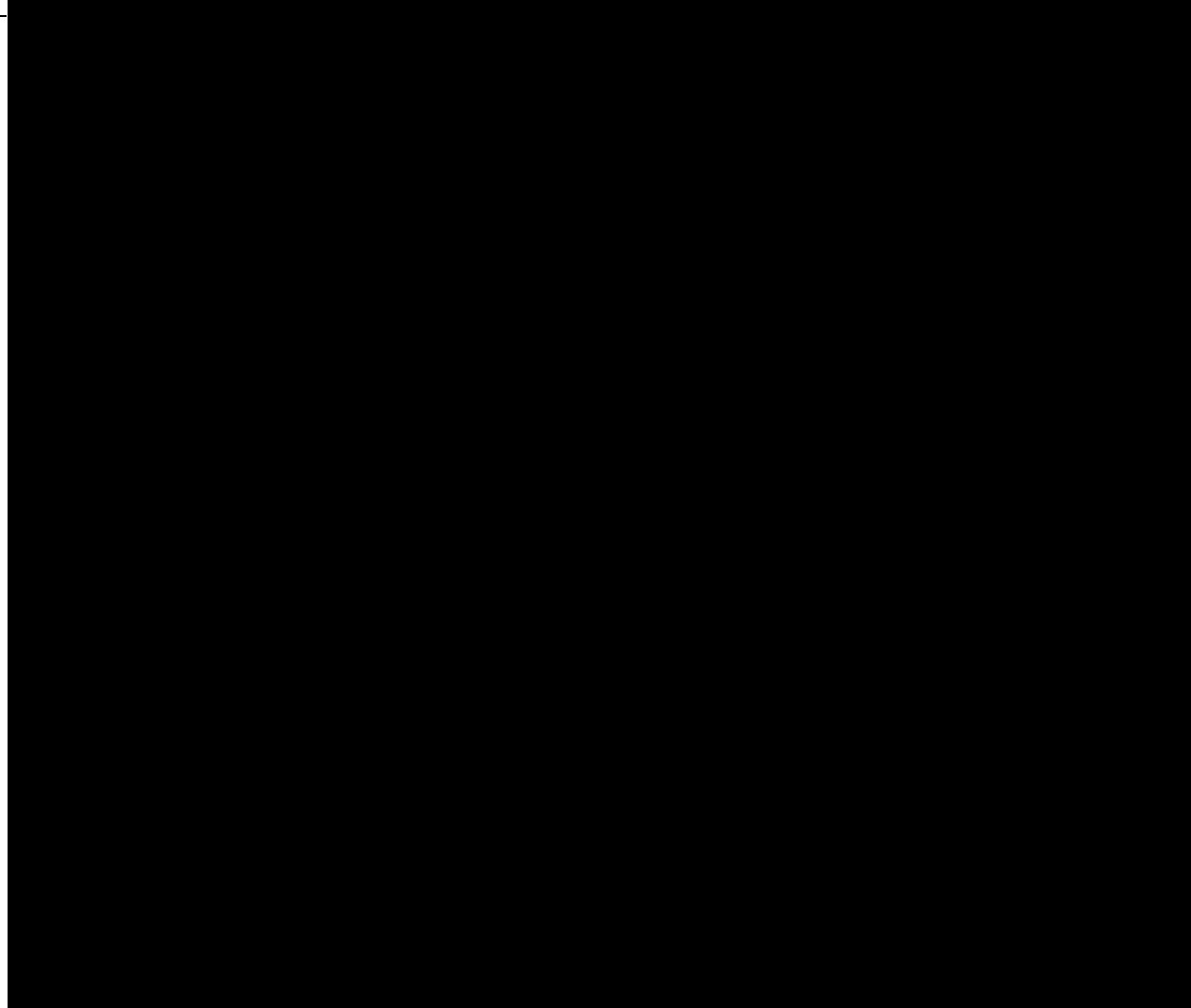
Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 11.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 12.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 10.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 11.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 10.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 11.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 12.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 10.

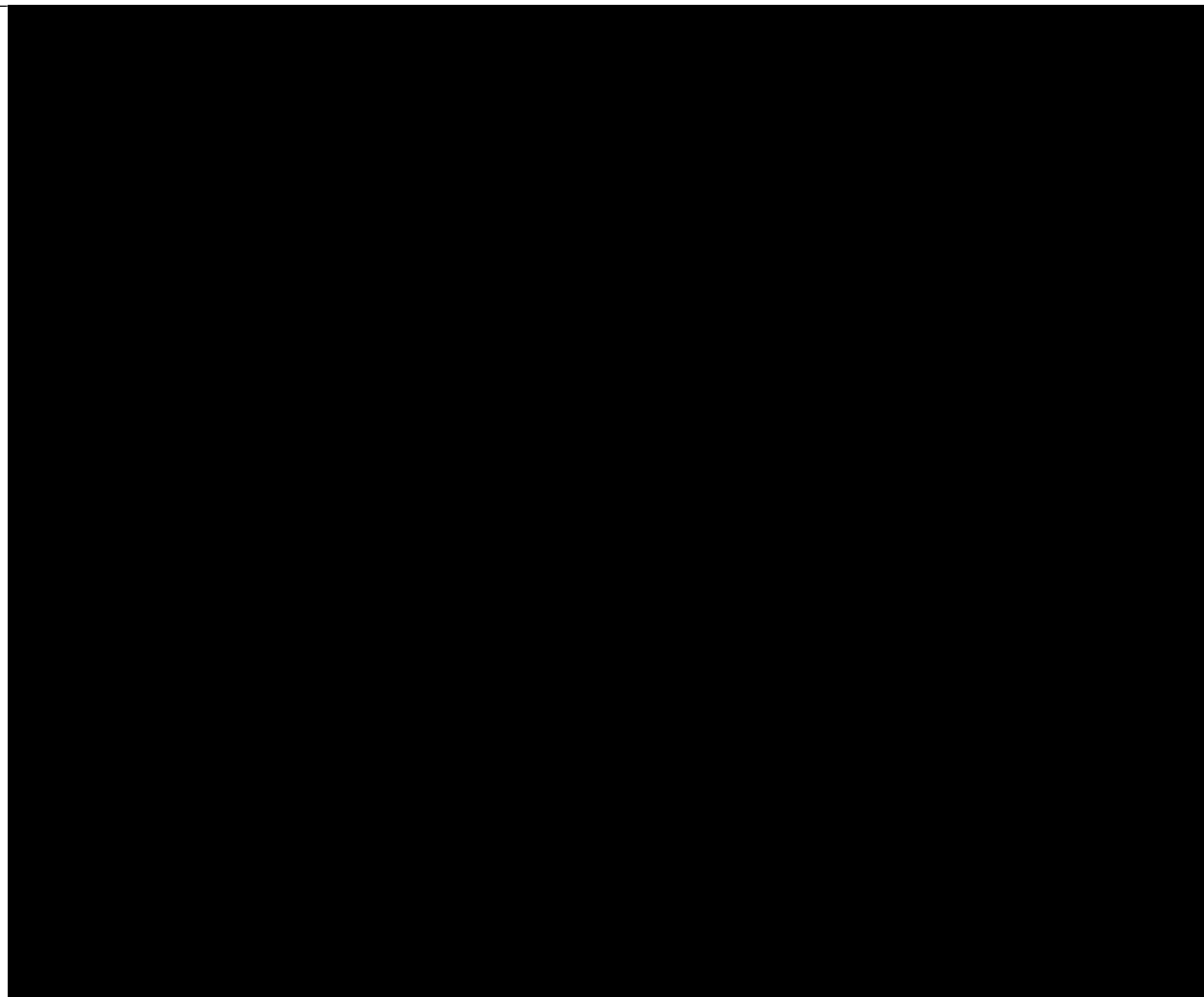
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 11.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

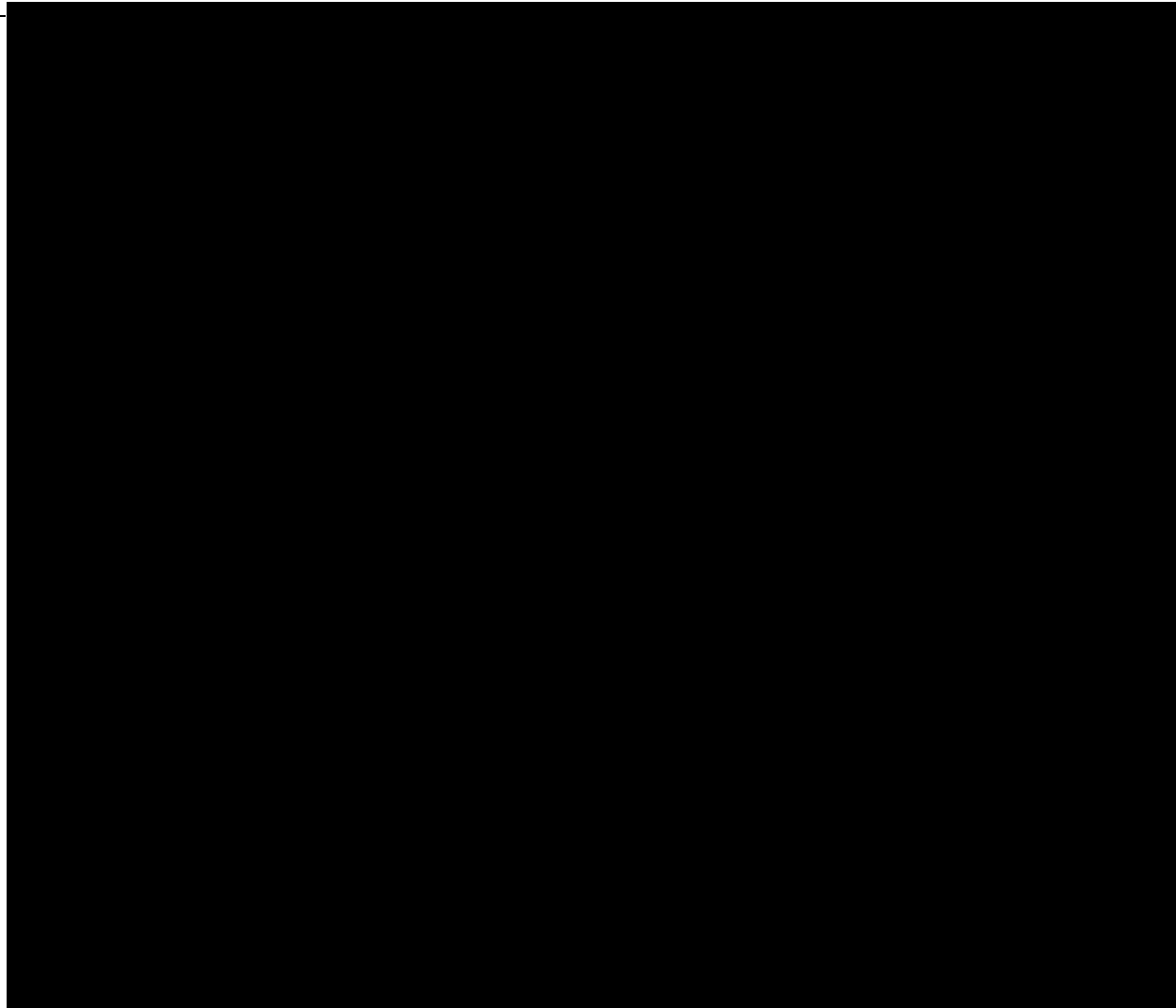
Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 10.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 11.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 12.

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

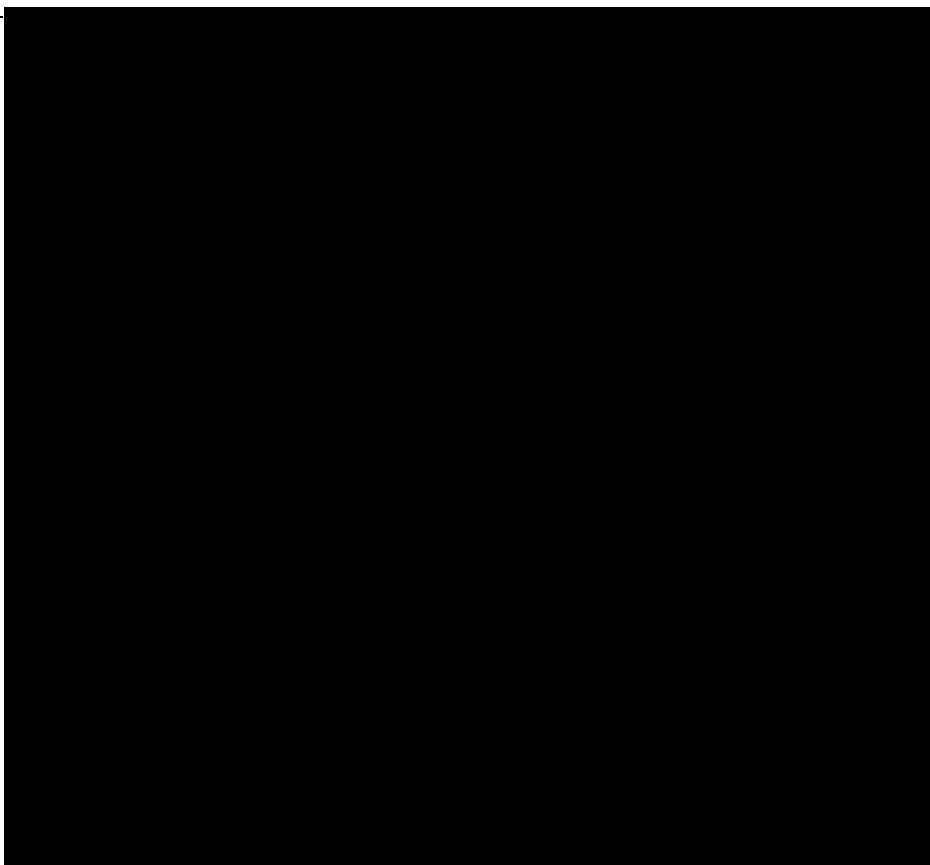
Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 4 [REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 9.

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 10.

5

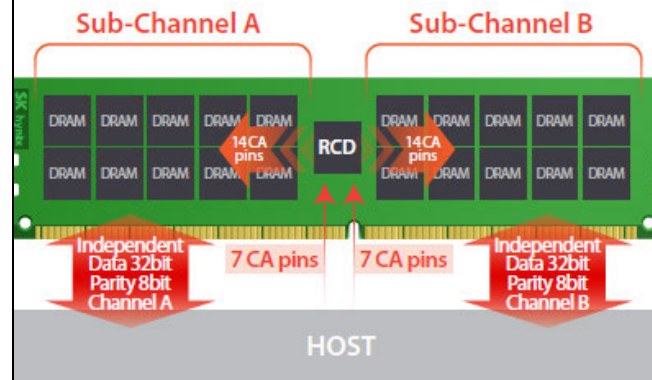
A double sub-channel DRAM architecture for 2x bandwidth and higher bus (signal) efficiency compared to a single-channel DDR4 DIMM

Ex. CC-40 (SK Hynix A New Standard) at 2.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

The latest DDR5 DRAM standard turbo-boosts RDIMM module performance and technology for the era of AI and Big Data. SK hynix expects its DDR5 RDIMMs to reach over 8,000Mbps in data rates with research into continued speed enhancements.

Features



| New Sub-channel DRAM Architecture

The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

Ex. CC-40 (SK Hynix A New Standard) at 5.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

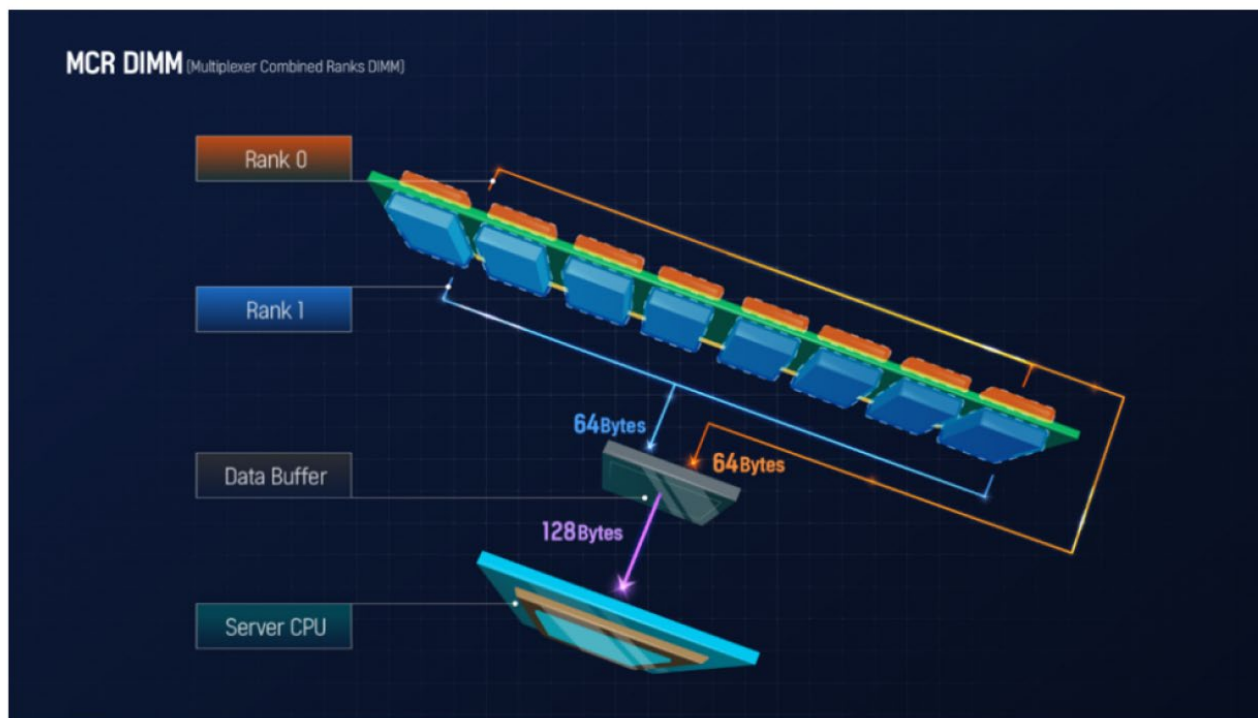
	In the server DRAM section, SK hynix presented a lineup of DDR5-based memory modules for servers. The spotlight was on the DDR5 RDIMM* and MRDIMM* products built using the 1c node,* the sixth generation of the 10 nm process technology. The company also showcased various DRAM solutions designed to maximize AI server performance and reduce power consumption, including its 96 GB DDR5 RDIMM, 256 GB 3DS* DDR5 RDIMM, and DDR5 MRDIMM in both 96 GB and 256 GB capacities. * Registered Dual In-line Memory Module (RDIMM): A server memory module product in which multiple DRAM chips are mounted on a substrate. * Multiplexed Rank Dual In-line Memory Module (MRDIMM): A server memory module product in which multiple DRAM chips are mounted on a substrate. Speed is enhanced by operating two ranks—the basic operating units of the module—simultaneously. * 10 nm DRAM process technology has progressed through six generations: 1x, 1y, 1z, 1a, 1b, and 1c. In August 2024, SK hynix became the first in the world to successfully develop 1c process technology. * 3D Stacked Memory (3DS): A high-bandwidth memory product in which two or more DRAM chips are packaged together and interconnected using TSV technology. Ex. CC-63 (SK Hynix Blog Post) at 5.
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Seoul, December 8, 2022 SK hynix Inc. (or “the company”, www.skhynix.com) announced today that it has developed working samples of DDR5* Multiplexer Combined Ranks* (MCR) Dual In-line Memory Module, the world’s fastest server DRAM product. The new product has been confirmed to operate at the data rate of minimum 8Gbps, and at least 80% faster than 4.8Gbps of the existing DDR5 products. MCR DIMM is an achievement coming from out-of-the-box thinking with an aim to improve the operation speed of DDR5. Challenging the prevailing concept that the operation speed of DDR5 relies on that of DRAM chip itself, engineers sought to find a way to improve the speed of modules instead of chips for development of the latest product. SK hynix designed the product in a way that enables simultaneous operation of two ranks by utilizing the data buffer* installed onto the MCR DIMM based on Intel’s MCR technology. <i>*Double Data Rate (DDR), a DRAM standard mainly used for servers and client applications, has been developed up to the fifth generation. The MCR DIMM is a module product with multiple DRAM chips attached to the board and improved speed as a result of two ranks operating simultaneously.</i> <i>*Rank: A collection of basic transfer units of the data sent to CPU from DRAM module. A rank typically refers to 64 bytes of data to be transferred to central processing unit as a bundle.</i> <i>*Buffer: A component that optimizes signal transmission performance between DRAM and CPU. Mainly installed onto modules for servers requiring high performance and reliability</i> Ex. CC-65 (SK Hynix Develops MCR DIMM) at 1.
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

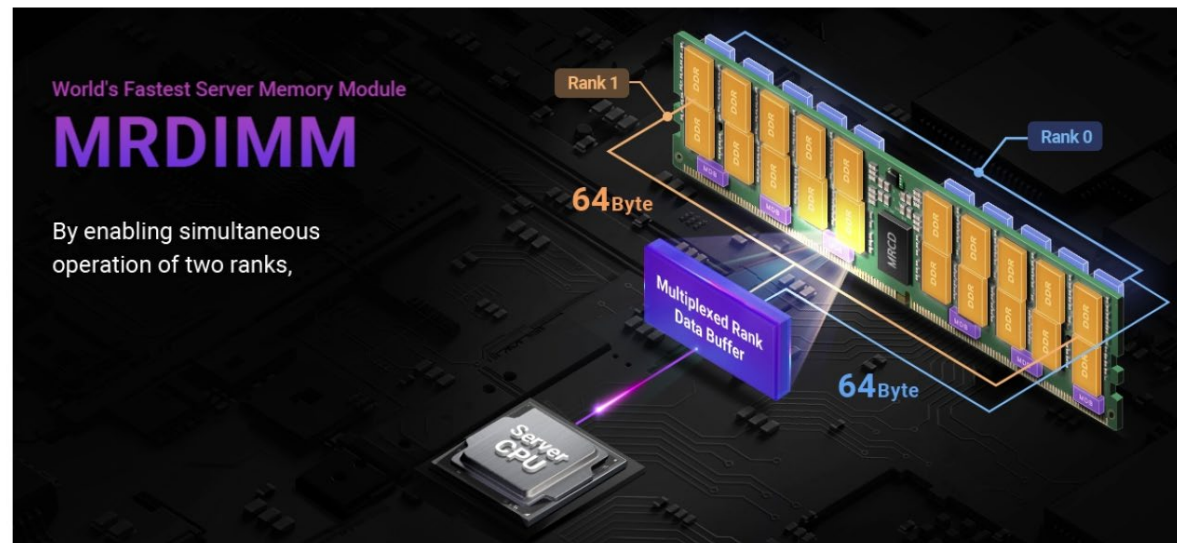
By enabling simultaneous operation of two ranks, MCR DIMM allows transmission of 128 bytes of data to CPU at once, compared with 64 bytes fetched generally in conventional DRAM module. An increase in the amount of data sent to the CPU each time supports the data transfer rate of minimum 8Gbps, twice as fast as a single DRAM.



Ex. CC-65 (SK Hynix Develops MCR DIMM) at 1 (illustrating separate groups of memory devices configurable to communicate data with the memory controller via corresponding subsets of data signal lines in blue and orange).

How MRDIMM Works

By facilitating simultaneous operation of two ranks, MRDIMM enables the transmission of 128 bytes of data to the CPU at once, compared to the 64 bytes typically fetched by conventional DRAM modules. This increase in the amount of data sent to the CPU each time supports twice the speed of a single DRAM.



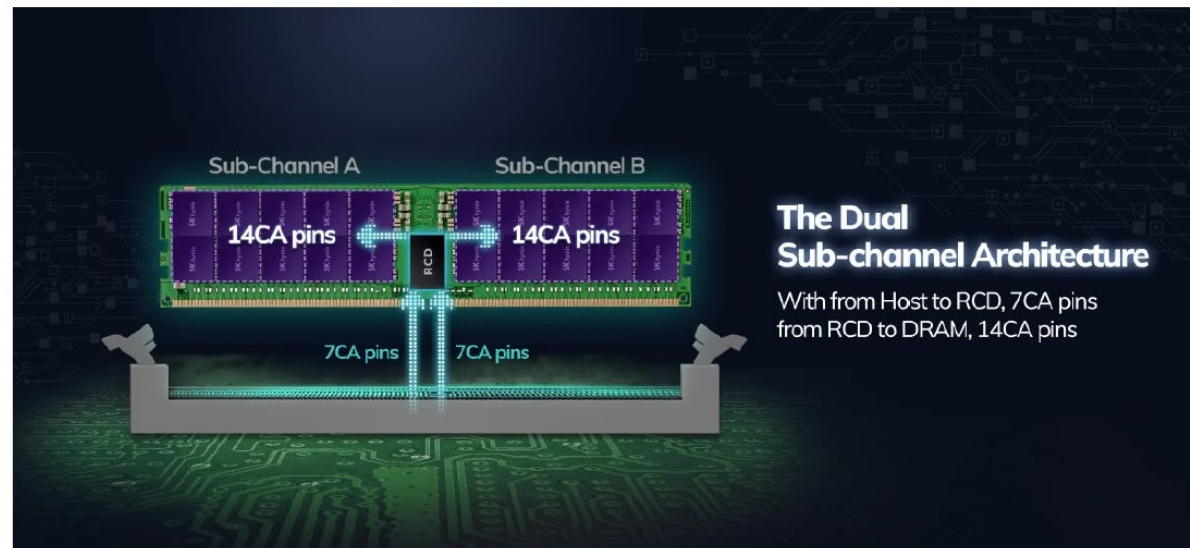
Ex. CC-42 (SK Hynix MRDIMM) at 2 (illustrating separate groups of memory devices configurable to communicate data with the memory controller via corresponding subsets of data signal lines in blue and orange).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Kim Yeong Jun provides a behind-the-scenes explanation of the newly attempted techniques that went into the product development. "In the process of developing the MCR DIMM, we introduced a new concept to increase operating speeds," he explained. "While not changing the way the existing DDR5 DRAM units are used, we designed two ranks—the operation unit of the module—to operate at the same time. While 64 bytes of data are transmitted in one rank, 128 bytes of data can be transmitted at once by operating two ranks simultaneously. A data buffer* is used for this operation as it combines the 64 bytes of data transmitted in two ranks and creates the collective transmit of 128 bytes." He added: "In particular, collaboration with global companies—Intel (U.S.) and Renesas (Japan)—played an important role in this development." Intel, which develops server CPUs, and Renesas, which develops data buffers for the MCR DIMM, collaborated with SK hynix to create synergy." *Buffer: A component that is mounted on top of the DRAM module to optimize the performance of the signal transmission between the DRAM and the CPU. It is mainly installed in server DRAM modules that require high performance and stability. Ex. CC-64 (SK hynix Newsroom Interview) at 6.
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Double Sub-channel DRAM Architecture

With a dual sub-channel structure, DDR5 RDIMM increases the number of CA (Command-Address) pins to seven double-data-rate CA buses each from host to RCD and 14 single-data-rate buses each on RCD to DRAM. Having two independent 40-bit channels thus enables 2x bandwidth and higher bus (signal) efficiency compared to a single-channel DDR4 DIMM.



Ex. CC-41 (SK hynix RDIMM) at 2.

As explained *supra*, the Practicing '407 Products use a JEDEC compliant 288-pin edge connector. The JEDEC DDR5 RDIMM standard further demonstrates the Practicing '407 Products' edge connectors allow each respective group of the memory devices to be configured to communicate data with the memory controller via a corresponding subset of the data signal lines. *See, e.g.,*

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Table 3 — Pin Definitions

Pin Name	Description	Pin Name	Description
CA[6:0]_A CA[6:0]_B	Command and Address (CA) Bus	DQ[31:0]_A DQ[31:0]_B	DIMM memory Data bus channel A and B
CS[1:0]_A_n CS[1:0]_B_n	Control (Chip Select) (CS_n)	CB[7:0]_A CB[7:0]_B	DIMM ECC Checkbits (CB) channel A and B

Ex. CC-88 (JEDEC 305B) at 3 (showing each channel has separate CA Bus and CS commands).

DQ[31:0]_A DQ[31:0]_B	Input/ Output	VDD	Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4. A4=High. Refer to vendor specific data sheets to determine which DQ is used.
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Ex. CC-88 (JEDEC 305B) at 4 (showing each channel has separate data lines).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

6.9.1 DIMM Connectivity Wiring

2Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Top	Upper	QBCK	QBCS0	QBCA	0
Bottom	Lower	QCCK	QACS1	QACA	1
Bottom	Upper	QDCK	QBCS1	QBCA	1

2Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Top		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]
Bottom	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

2Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Top	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]
Bottom		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

1Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QBCS0	QACA	0

1Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

1Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

2Rx8					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QACS1	QACA	1

2Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]	
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

2Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]	CB[7:0]	
Bottom	[7:0]		[15:8]		[23:16]		[31:24]			CB[7:0]

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6.9.1 DIMM Connectivity Wiring (cont'd)

1Rx8 (less than 8000 Mbps)										
DIMM Side	Row	Clock	CS	CA	Rank					
Top	Lower	QACK	QACS0	QACA	0					
Bottom	Lower	QACK	QACS0	QACA	0					
1Rx8 (8000 Mbps and Faster)										
DIMM Side	Row	Clock	CS	CA	Rank					
Top	Lower	QCCK	QACS0	QACA	0					
Bottom	Lower	QCCK	QACS0	QACA	0					
1Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top				[15:8]					[31:24]	
Bottom	[7:0]					[23:16]		CB[7:0]		
1Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[7:0]					[23:16]				
Bottom			[15:8]					[31:24]		CB[7:0]

Figure 12 — DIMM Connectivity Wiring

Ex. CC-88 (JEDEC 305B) at 31-32 (showing groups of memory devices on various sides and in various rows on the module configured to receive CA, CS, and data signals via different subsets of CA, CS, and data signal lines).

[1.c] circuitry mounted on the PCB and configurable to:

The Practicing '407 Products comprise circuitry mounted on the PCB. For example:

For example, the Practicing '407 Products include circuitry mounted on the PCB, including RCD (Registering Clock Driver) circuitry. *See, e.g.,*



SERVER DDR5 RDIMM

Ex. CC-34 (SK hynix Products).

RDIMM

[VIEW MORE](#)



DDR5
Generation

16-256GB
Module Density

4800-6400Mbps and higher
Speed

DDR4
Generation

16-256GB
Module Density

2666-3200Mbps
Speed

Application



Server



Networking

- DIMM with a register (Register Clock Driver, RCD) for command/address signals between the DRAM and system memory controller
- Supports x4, x8 DRAM and x4 3DS stacks

MRDIMM

[VIEW MORE](#)



DDR5
Generation

32-128GB
Module Density

up to 8800Mbps
Speed

Application



Server

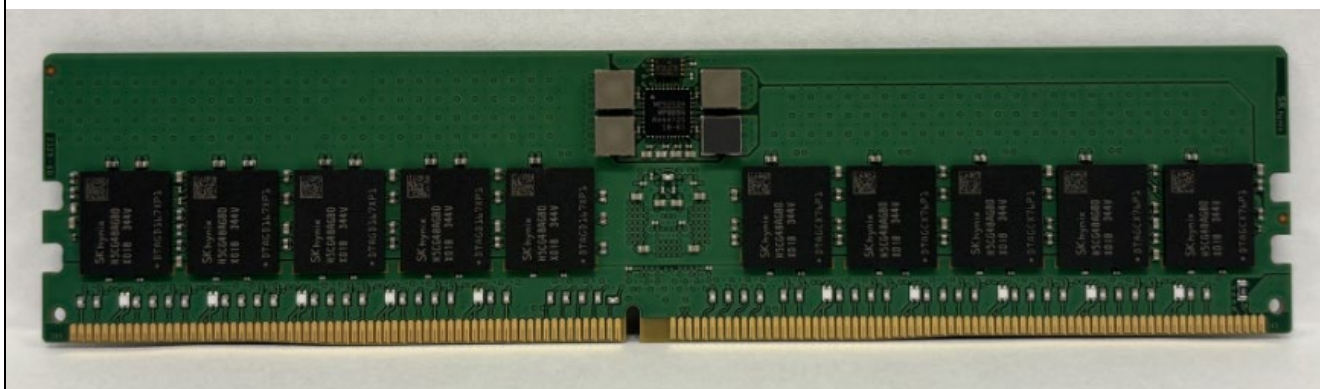
- DIMM with a register (Register Clock Driver, Data Buffer) for enabling simultaneous operation of two ranks, which increases bandwidth.
- Supports x4, x8 DRAM

Ex. CC-35 (SK hynix Module).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



Ex. CC-36 at IMG_5114 (showing SK hynix DDR5 RDIMM Product).

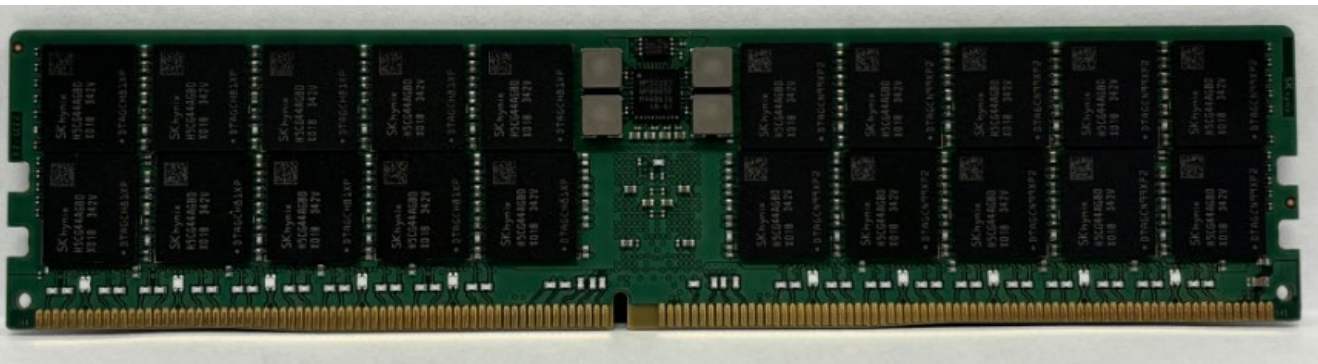


Ex. CC-36 at IMG_5117 (showing SK hynix DDR5 RDIMM Product).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

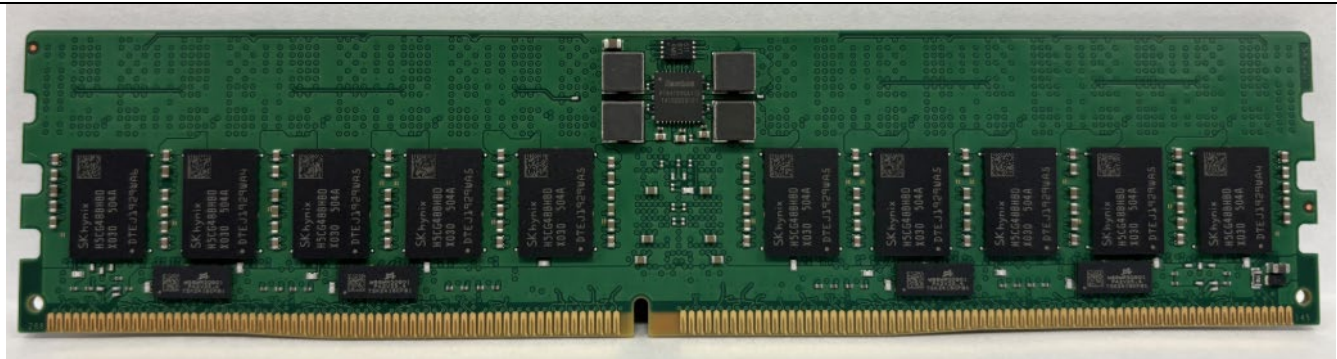


Ex. CC-36 at IMG_5112 (showing Netlist DDR5 RDIMM Overclock Product).



Ex. CC-36 at IMG_5109 (showing Netlist DDR5 RDIMM Overclock Product).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

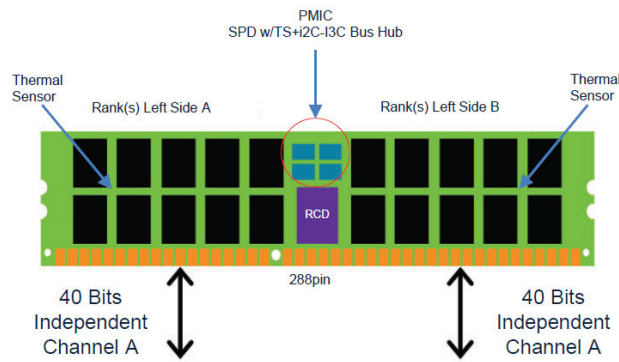


Ex. CC-36 at IMG_5126 (showing Netlist DDR5 MRDIMM Product).



Ex. CC-36 at IMG_5122 (showing Netlist DDR5 MRDIMM Product).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



DIMM Changes from DDR4 to DDR5

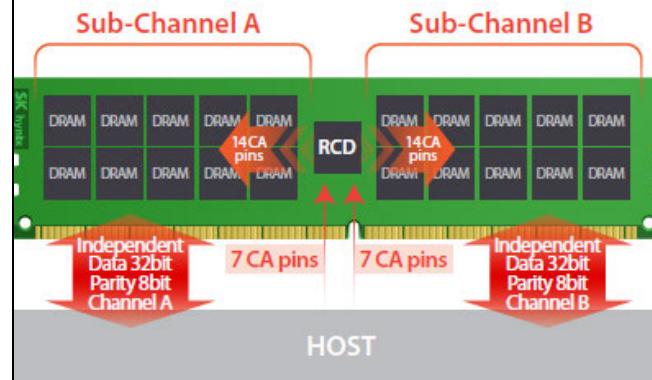
Feature	DDR4	DDR5
Frequency (GT/s)	1.6~3.2	4.8~5.6
Voltage (V)	1.2	1.1
Power Supply	PMIC on MB	PMIC on DIMM
Temp sensor device	1 per DIMM	2 or 3 per DIMM
Side bus protocol	I2C bus	Legacy SMBus, I2C, I3C
Sideband bus access	Direct Access	1xC Hub in SPD
Channel Architecture	72-bit data channel 64 data + 8 ECC 1 channel per DIMM	40-bit data channel 32 data+ 8 ECC 2 channel per DIMM
DFE Support	DB Only	RCD, DB, and DRAM

Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

The latest DDR5 DRAM standard turbo-boosts RDIMM module performance and technology for the era of AI and Big Data. SK hynix expects its DDR5 RDIMMs to reach over 8,000Mbps in data rates with research into continued speed enhancements.

Features



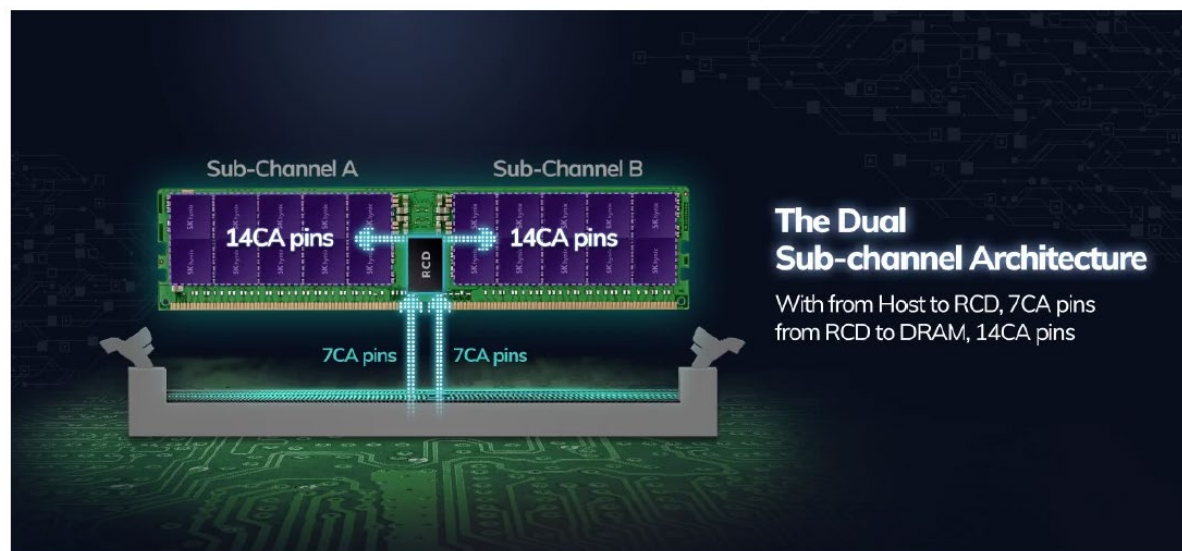
| New Sub-channel DRAM Architecture

The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

Ex. CC-40 (SK Hynix A New Standard) at 5.

Double Sub-channel DRAM Architecture

With a dual sub-channel structure, DDR5 RDIMM increases the number of CA (Command-Address) pins to seven double-data-rate CA buses each from host to RCD and 14 single-data-rate buses each on RCD to DRAM. Having two independent 40-bit channels thus enables 2x bandwidth and higher bus (signal) efficiency compared to a single-channel DDR4 DIMM.



Ex. CC-41 (SK hynix RDIMM) at 2.

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 1

[REDACTED]

[REDACTED]

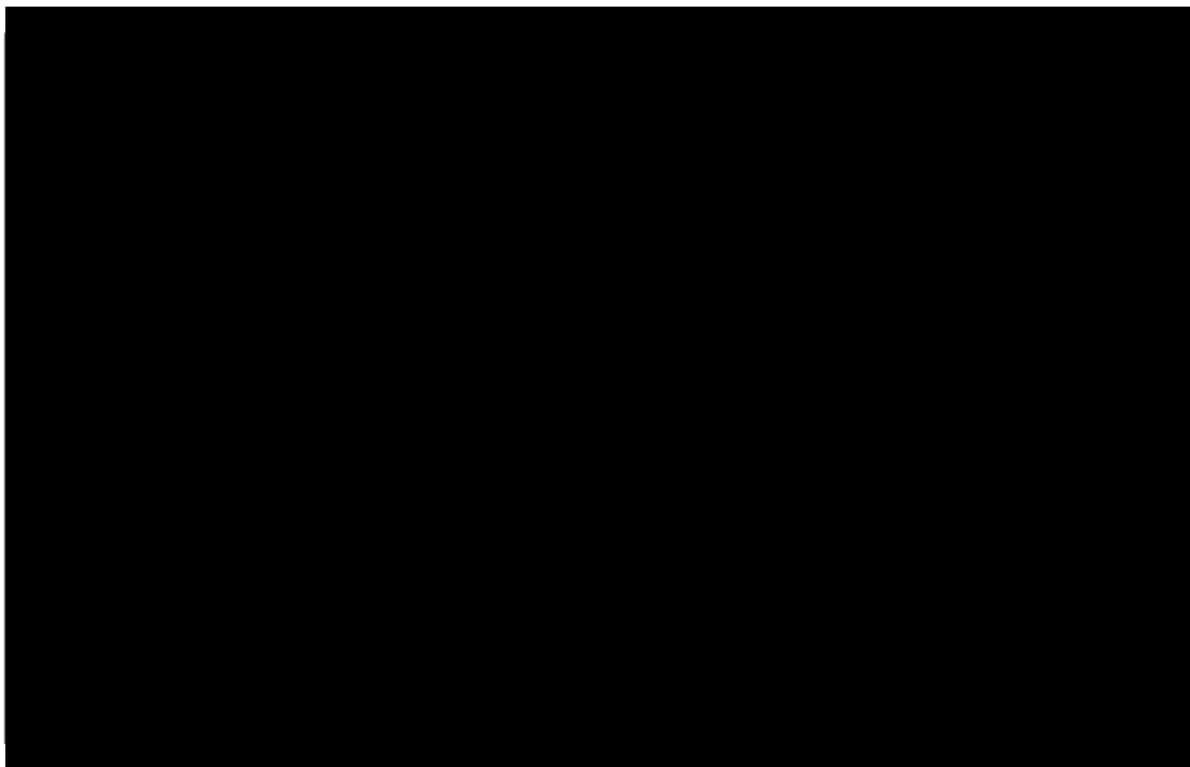
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

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[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 10.



Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 1

[REDACTED]

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

[REDACTED]

Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 9 [REDACTED]
[REDACTED].

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

[1.c.1] receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A signal lines;	The Practicing '407 Products comprise circuitry mounted on the PCB and configurable to receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A signal lines. For example: The circuitry mounted on the PCB is configurable to receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A signal lines. <i>See, e.g.,</i>
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[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 4

[REDACTED]

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 5

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 5.

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

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[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 7 [REDACTED] [REDACTED]
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[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

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[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 8	[REDACTED]
	[REDACTED]	[REDACTED]
	[REDACTED]	
	[REDACTED]	
	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 9	[REDACTED]
	[REDACTED]	[REDACTED]
	[REDACTED]	

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

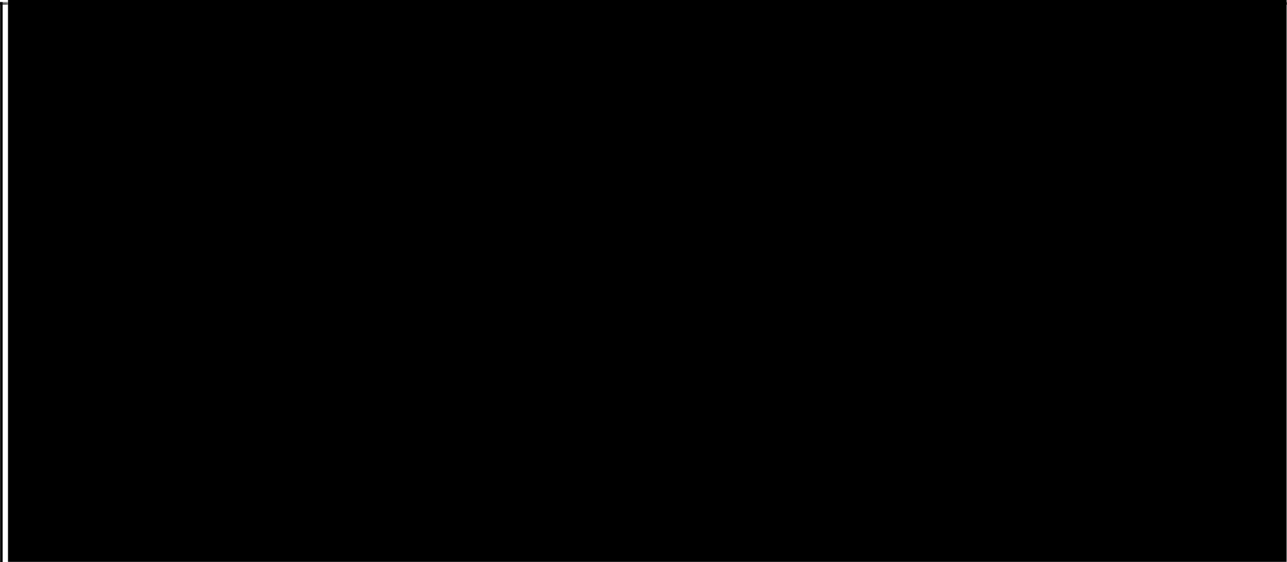
Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 10

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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

		
	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 10	[REDACTED] [REDACTED]

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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 4

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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 5 [REDACTED]
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 5 [REDACTED]
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

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	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 6 [REDACTED]
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	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 7 [REDACTED] [REDACTED] Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 8.
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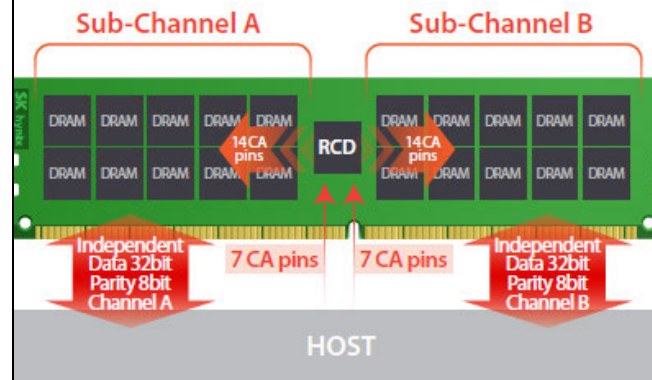
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	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 9	

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The latest DDR5 DRAM standard turbo-boosts RDIMM module performance and technology for the era of AI and Big Data. SK hynix expects its DDR5 RDIMMs to reach over 8,000Mbps in data rates with research into continued speed enhancements.

Features



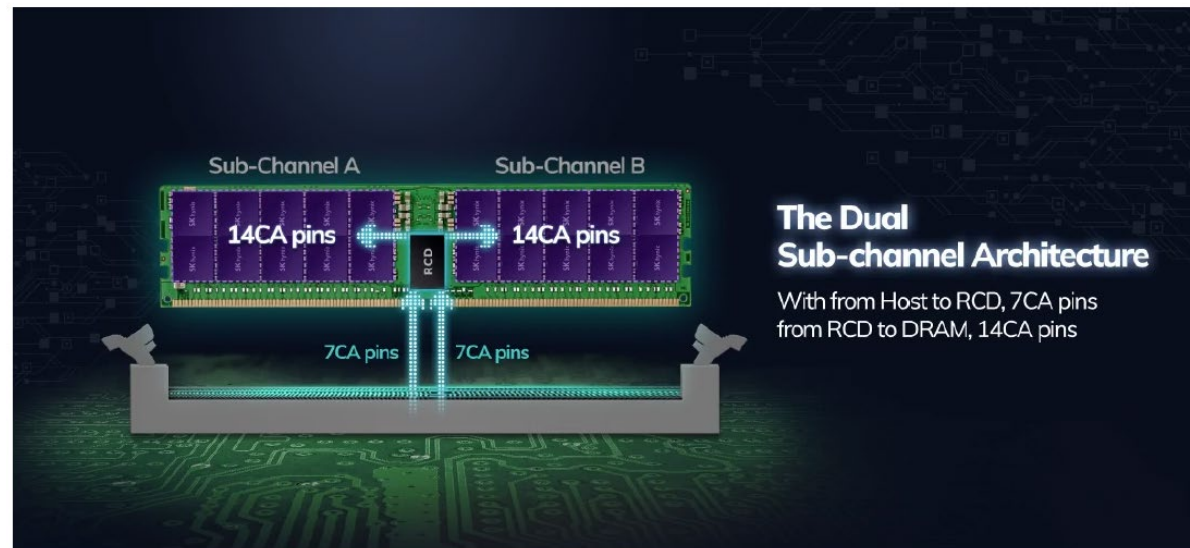
| New Sub-channel DRAM Architecture

The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

Ex. CC-40 (SK Hynix A New Standard) at 5 (demonstrating the RCD receiving C/A signals through "CA Pins" from the memory controller).

Double Sub-channel DRAM Architecture

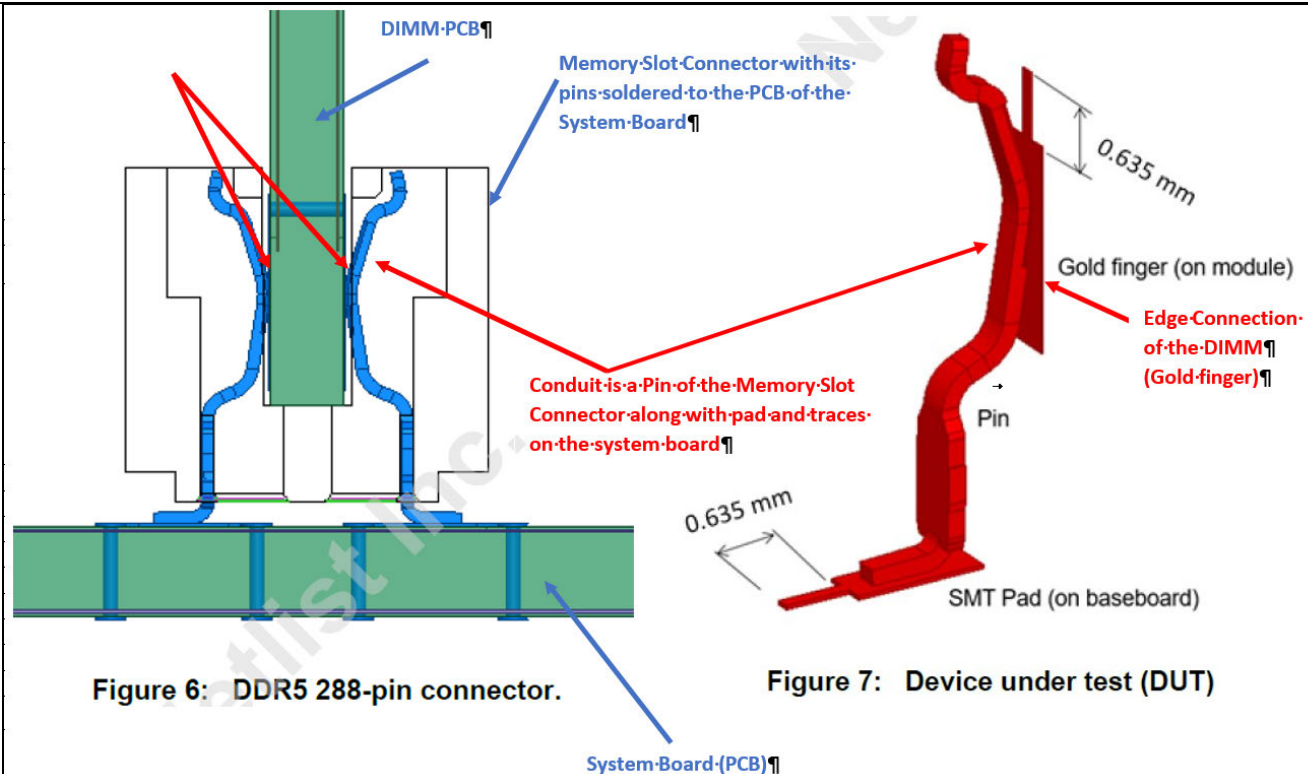
With a dual sub-channel structure, DDR5 RDIMM increases the number of CA (Command-Address) pins to seven double-data-rate CA buses each from host to RCD and 14 single-data-rate buses each on RCD to DRAM. Having two independent 40-bit channels thus enables 2x bandwidth and higher bus (signal) efficiency compared to a single-channel DDR4 DIMM.



Ex. CC-41 (SK hynix RDIMM) at 2 (demonstrating the RCD receiving C/A signals through “7CA Pins” from the memory controller).

As explained *supra*, the Practicing '407 Products are JEDEC compliant. The DCA, DCS and DCK signals are therefore received via the corresponding C/A, C/S and CK signals lines in the system bus because the DCA, DCS and DCK pins of an RDIMM each mate with a corresponding connector in the memory slot to form electrical connection to the corresponding slot pin and signal traces, as illustrated below:

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Ex. CC-73 (JEDEC PS-005B) at 13, Figs. 6–7 (annotated) (illustrating an edge connection on a DIMM being electrically connected with a corresponding pin in the memory slot and the corresponding signal trace/line on the system board).

The JEDEC DDR5 RDIMM standard further demonstrates the RCD circuitry is configurable to receive from the memory controller a system clock via the one or more clock signal lines (e.g., “CK_t” and “CK_c” / “DCK_t” and “DCK_c”) and input control and address (C/A) signals via the C/A signal lines (e.g., “CA[]_A” and “CA[]_B” / “DCA[]_A” and “DCA[]_B” / “CS[]_A_n” and “CS[]_B_n” / “DCS[]_A_n” and “DCS[]_B_n”). See, e.g.,

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Table 3 — Pin Definitions

Pin Name	Description	Pin Name	Description
CA[6:0]_A CA[6:0]_B	Command and Address (CA) Bus	DQ[31:0]_A DQ[31:0]_B	DIMM memory Data bus channel A and B
CS[1:0]_A_n CS[1:0]_B_n	Control (Chip Select) (CS_n)	CB[7:0]_A CB[7:0]_B	DIMM ECC Checkbits (CB) channel A and B
PAR_A PAR_B	Parity input	DQS[9:0]_A_t DQS[9:0]_B_t	Data Strobes (positive line of differential pair)
CK_t	Clock (true/positive)	DQS[9:0]_A_c DQS[9:0]_B_c	Data Strobes (negative line of differential pair)
CK_c	Clock (complement/negative)	TDQS[9:5]_A_t TDQS[9:5]_B_t	Not valid for x4 operation. Enabled via Mode Register.

Ex. CC-88 (JEDEC 305B) at 3 (illustrating circuitry inputs for receiving clock signal lines (*e.g.*, “CK_t” and “CK_c”) and control/address (C/A) signal lines (*e.g.*, “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”) from the memory controller).

Table 4 — Input/Output Functional Description

Symbol	Type	I/O Levels	Function
CK_t, CK_c.	Input	VDD	Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.
CA[6:0]_A CA[6:0]_B	Input	VDD	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during Mode Register Set commands.

Ex. CC-88 (JEDEC 305B) at 4 (illustrating circuitry inputs for receiving clock signal lines (*e.g.*, “CK_t” and “CK_c”) and control/address (C/A) signal lines (*e.g.*, “CA[]_A” and “CA[]_B”) from the memory controller).

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DQ[31:0]_A DQ[31:0]_B	Input/ Output	VDD	Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.
CB[7:0]_A CB[7:0]_B	Input/ Output	VDD	ECC Checkbits Input/ Output: Bi-directional data bus - for 8-bit ECC (EC8) all 8 bits are used - for 4-bit ECC (EC4) [3:0] bits are used; [7:4] bits are floating

Ex. CC-88 (JEDEC 305B) at 4.

Table 4 — Input/Output Functional Description (cont'd)

Symbol	Type	I/O Levels	Function
DQS[9:0]_A_t DQS[9:0]_B_t	Input/ Output	VDD	Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. The Data Strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential Data Strobe only and does not support single-ended.
DQS[9:0]_A_c DQS[9:0]_B_c			
TDQS[9:5]_A_t TDQS[9:5]_B_t	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.
TDQS[9:5]_A_c TDQS[9:5]_B_c	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.

Ex. CC-88 (JEDEC 305B) at 5.

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	6.1 Signal Groups This standard categorizes DDR5 SDRAM timing-critical signals into specific groups. Figure 2 — (Example) RDIMM 2Rx4 Topology summarizes the high-speed signal groups for RDIMM. Signal groups, except Data and Strobe, implement a fly-by topology. Actual signal names in Annex may vary from individual reference design. The signal groups are: Pre-RCD - Pre-RCD CA and CS group includes: DCA[6:0]_[B:A], DCS[1:0]_[B:A]_n, DPAR_[B:A] - Pre-RCD Clock group includes: DCK_t and DCK_c - Pre-RCD Other group includes: ALERT_n, DRST_n - Pre-RCD Loopback group includes: DLBDQ_[B:A], DLBDQS_[B:A] Post-RCD to SDRAM - Post-RCD CA group includes Q[B:A]CA[13:0]_[B:A] - Post-RCD CS group includes Q[B:A]CS[1:0]_[B:A]_n - Post-RCD Clock group includes Q[D:A]CK_[B:A]_t, Q[D:A]CK_[B:A]_c - Post-RCD Other group includes QRST_[B:A]_n, DERROR_[B:A]_n - Post-RCD Other group includes QLBDQ_[B:A], QLBDQS_[B:A] DQ and DQS - edge connector pin to SDRAM with series resistor SideBand - Host to Hub group: HSDA, HSCL, HSA - Hub local bus group: LSDA, LSCL (star topology from Hub) Ex. CC-88 (JEDEC 305B) at 12 (illustrating circuitry inputs for receiving clock signal lines (e.g., “DCK_t” and “DCK_c”) and control/address (C/A) signal lines (e.g., “DCA[]_A,” “DCA[]_B,” “DCS[]_A_n,” and “DCS[]_B_n”) from the memory controller).
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

6.1 Signal Groups (cont'd)

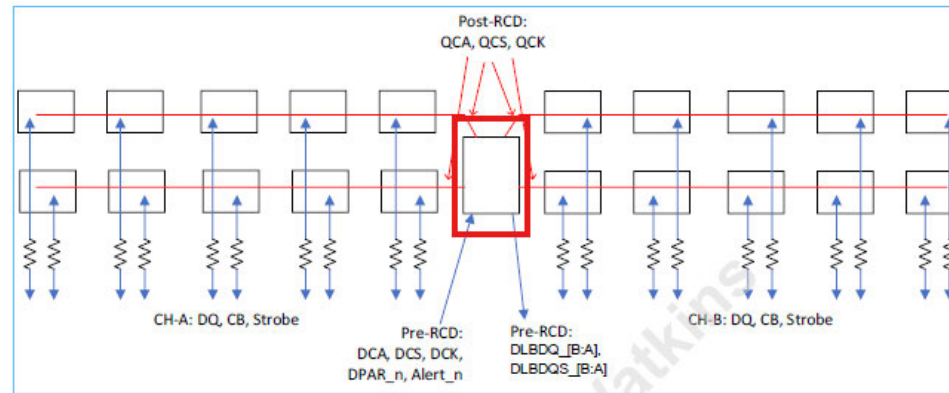


Figure 2 — (Example) RDIMM 2Rx4 Topology

Ex. CC-88 (JEDEC 305B) at 13 (illustrating RCD, in red, receiving “Pre-RCD” control and address signals (“DCA” and “DCS”) and clock signals (“DCK”) from the memory controller).

As explained *supra*, the Practicing '407 Products are JEDEC compliant. The JEDEC DDR5 RCD definition further demonstrates the RCD circuitry is configurable to receive from the memory controller a system clock via the one or more clock signal lines (*e.g.*, “DCK_t” and “DCK_c”) and input control and address (C/A) signals via the C/A signal lines (*e.g.*, “DCA[]_A,” “DCA[]_B,” “DCS[]_A_n,” and “DCS[]_B_n”). *See, e.g.*,

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3.4.1 Double Data Rate DCA

The DDR5 RCD will support DDR (Double Data Rate) on the host-interface DCA signals. The host-interface CA bus is 7 bits, plus parity. The DRAM-interface CA bus remains SDR (Single Data Rate) and is 14 bits and without parity. The RCD will expand the incoming host-interface 7-bit DCA bus to 14 bits on the DRAM interface. Table 8 shows the mapping of the incoming DCA bits to the output QCA. The Chip Selects remain SDR on both the host-interface and DRAM interface. The two channel DDR5RCD has six CA ports, two DCA input ports and four QCA output ports as shown in Figure 6.

Table 8 — DDR CA Mapping DCA to QCA

UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6
0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6
1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13

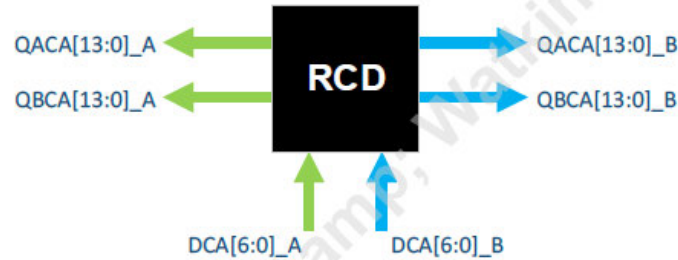
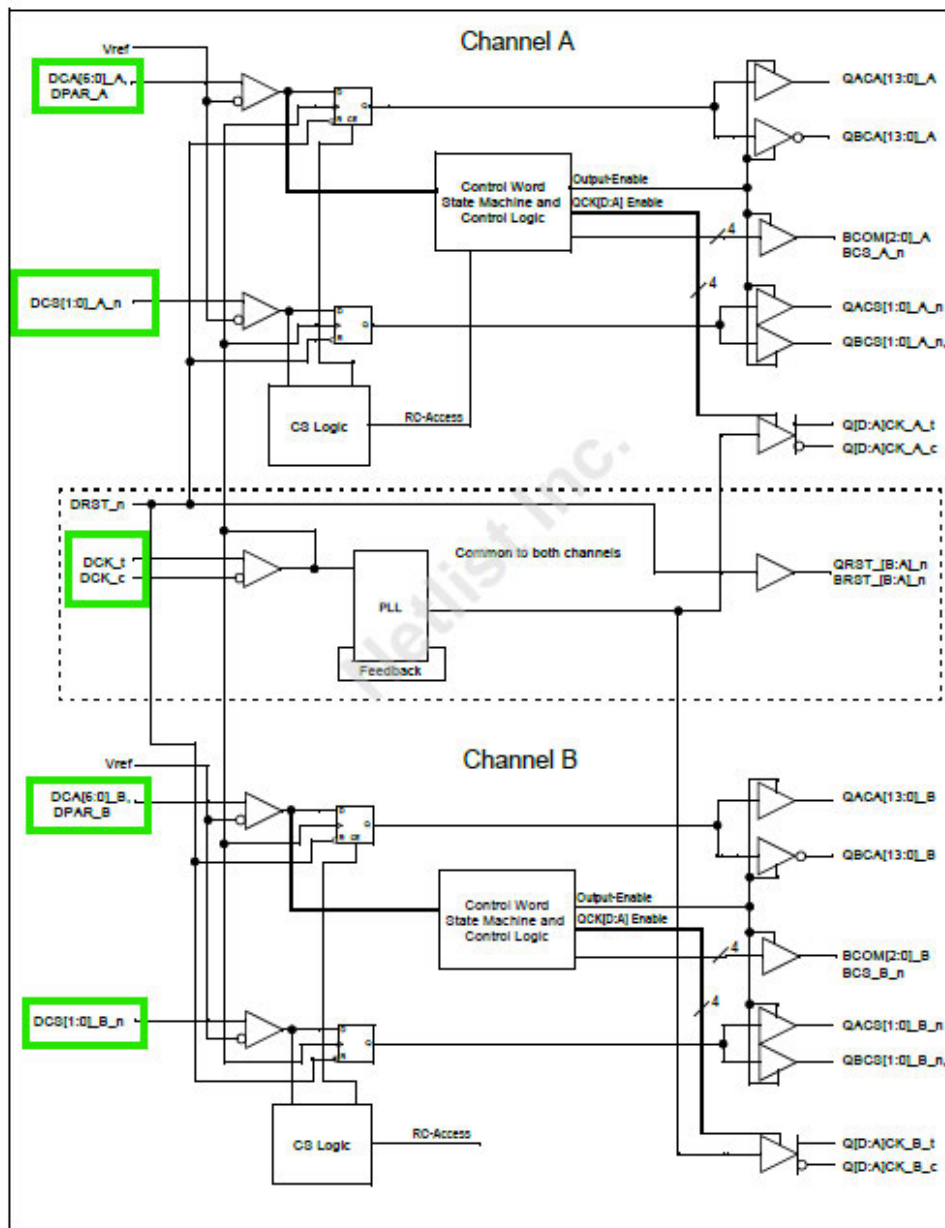


Figure 6 — RCD DCA and QCA Ports

Ex. CC-32 (JEDEC 82-512) at 15 (illustrating circuitry inputs for receiving control/address (C/A) signal lines (e.g., “DCA[]_A” and “DCA[]_B”) from the memory controller). *See also* Ex. CC-30 (JEDEC 82-513) at 15.

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Ex. CC-32 (JEDEC 82-512) at 7 (illustrating circuitry inputs for receiving clock signal lines (e.g., “DCK_t” and “DCK_c”) and control/address (C/A) signal lines (e.g., “DCA[]_A,” “DCA[]_B,” “DCS[]_A_n,” and “DCS[]_B_n”) from the memory controller). *See also* Ex. CC-30 (JEDEC 82-513) at 7.

Signal Group	Signal Name	Type	Description
Input Control bus	DCS[1:0]_[B:A]_n	POD V _{REF} based	Chip Select inputs to the RCD. Two inputs per DRAM channel.
Input Address and Command bus	DCA[6:0]_[B:A]	POD V _{REF} based	Command/Address bus inputs to the RCD. Separate sets per channel.
Parity input	DPAR_[B:A]	POD V _{REF} based	Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.
Clock inputs	DCK_t/DCK_c	POD differential	Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.

Ex. CC-32 (JEDEC 82-512) at 4 (illustrating circuitry inputs for receiving clock signal lines (e.g., “DCK_t” and “DCK_c”) and control/address (C/A) signal lines (e.g., “DCA[]_A,” “DCA[]_B,” “DCS[]_A_n,” and “DCS[]_B_n”) from the memory controller). *See also* Ex. CC-30 (JEDEC 82-513) at 4.

3.1 Description

The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs.

Ex. CC-32 (JEDEC 82-512) at 6. *See also* Ex. CC-30 (JEDEC 82-513) at 6.

[1.c.2] generate module C/A signals in response to the input C/A signals;

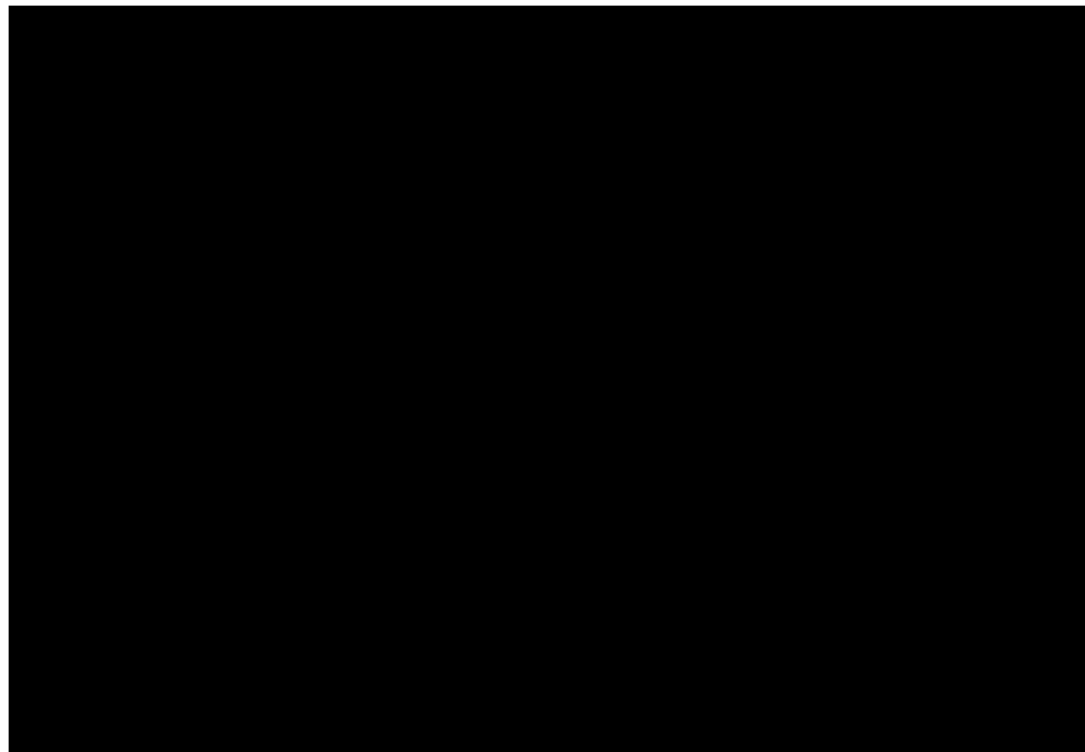
The Practicing '407 Products comprise circuitry mounted on the PCB and configurable to generate module C/A signals in response to the input C/A signals.

For example, the RCD circuitry mounted on the PCB is configurable to generate module C/A signals (in response to input C/A signals (e.g.,). *See, e.g.,*

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Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 4.

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Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 5.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 5.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 6.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

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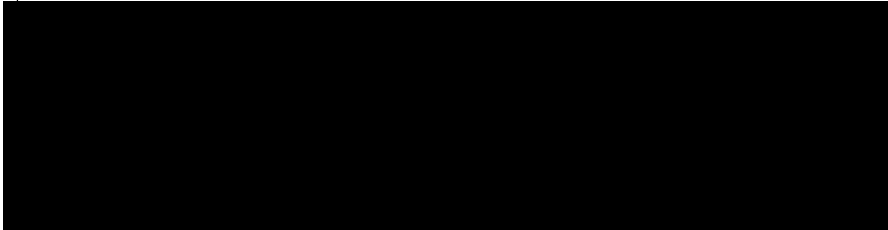
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 7.
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Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 8.

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	 Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 9.
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 10.

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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

[REDACTED]

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 10 [REDACTED]
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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 4.

Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 5.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

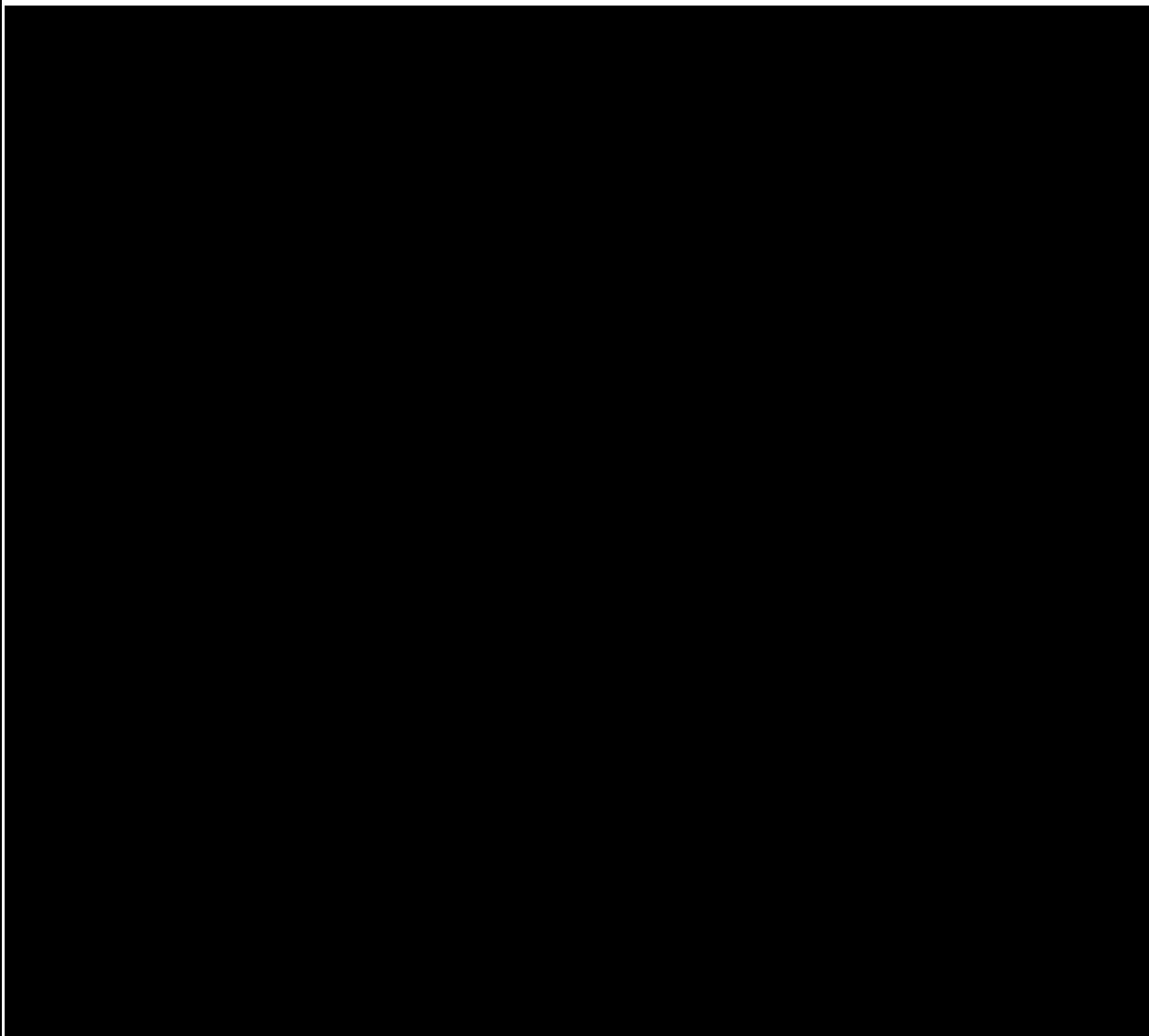
Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 5.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

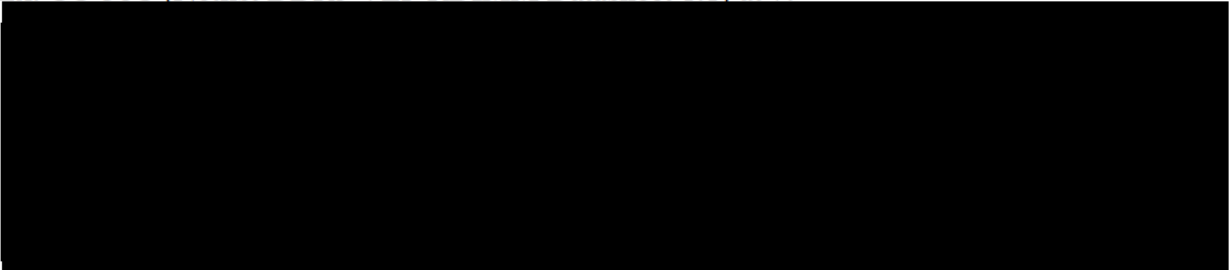
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

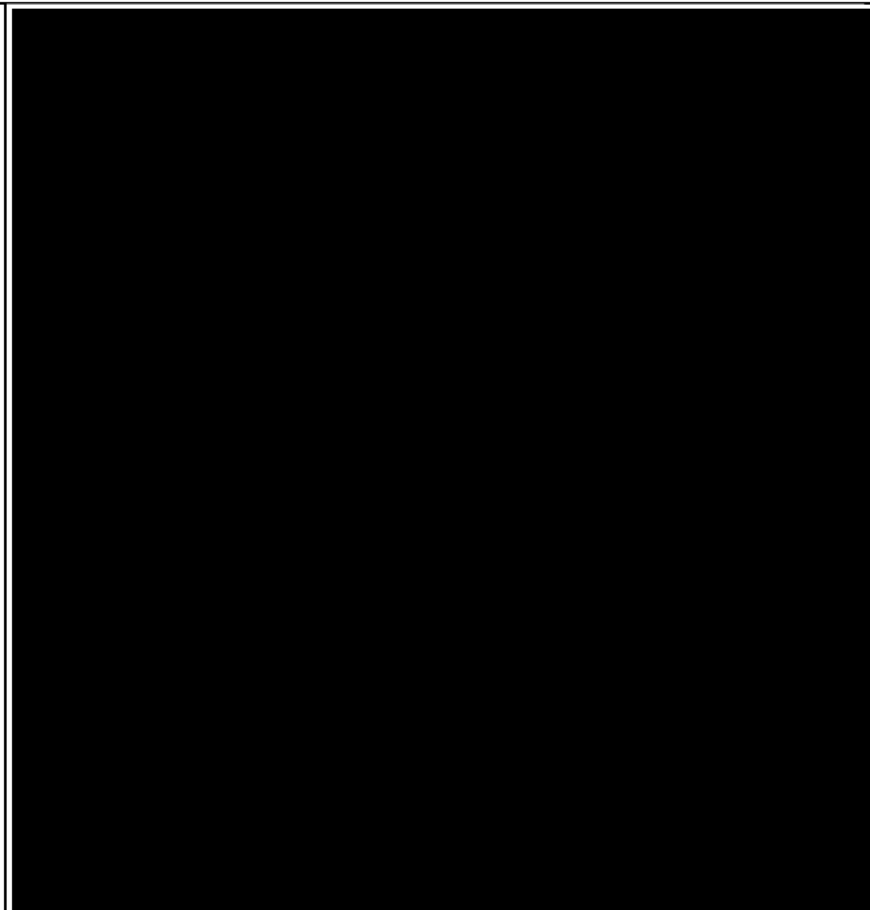
Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 6.



U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 7.  Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 8.
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

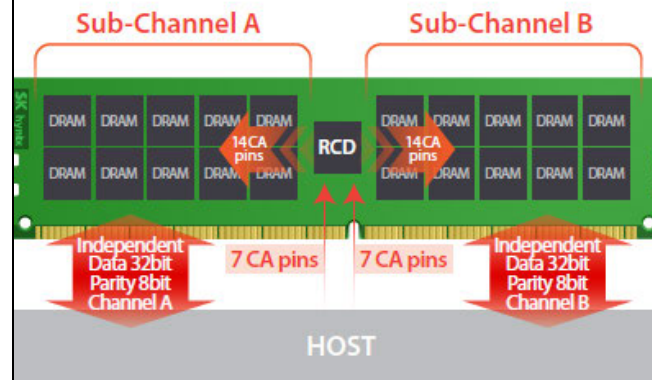


Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 9.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

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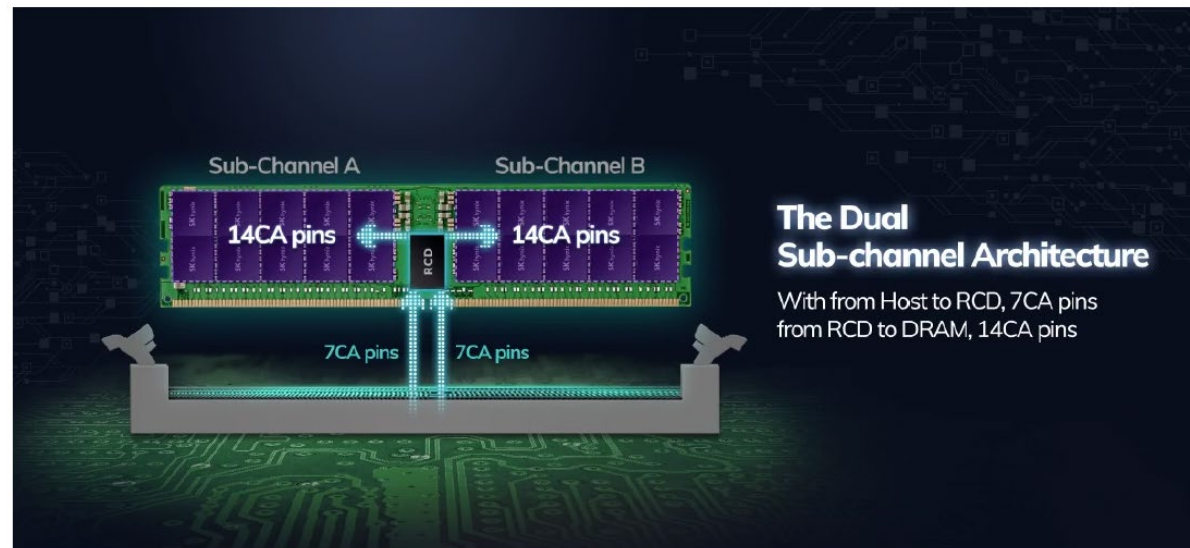
New Sub-channel DRAM Architecture

The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

Ex. CC-40 (SK Hynix A New Standard) at 5 (demonstrating (1) the RCD circuitry generating module C/A signals in response to the input C/A signals from the “HOST” and (2) the RCD circuitry outputting those module C/A signals to DRAM memory devices).

Double Sub-channel DRAM Architecture

With a dual sub-channel structure, DDR5 RDIMM increases the number of CA (Command-Address) pins to seven double-data-rate CA buses each from host to RCD and 14 single-data-rate buses each on RCD to DRAM. Having two independent 40-bit channels thus enables 2x bandwidth and higher bus (signal) efficiency compared to a single-channel DDR4 DIMM.



Ex. CC-41 (SK hynix RDIMM) at 2 (demonstrating (1) the RCD circuitry generating module C/A signals through "14CA Pins" in response to the input C/A signals from the "Host" and (2) the RCD circuitry outputting those module C/A signals to DRAM memory devices).

As explained *supra*, the Practicing '407 Products are JEDEC compliant. The JEDEC DDR5 RDIMM standard further demonstrates that the Practicing '407 Products generate module C/A signals (*e.g.*, "QACA[]_A," "QBCA[]_A," "QACA[]_B," and "QBCA[]_B") in response to the input C/A signals (*e.g.*, "DCA[]_A" and "DCA[]_B"). *See, e.g.*,

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6.1 Signal Groups

This standard categorizes DDR5 SDRAM timing-critical signals into specific groups. Figure 2 — (Example) RDIMM 2Rx4 Topology summarizes the high-speed signal groups for RDIMM. Signal groups, except Data and Strobe, implement a fly-by topology. Actual signal names in Annex may vary from individual reference design. The signal groups are:

Pre-RCD

- Pre-RCD CA and CS group includes: DCA[6:0]_[B:A], DCS[1:0]_[B:A]_n, DPAR_[B:A]
- Pre-RCD Clock group includes: DCK_t and DCK_c
- Pre-RCD Other group includes: ALERT_n, DRST_n
- Pre-RCD Loopback group includes: DLBDQ_[B:A], DLBDQS_[B:A]

Post-RCD to SDRAM

- Post-RCD CA group includes Q[B:A]CA[13:0]_[B:A]
- Post-RCD CS group includes Q[B:A]CS[1:0]_[B:A]_n
- Post-RCD Clock group includes Q[D:A]CK_[B:A]_t, Q[D:A]CK_[B:A]_c
- Post-RCD Other group includes QRST_[B:A]_n, DERROR_[B:A]_n
- Post-RCD Other group includes QLBDQ_[B:A], QLBDQS_[B:A]

DQ and DQS - edge connector pin to SDRAM with series resistor

SideBand

- Host to Hub group: HSDA, HSCL, HSA
- Hub local bus group: LSDA, LSCL (star topology from Hub)

Ex. CC-88 (JEDEC 305B) at 12 (that RCD receives from the memory controller Pre-RCD command and address signals and generate Post-RCD command and address signals).

As explained *supra*, the Practicing '407 Products are JEDEC compliant. The JEDEC DDR5 RCD definition further demonstrates that the Practicing '407 Products generate module C/A signals (*e.g.*, “QACA[]_A,” “QBCA[]_A,” “QACA[]_B,” and “QBCA[]_B”) in response to the input C/A signals (*e.g.*, “DCA[]_A” and “DCA[]_B”). *See, e.g.*,

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3.4.1 Double Data Rate DCA

The DDR5 RCD will support DDR (Double Data Rate) on the host-interface DCA signals. The host-interface CA bus is 7 bits, plus parity. The DRAM-interface CA bus remains SDR (Single Data Rate) and is 14 bits and without parity. The RCD will expand the incoming host-interface 7-bit DCA bus to 14 bits on the DRAM interface. Table 8 shows the mapping of the incoming DCA bits to the output QCA. The Chip Selects remain SDR on both the host-interface and DRAM interface. The two channel DDR5RCD has six CA ports, two DCA input ports and four QCA output ports as shown in Figure 6.

Table 8 — DDR CA Mapping DCA to QCA

UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6
0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6
1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13

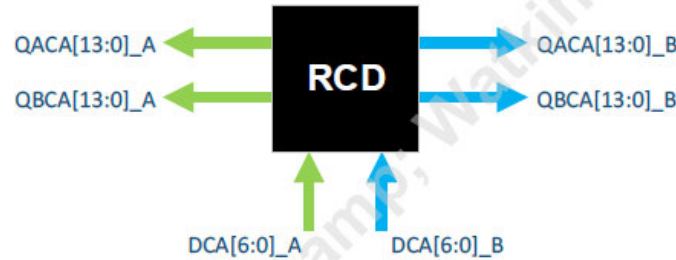


Figure 6 — RCD DCA and QCA Ports

Ex. CC-32 (JEDEC 82-512) at 15. *See also* Ex. CC-30 (JEDEC 82-513) at 15.

	DCA Channel	POD	DRAM Channel
Output Control bus	Q[B:A]CS[1:0]_B[A]_n	POD	Chip Select signals to the DRAMs. 2 copies of each signal.
Output Address and Command bus	Q[B:A]CA[13:0]_B[A]	POD	Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).
Clock outputs	Q[D:A]CK_B[A]_t Q[D:A]CK_B[A]_c	POD differential	Clock outputs to the DRAMs. Four copies per channel.

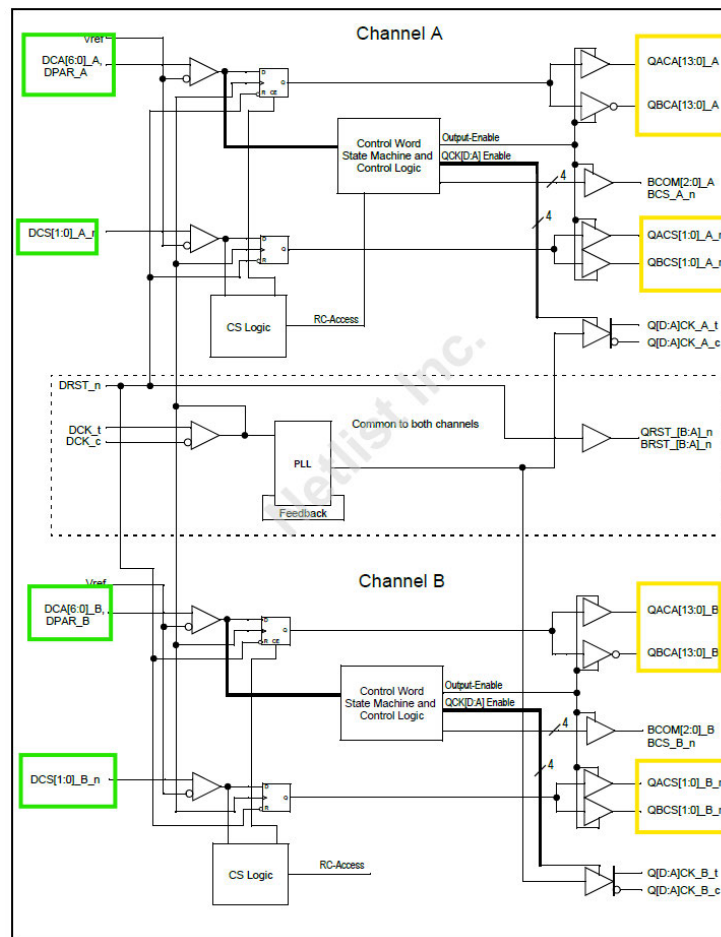
Ex. CC-32 (JEDEC 82-512) at 4. *See also* Ex. CC-30 (JEDEC 82-513) at 4.

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	3.1 Description The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-32 (JEDEC 82-512) at 6. <i>See also</i> Ex. CC-30 (JEDEC 82-513) at 6.
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Ex. CC-32 (JEDEC 82-512) at 7 (showing the RCD circuitry generates module C/A signals shown in yellow in response to the input C/A signals shown in green). *See also* Ex. CC-30 (JEDEC 82-513) at 7.

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2.2 Terminal Functions

Table 2 — Terminal Functions

Signal Group	Signal Name	Type	Description
Input Control bus	DCS[1:0]_[B:A]_n	POD V_{REF} based	Chip Select inputs to the RCD. Two inputs per DRAM channel.
Input Address and Command bus	DCA[6:0]_[B:A]	POD V_{REF} based	Command/Address bus inputs to the RCD. Separate sets per channel.
Parity input	DPAR_[B:A]	POD V_{REF} based	Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.
Clock inputs	DCK_t/DCK_c	POD differential	Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.
Loopback	DLBD_[B:A] DLBS_[B:A]	POD V_{REF} based	Loopback Inputs to RCD DLBD - loopback Data DLBS - loopback Strobe
Reset input	DRST_n	CMOS input	Active LOW asynchronous reset input. When LOW, it causes a reset of the internal latches and disables the outputs, thereby forcing the outputs to float.
Error input	DError_IN_[B:A]_n	Low voltage swing POD input	DRAM CRC ALERT_n output is connected to this input pin, which in turn is buffered and redriven to the ALERT_n output of the register. There is a separate signal per channel.
Data buffer control and communication outputs	BCS_[B:A]_n	POD	Data buffer chip selects. Indicate a valid command on the BCOM bus, and used to signal non-target termination. Separate per DRAM channel.
	BCOM[2:0]_[B:A]	POD	Data buffer command. The start of each command is signaled by an active chip select. Separate per DRAM channel.
	BRST_[B:A]_n	CMOS	Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD BRST to reset the DDR5 Data Buffer on all DIMM topologies.
	BCK_[B:A]_t BCK_[B:A]_c	POD differential	Differential clock output pair to the data buffer. Separate per DRAM channel.
Output Control bus	Q[B:A]CS[1:0]_[B:A]_n	POD	Chip Select signals to the DRAMs. 2 copies of each signal.
Output Address and Command bus	Q[B:A]CA[13:0]_[B:A]	POD	Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).
Clock outputs	Q[B:A]CLK_[B:A]_t + Q[B:A]CLK_[B:A]_c	POD differential	Clock outputs to the DRAMs. Four copies per channel.

Ex. CC-32 (JEDEC 82-512) at 4. *See also* Ex. CC-30 (JEDEC 82-513) at 4.

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3.3.3 QCA Bus Operation

Under normal operation, the RCD passes commands through from the D inputs to the Q outputs when a command is present on the bus, and drives HIGH on the QCA outputs when no command is being sent. The RCD passes the DCA bus through when either DCS input is active but there are exceptions to this:

- If the register detects a parity error and parity checking is enabled, the DCA bus is still passed through to DRAM and QCS is blocked.
- During the 2nd UI of a 2 UI command, the DCA bus must be passed through. There will only be a DCS signal active for the 2nd UI if a non-target termination is being signaled to a rank.
- When DRAM interface Block MR Mode is enabled $RW01[1] = 0$.
- When CA Pass-through mode is enabled $RW00[2] = 1$.

Ex. CC-32 (JEDEC 82-512) at 9. *See also* Ex. CC-30 (JEDEC 82-513) at 9.

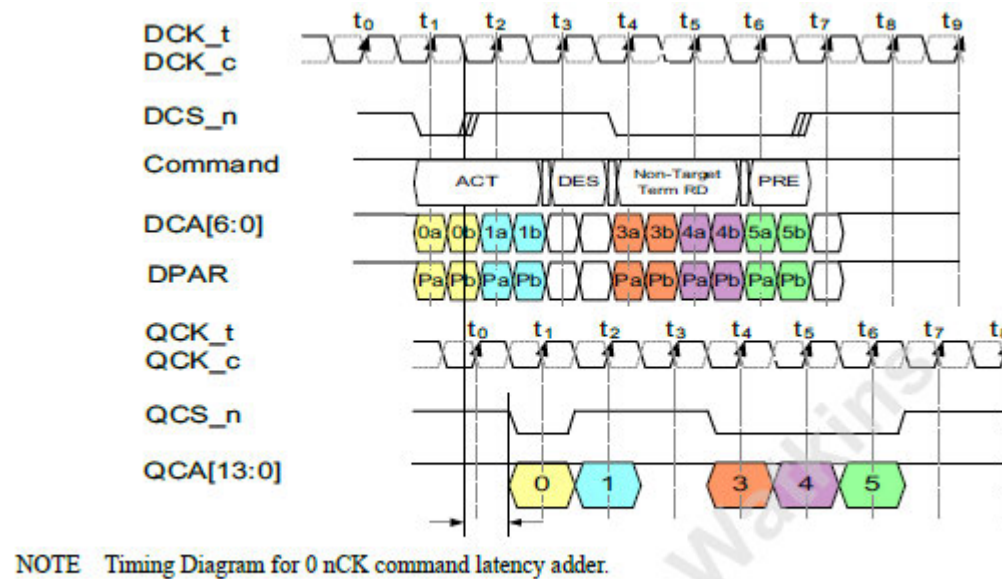


Figure 9 — DDR CA Multi-Command Timing Diagram Example

Ex. CC-32 (JEDEC 82-512) at 17 (timing diagram showing the generation of module C/A signals (QCA[13:0]) in response to the input C/A signals (DCA[6:0]). *See also* Ex. CC-30 (JEDEC 82-513) at 17.

[illegible]

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Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 10

[REDACTED]

[REDACTED]

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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 9

As explained *supra*, the Practicing '407 Products are JEDEC compliant. The JEDEC DDR5 RDIMM standard further demonstrates each Practicing '407 Products' RCD generates multiple local clock signals corresponding, respectively, to the plurality of groups of the memory devices. For example, the RCD generates multiple QCK clock signals.

6.1 Signal Groups (cont'd)

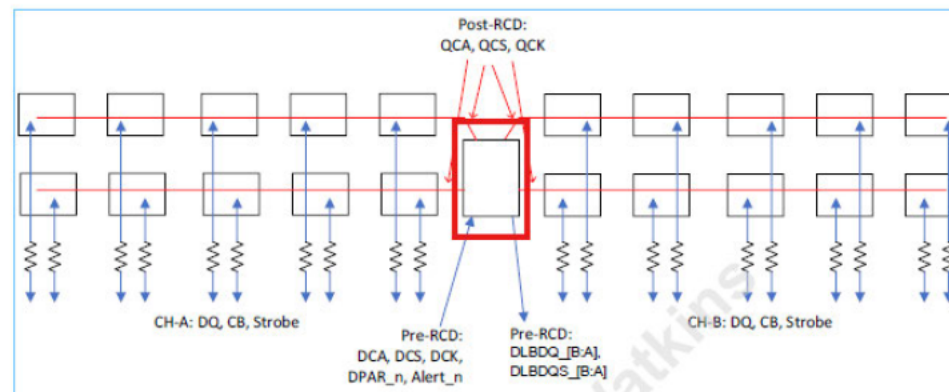


Figure 2 — (Example) RDIMM 2Rx4 Topology

Ex. CC-88 (JEDEC 305B) at 13 (illustrating RCD, in red, receiving “Pre-RCD” clock (“DCK”) and generating “Post-RCD” control and address signals (“QCK”).

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6.9.1 DIMM Connectivity Wiring

2Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Top	Upper	QBCK	QBCS0	QBCA	0
Bottom	Lower	QCCK	QACS1	QACA	1
Bottom	Upper	QDCK	QBCS1	QBCA	1

2Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Top		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]
Bottom	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

2Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Top	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]
Bottom		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

1Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QBCS0	QACA	0

1Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

1Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

2Rx8					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QACS1	QACA	1

2Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]	
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

2Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom	[7:0]		[15:8]		[23:16]		[31:24]			CB[7:0]

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6.9.1 DIMM Connectivity Wiring (cont'd)

1Rx8 (less than 8000 Mbps)										
DIMM Side	Row	Clock	CS	CA	Rank					
Top	Lower	QACK	QACS0	QACA	0					
Bottom	Lower	QACK	QACS0	QACA	0					
1Rx8 (8000 Mbps and Faster)										
DIMM Side	Row	Clock	CS	CA	Rank					
Top	Lower	QCKK	QACS0	QACA	0					
Bottom	Lower	QCKK	QACS0	QACA	0					
1Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[15:8]					[31:24]				
Bottom	[7:0]					[23:16]				
1Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[7:0]					[23:16]				
Bottom	[15:8]					[31:24]				

Figure 12 — DIMM Connectivity Wiring

Ex. CC-88 (JEDEC 305B) at 31-32 (showing groups of memory devices on various sides and in various rows on the module configured to receive corresponding local clocks, e.g., QACK, QBCK, QCKK, QDCK, generated by the RCD).

As explained *supra*, the Practicing '407 Products are JEDEC compliant. The JEDEC DDR5 RCD definition further demonstrates each Practicing '407 Products' RCD generates multiple local clock signals corresponding, respectively, to the plurality of groups of the memory devices. For example, the RCD generates multiple CK_t and CK_c clock signals. *See, e.g.,*

Clock outputs		POD differential	
Q[D:A]CK _{[B:A]_t} Q[D:A]CK _{[B:A]_c}		Clock outputs to the DRAMs. Four copies per channel.	
Feedback	OT RD	POD	Feedback outputs to Host

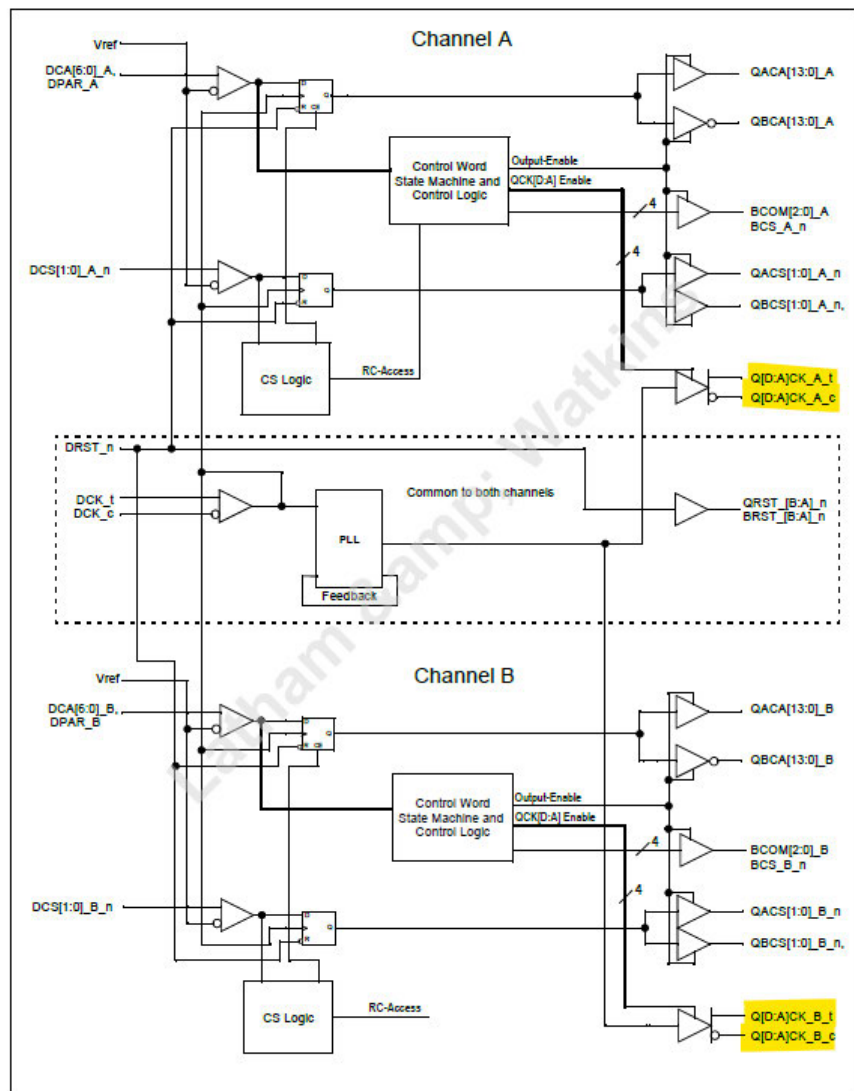
Ex. CC-32 (JEDEC 82-512) at 4 (“Four copies per channel.”). *See also* Ex. CC-30 (JEDEC 82-513) at 4.

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	3.1 Description The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-32 (JEDEC 82-512) at 6. <i>See also</i> Ex. CC-30 (JEDEC 82-513) at 6.
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3.2 Logic Diagram Positive Logic



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	Ex. CC-32 (JEDEC 82-512) at 7. <i>See also</i> Ex. CC-30 (JEDEC 82-513) at 7.
[1.c.3.i] the plurality of local clocks having respective phase relationships with the system clock,	In the Practicing '407 Products, the plurality of local clocks have respective phase relationships with the system clock. For example: As explained <i>supra</i>, the Practicing '407 Products are JEDEC compliant. The JEDEC DDR5 RCD definition demonstrates the RCD of the Practicing '407 Products supports “output delay control,” where the local clocks have respective phase relationships with the system clock. <i>See, e.g.,</i> 3.3.8 Per-group Output Delay Control The RCD supports output delay control for output signal groups defined and located in RW12 to RW1E. The output delay settings are incremental CK fractions of 1/64 tCK. Each signal group can be enabled or disabled independently through RCD Control Words. The Host must wait t_{ODU} after the Output Delay Control Word has been written to a signal group before the output becomes stable. The output delay feature controlled by RW12 to RW1E is not supported when the RCD is configured in Test Frequency mode (Band 2, RW05[7] = 1) or in PLL bypass mode (RW05[3:0] = 1111). Ex. CC-32 (JEDEC 82-512) at 12. <i>See also</i> Ex. CC-30 (JEDEC 82-513) at 11-12. For example, local clocks (QxCK) shown below have respective phase relationships with respect to the system clock (DCK).

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14.7 Output Clock Driver Characteristics (cont'd)

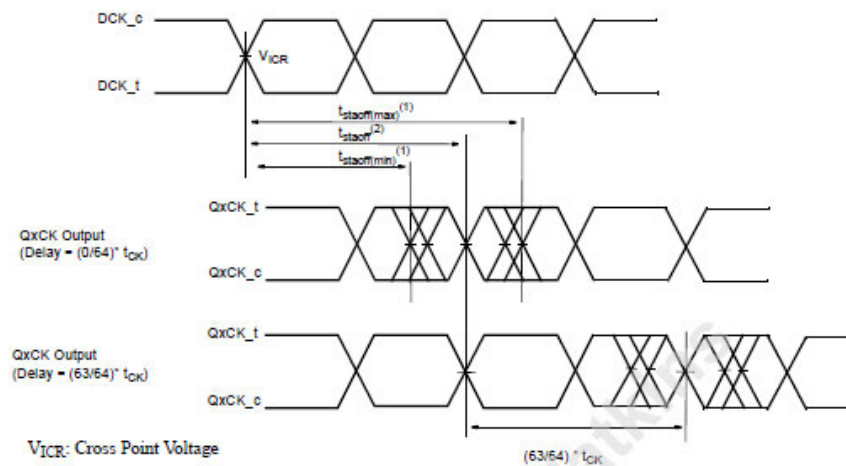


Figure 112 — Definition for t_{staoff} and t_{dynoff} in SDR Mode

Ex. CC-32 (JEDEC 82-512) at 242. See also Ex. CC-30 (JEDEC 82-513) at 253.

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Table 23 — Control Words Duplicated for Frequency Context Support

Register Address	Description
RW0A	QCK Signals Driver Characteristics Control Word
RW0C	QxCA and QxCS _n Signals Driver Characteristics Control Word
RW0D	Data Buffer Interface Driver Characteristics Control Word
RW10	IBT Control Word
RW11[3:0]	Command Latency Adder Configuration Control Word
RW12	QACK Output Delay Control Word
RW13	QBCK Output Delay Control Word
RW14	QCKK Output Delay Control Word
RW15	QDCK Output Delay Control Word
RW17 .. RW1A	QxCS _n Output Delay Control Word
RW1B, RW1C	QxCA Output Delay Control Word
RW1D	BCS _n and BCOM[2:0] Output Delay Control Word
RW1E	BCK Output Delay Control Word
RW31[7:4, 0]	DfE Configuration Control Word
RW40 .. RW47	Internal VrefCA Control Word
RW48, RW49	Internal VrefCS Control Word
RW50[0], RW51, RW52[3:0], RW53[3:0], RW54	CTLE Control Words
PG[00, 01]	DfE Gain and Tap 1, Tap 2, Tap 3, Tap 4 Coefficients per Input Receiver
PG[5]RW[7B:60]	Per-bit QCA Output Delay Control Words

Ex. CC-32 (JEDEC 82-512) at 68 (showing local clocks QACK, QBCK, QCKK, and QDCK that have respective phase relationships with the system clock). *See also* Ex. CC-30 (JEDEC 82-513) at 70.

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8.13.2 RW12 - QACK Output Delay Control Word

Table 118 — RW12: QACK Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QACK _t / QACK _c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by +(1/64) * t _{CK}
x	x	0	0	0	0	1	0		Delay Outputs by +(2/64) * t _{CK}
x	x								...
x	x	1	1	1	1	0	1		Delay Outputs by +(61/64) * t _{CK}
x	x	1	1	1	1	1	0		Delay Outputs by +(62/64) * t _{CK}
x	x	1	1	1	1	1	1		Delay Outputs by +(63/64) * t _{CK}
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x	Reserved	Reserved
0	x	x	x	x	x	x	x	Output Delay Feature Enable for	(Default) Feature Disabled
1	x	x	x	x	x	x	x	QACK _t /QACK _c ²	Feature Enabled ³

NOTE 1 These control bits do not have any effect unless RW12[7] = 1.
 NOTE 2 RW12[7:0] will be sticky, cleared by power cycle not Reset.
 NOTE 3 When feature is enabled the delay settings in RW12[5:0] require a time of t_{OPR} for the delay to become stable on the outputs.

Ex. CC-32 (JEDEC 82-512) at 146; *see also id.* at 147-148 (showing tables 119, 120, and 121). *See also* Ex. CC-30 (JEDEC 82-513) at 150-152.

[1.c.3.ii] the respective phase relationships being programmable independently of each other; and

In the Practicing '407 Products, the respective phase relationships are programmable independently of each other.
 As explained *supra*, the Practicing '407 Products are JEDEC compliant. The JEDEC DDR5 RCD definition demonstrates the RCD of the Practicing '407 Products supports “output delay control,” where the local clocks have respective phase relationships with the system clock. The delay control of the local clocks (QxCK) with respect to the system clock (DCK) can be independently programmed. *See, e.g.,*

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3.3.8 Per-group Output Delay Control

The RCD supports output delay control for output signal groups defined and located in RW12 to RW1E. The output delay settings are incremental CK fractions of 1/64 tCK. Each signal group can be enabled or disabled independently through RCD Control Words. The Host must wait t_{ODU} after the Output Delay Control Word has been written to a signal group before the output becomes stable. The output delay feature controlled by RW12 to RW1E is not supported when the RCD is configured in Test Frequency mode (Band 2, RW05[7] = 1) or in PLL bypass mode (RW05[3:0] = 1111).

Ex. CC-32 (JEDEC 82-512) at 12 (showing “output delay” for output signal groups). *See also* Ex. CC-30 (JEDEC 82-513) at 11-12.

Table 23 — Control Words Duplicated for Frequency Context Support

Register Address	Description
RW0A	QCK Signals Driver Characteristics Control Word
RW0C	QxCA and QxCS_n Signals Driver Characteristics Control Word
RW0D	Data Buffer Interface Driver Characteristics Control Word
RW10	IBT Control Word
RW11[3:0]	Command Latency Adder Configuration Control Word
RW12	QACK Output Delay Control Word
RW13	QBCK Output Delay Control Word
RW14	QCK Output Delay Control Word
RW15	QDCK Output Delay Control Word
RW17 .. RW1A	QxCS_n Output Delay Control Word
RW1B, RW1C	QxCA Output Delay Control Word
RW1D	BCS_n and BCOM[2:0] Output Delay Control Word
RW1E	BCK Output Delay Control Word
RW31[7:4, 0]	DFE Configuration Control Word
RW40 .. RW47	Internal VrefCA Control Word
RW48, RW49	Internal VrefCS Control Word
RW50[0], RW51, RW52[3:0], RW53[3:0], RW54	CTLE Control Words
PG[00, 01]	DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4 Coefficients per Input Receiver
PG[5]RW[7B:60]	Per-bit QCA Output Delay Control Words

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Ex. CC-32 (JEDEC 82-512) at 68 (showing delays in QACK, QBCK, QCK, and QDCK can be set independently). *See also* Ex. CC-30 (JEDEC 82-513) at 70.

8.13.2 RW12 - QACK Output Delay Control Word

Table 118 — RW12: QACK Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QACK_t/QACK_c Output Signals ^{1,2}	Delay Outputs by $+(0/64) \cdot t_{CK}$ (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by $+(1/64) \cdot t_{CK}$
x	x	0	0	0	0	1	0		Delay Outputs by $+(2/64) \cdot t_{CK}$
x	x								...
x	x	1	1	1	1	0	1		Delay Outputs by $+(61/64) \cdot t_{CK}$
x	x	1	1	1	1	1	0		Delay Outputs by $+(62/64) \cdot t_{CK}$
x	x	1	1	1	1	1	1		Delay Outputs by $+(63/64) \cdot t_{CK}$
x	0	x	x	x	x	x	x		Reserved
x	1	x	x	x	x	x	x	Output Delay Feature Enable for QACK_t/QACK_c ²	Reserved
0	x	x	x	x	x	x	x		(Default) Feature Disabled
1	x	x	x	x	x	x	x		Feature Enabled ³

NOTE 1 These control bits do not have any effect unless RW12[7] = 1.
 NOTE 2 RW12[7:0] will be sticky, cleared by power cycle not Reset.
 NOTE 3 When feature is enabled the delay settings in RW12[5:0] require a time of t_{OPU} for the delay to become stable on the outputs.

Ex. CC-32 (JEDEC 82-512) at 146; *see also id.* at 147-148 (showing how to program the delays independently, in tables 119, 120, and 121). *See also* Ex. CC-30 (JEDEC 82-513) at 150-152.

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Table 232 — Output Clock Driver Characteristics at Application Mode (Frequency Band 1) (cont'd)

Symbol	Parameter	Conditions	DDR5-3200		DDR5-3600		DDR5-4000/ 4400/4800		DDR5-5200/ 5600		Unit	Note
			Min	Max	Min	Max	Min	Max	Min	Max		
t _{STAB_DBDATA}	LRDIMM stabilization time for commands requiring use of data path	DCK_t/DCK_c stable	-	4.14	-	4.14	-	4.14	-	4.14	μs	4,5
t _{CKsk}	Fractional Clock Output Skew		-	20	-	20	-	20	-	20	ps	6
t _{stao}	Clock delay through the register between the input clock and output clock		Min: (t _{PDM} (min) + 1/2 tCK); Max: (t _{PDM} (max) + 1/2 tCK)								ps	7,8
t _{dynoff}	Maximum re-driven dynamic clock offset		-	40	-	35	-	35	-	35	ps	9,10
NOTE 1 At Down-bin Data Rate (1980 MT/s ≤ f ≤ 2100 MT/s), the max value of t_{STAB01} is 10μs. NOTE 2 This parameter applies to control word writes to RW05/RW06 and Bit OP0 in RW00. Since application of RW05/RW06 is postponed in the frequency change sequence defined in Section 5.4.2, a t_{STAB01} waiting time is always sufficient after exit from Self Refresh with Clock Stop (i.e., t_{STAB02} is not required). See timing diagrams in Figure 35 on page 48 and Figure 36 on page 49. NOTE 3 Measured from start of DCK after Exit from Self Refresh with Clock Stop to the first DCA Command that gets forwarded to the LRDIMM BCOM bus interface but does not require the use of data path (e.g., SRX, NOP, etc.). NOTE 4 Applies only to Self Refresh Exit without frequency change. NOTE 5 Measured from start of DCK after Exit from Self Refresh with Clock Stop to the first DCA Command that gets forwarded to the LRDIMM BCOM bus interface and requires the use of data path (e.g., WR, RD). NOTE 6 This skew represents the absolute output clock skew and contains the pad skew and package skew (See Figure 113). This parameter is specified for the clock pairs on each side of the register independently. The skew is applicable to left side clock pairs between QACK_A_t/QACK_A_c and QCCK_A_t/QCCK_A_c, as well as right side of the clock pairs between QACK_B_t/QACK_B_c and QCCK_B_t/QCCK_B_c. Similarly, to left side clock pairs between QBCK_A_t/QBCK_A_c and QDCK_A_t/QDCK_A_c, as well as right side of the clock pairs between QBCK_B_t/QBCK_B_c and QDCK_B_t/QDCK_B_c. This is not a tested parameter and has to be considered as a design goal only. NOTE 7 RW05/RW06 must be set to correct speed and the device must run at one of the speed nodes defined in Table 199 on page 191. See Figure 111 and Figure 112. NOTE 8 This measures the delay from the rising differential input clock which samples incoming DCA to the rising differential output clock that will be used to sample the same QCA data. The parameter describes the maximum and minimum QxCK clock t_{pp}. NOTE 9 V_{DD} measurement DC bandwidth is limited to 20 MHz. NOTE 10 See Figure 111 and Figure 112.												

Ex. CC-32 (JEDEC 82-512) at 240 (describing “output clock skew” that “is specified for the clock pairs on each side of the register independently”). *See also* Ex. CC-30 (JEDEC 82-513) at 250.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

14.7 Output Clock Driver Characteristics (cont'd)

Table 233 — Output Clock Driver Characteristics at Test Mode (Frequency Band 2)

Symbol	Parameter	Conditions	Test Frequency (140 to 990 MHz)		Unit	Note
			Min	Max		
t _{STAB01}	RCD Stabilization time from start of DCK toggle to PLL lock	DCK_t/DCK_c stable	-	50	μs	
t _{STAB02}	RCD Stabilization time from control word write to PLL lock	DCK_t/DCK_c stable	-	50	μs	1
t _{CKsk}	Fractional Clock Output Skew		-	100	ps	2
t _{staoff}	Clock delay through the register between the input clock and output clock		Min: (t _{PDm} (min) + 1/2 tCK); Max: (t _{PDm} (max) + 1/2 tCK)		ps	3,4
t _{dynoff}	Maximum re-driven dynamic clock offset		-	500	ps	5,6

NOTE 1 This parameter applies to control word writes to RW05 and Bit OP0 in RW00.

NOTE 2 This skew represents the absolute output clock skew and contains the pad skew and package skew (See Figure 113). This parameter is specified for the clock pairs on each side of the register independently. The skew is applicable to left side clock pairs between QACK_A_t/QACK_A_c and QCKK_A_t/QCKK_A_c, as well as right side of the clock pairs between QACK_B_t/QACK_B_c and QCKK_B_t/QCKK_B_c. Similarly, to left side clock pairs between QBCK_A_t/QBCK_A_c and QDCK_A_t/QDCK_A_c, as well as right side of the clock pairs between QBCK_B_t/QBCK_B_c and QDCK_B_t/QDCK_B_c. This is not a tested parameter and has to be considered as a design goal only.

NOTE 3 RW05 must not be set to enable PLL bypass mode and the device must run within the Test Frequency range defined in Table 199 on page 191. See Figure 111 and Figure 112.

NOTE 4 This measures the delay from the rising differential input clock which samples incoming DCA to the rising differential output clock that will be used to sample the same QCA data. The parameter describes the maximum and minimum QxCK clock t_{PD}.

NOTE 5 V_{DD} measurement DC bandwidth is limited to 20 MHz.

NOTE 6 See Figure 111 and Figure 112.

Ex. CC-32 (JEDEC 82-512) at 241 (describing “output clock skew” that “is specified for the clock pairs on each side of the register independently”). *See also* Ex. CC-30 (JEDEC 82-513) at 251.

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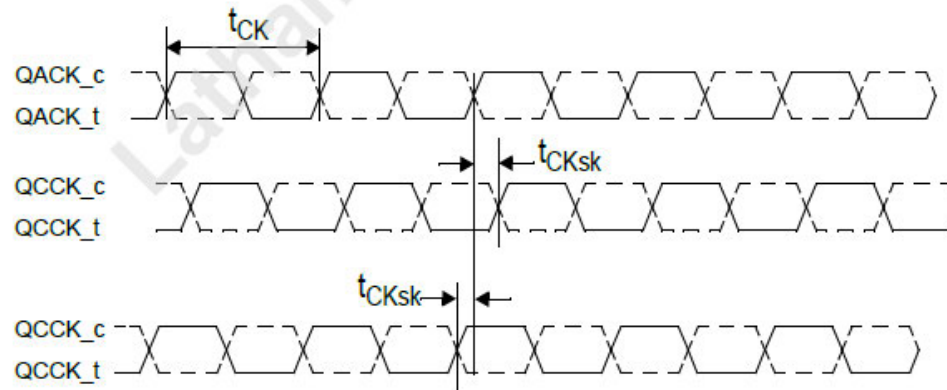


Figure 113 — Clock Output (QxCK) Skew

Ex. CC-32 (JEDEC 82-512) at 242. *See also* Ex. CC-30 (JEDEC 82-513) at 254.

[1.c.4] output the module C/A signals to the memory devices; and

See evidence cited for element [1.c.2].

The Practicing '407 Products comprise circuitry mounted on the PCB and configurable to output the module C/A signals to the memory devices.

For example, each Practicing '407 Products' RCD outputs module C/A signals ([REDACTED]) to SDRAM and DRAM memory devices as shown below.

See, e.g.,

[REDACTED]

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[REDACTED]

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 10 [REDACTED]

[REDACTED]

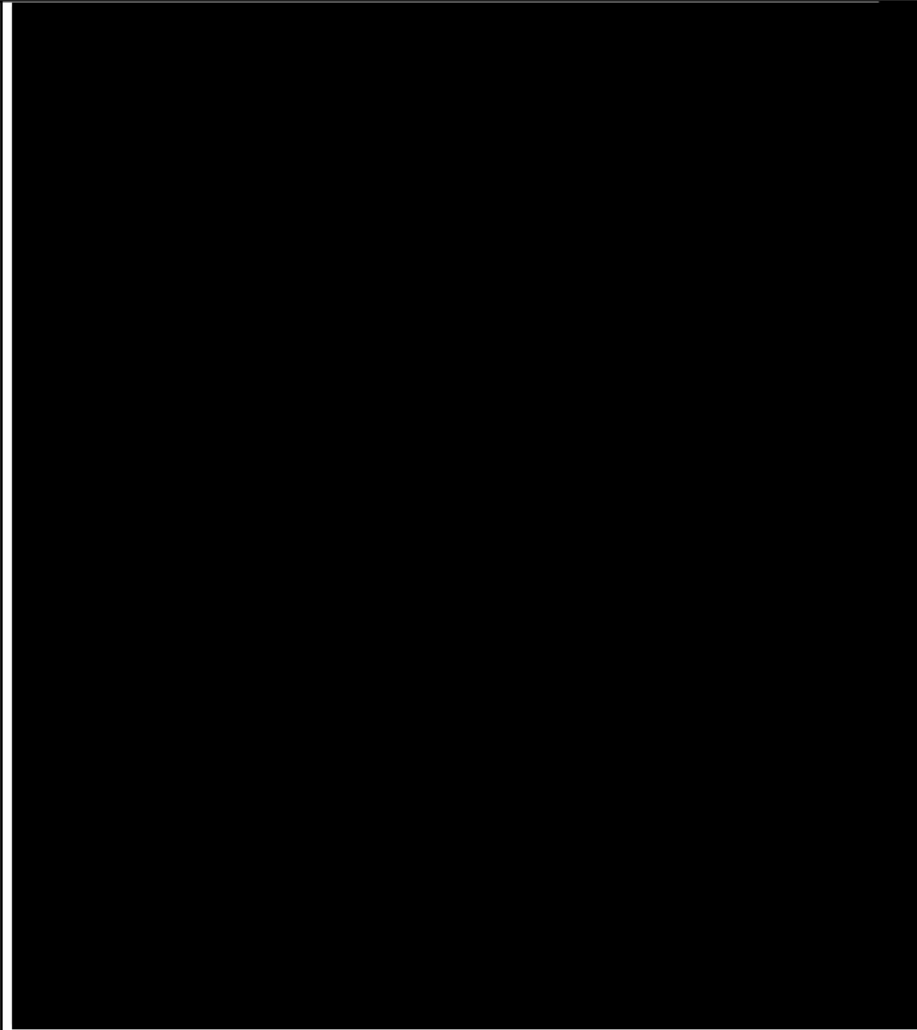
[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	[REDACTED]	
	[REDACTED]	.
	[REDACTED]	
	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 18 [REDACTED]	
	[REDACTED]	

[REDACTED]

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	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 9	

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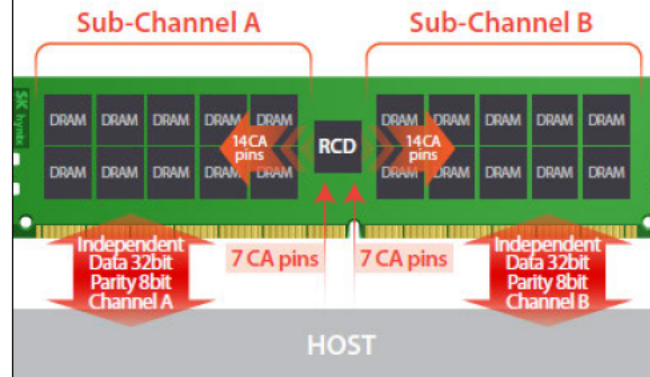
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 10

The latest DDR5 DRAM standard turbo-boosts RDIMM module performance and technology for the era of AI and Big Data. SK hynix expects its DDR5 RDIMMs to reach over 8,000Mbps in data rates with research into continued speed enhancements.

Features



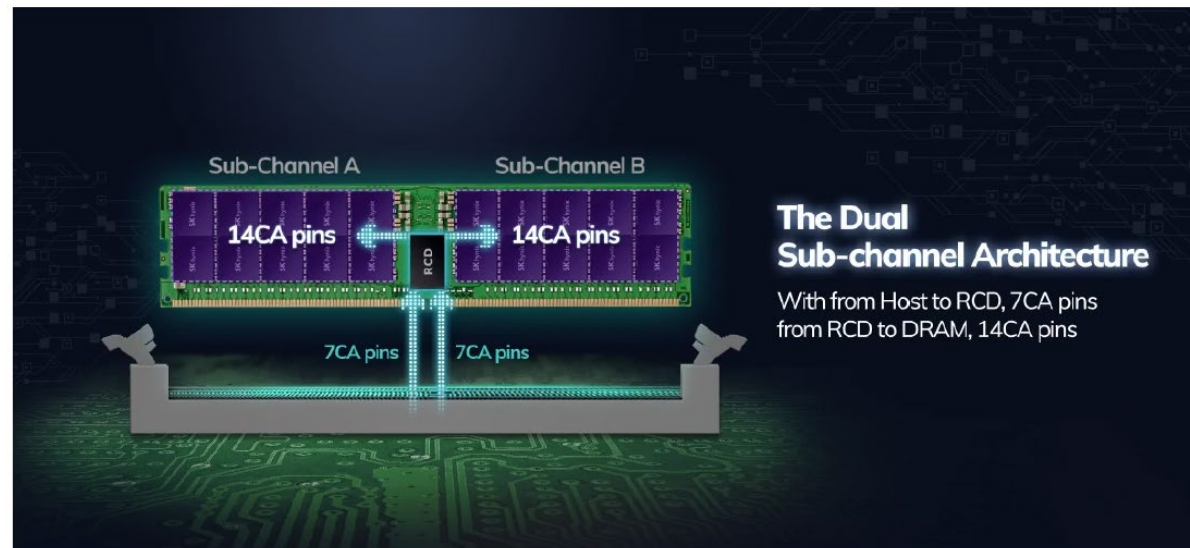
New Sub-channel DRAM Architecture

The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

Ex. CC-40 (SK Hynix A New Standard) at 5 (demonstrating (1) the RCD circuitry generating module C/A signals in response to the input C/A signals from the “HOST” and (2) the RCD circuitry outputting those module C/A signals to DRAM memory devices).

Double Sub-channel DRAM Architecture

With a dual sub-channel structure, DDR5 RDIMM increases the number of CA (Command-Address) pins to seven double-data-rate CA buses each from host to RCD and 14 single-data-rate buses each on RCD to DRAM. Having two independent 40-bit channels thus enables 2x bandwidth and higher bus (signal) efficiency compared to a single-channel DDR4 DIMM.



Ex. CC-41 (SK hynix RDIMM) at 2 (demonstrating (1) the RCD circuitry generating module C/A signals through "14CA Pins" in response to the input C/A signals from the "Host" and (2) the RCD circuitry outputting those module C/A signals to DRAM memory devices).

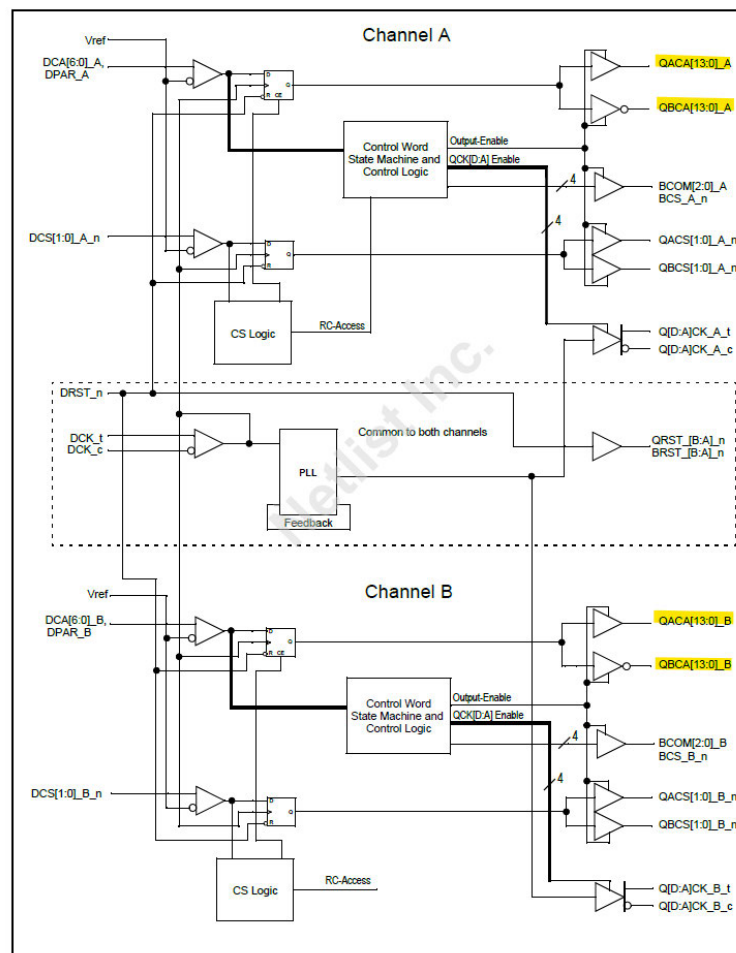
As explained *supra*, the Practicing '407 Products are JEDEC compliant. The JEDEC DDR5 RCD definition demonstrates each Practicing '407 Products' RCD outputs module C/A signals to memory devices. *See, e.g.,*

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	3.1 Description The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-32 (JEDEC 82-512) at 6. <i>See also</i> Ex. CC-30 (JEDEC 82-513) at 6.
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



Ex. CC-32 (JEDEC 82-512) at 7. *See also* Ex. CC-30 (JEDEC 82-513) at 7.

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Data buffer control and communication outputs	BCS_[B:A]_n	POD	the register. There is a separate signal per channel.
	BCOM[2:0]_[B:A]	POD	Data buffer chip selects. Indicate a valid command on the BCOM bus, and used to signal non-target termination. Separate per DRAM channel.
	BRST_[B:A]_n	CMOS	Data buffer command. The start of each command is signaled by an active chip select. Separate per DRAM channel.
	BCK_[B:A]_t BCK_[B:A]_c	POD differential	Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD BRST to reset the DDR5 Data Buffer on all DIMM topologies.
Output Control bus	Q[B:A]CS[1:0]_[B:A]_n	POD	Differential clock output pair to the data buffer. Separate per DRAM channel.
Output Address and Command bus	Q[B:A]CA[13:0]_[B:A]	POD	Chip Select signals to the DRAMs. 2 copies of each signal.
Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).
Clock outputs to the DRAMs. Four copies per channel.			

Ex. CC-32 (JEDEC 82-512) at 4. *See also* Ex. CC-30 (JEDEC 82-513) at 4.

3.3.3 QCA Bus Operation

Under normal operation, the RCD passes commands through from the D inputs to the Q outputs when a command is present on the bus, and drives HIGH on the QCA outputs when no command is being sent. The RCD passes the DCA bus through when either DCS input is active but there are exceptions to this:

- If the register detects a parity error and parity checking is enabled, the DCA bus is still passed through to DRAM and QCS is blocked.
- During the 2nd UI of a 2 UI command, the DCA bus must be passed through. There will only be a DCS signal active for the 2nd UI if a non-target termination is being signaled to a rank.
- When DRAM interface Block MR Mode is enabled RW01[1] = 0.
- When CA Pass-through mode is enabled RW00[2] = 1.

Ex. CC-32 (JEDEC 82-512) at 9. *See also* Ex. CC-30 (JEDEC 82-513) at 9.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	3.4.1 Double Data Rate DCA The DDR5 RCD will support DDR (Double Data Rate) on the host-interface DCA signals. The host-interface CA bus is 7 bits, plus parity. The DRAM-interface CA bus remains SDR (Single Data Rate) and is 14 bits and without parity. The RCD will expand the incoming host-interface 7-bit DCA bus to 14 bits on the DRAM interface. Table 8 shows the mapping of the incoming DCA bits to the output QCA. The Chip Selects remain SDR on both the host-interface and DRAM interface. The two channel DDR5RCD has six CA ports, two DCA input ports and four QCA output ports as shown in Figure 6. Table 8 — DDR CA Mapping DCA to QCA <table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <th>UI</th> <th>DCA0</th> <th>DCA1</th> <th>DCA2</th> <th>DCA3</th> <th>DCA4</th> <th>DCA5</th> <th>DCA6</th> </tr> <tr> <td>0</td> <td>QCA0</td> <td>QCA1</td> <td>QCA2</td> <td>QCA3</td> <td>QCA4</td> <td>QCA5</td> <td>QCA6</td> </tr> <tr> <td>1</td> <td>QCA7</td> <td>QCA8</td> <td>QCA9</td> <td>QCA10</td> <td>QCA11</td> <td>QCA12</td> <td>QCA13</td> </tr> </table> Figure 6 — RCD DCA and QCA Ports Ex. CC-32 (JEDEC 82-512) at 15. <i>See also</i> Ex. CC-30 (JEDEC 82-513) at 15.	UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6	0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6	1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13
UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6																		
0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6																		
1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13																		
[1.c.5] output the plurality of local clocks to the memory devices,	<i>See</i> evidence cited for element [1.c.3]. The Practicing '407 Products comprise circuitry mounted on the PCB and configurable to output the plurality of local clocks to the memory devices. For example: The DDR5 RCD circuitry in the Practicing '407 Products is configurable to output the plurality of local clocks (e.g., [REDACTED] to SDRAM and DRAM memory devices as shown below. <i>See, e.g.,</i>																								

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 10 [REDACTED]

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	[REDACTED]	
	[REDACTED].	
	[REDACTED]	
	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 18 [REDACTED]	
	[REDACTED]	

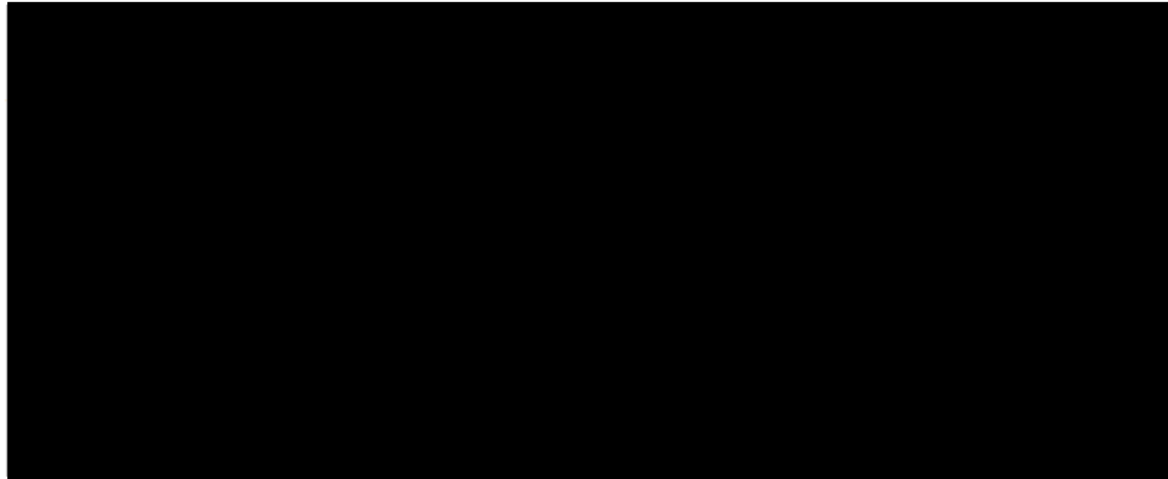
[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	A large rectangular area of the table is completely redacted with a solid black fill, obscuring all content within its boundaries. Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 9 [REDACTED]
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[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart



Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 14.

Furthermore, the RCD in the Practicing '407 Products comply with JEDEC DDR5 RCD specification. The JEDEC DDR5 RCD specification explains the RCD outputs the plurality of local clocks to the memory devices. *See, e.g.,*

Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	cycles (i.e., Deselect). Clock outputs to the DRAMs. Four copies per channel.
If nonback	OT RD	POD	If nonback outputs to Host

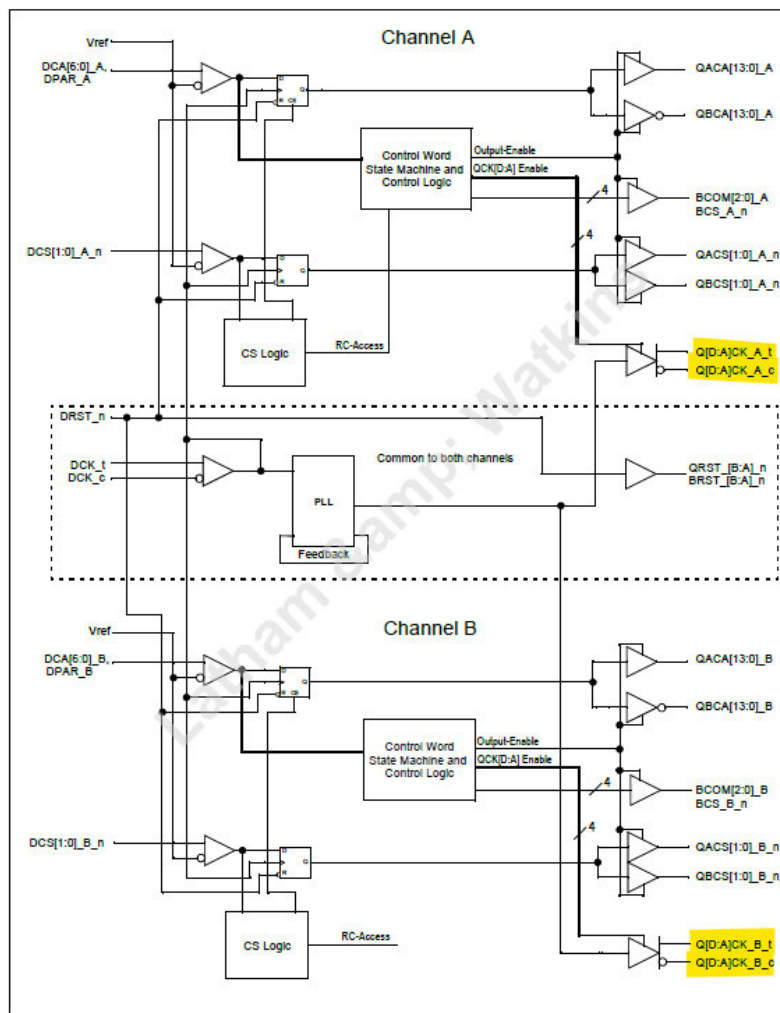
Ex. CC-32 (JEDEC 82-512) at 4 (showing the RCD includes clock outputs for the memory devices). *See also* Ex. CC-30 (JEDEC 82-513) at 4.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	3.1 Description The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-32 (JEDEC 82-512) at 6. <i>See also</i> Ex. CC-30 (JEDEC 82-513) at 6.
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

3.2 Logic Diagram Positive Logic



Ex. CC-32 (JEDEC 82-512) at 7. See also Ex. CC-30 (JEDEC 82-513) at 7.

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[1.c.5.i] wherein a respective local clock having a respective programmable phase relationship with the system clock is output to a corresponding group of the memory devices and not to any other group of the memory devices,	In the Practicing '407 Products, a respective local clock has a respective programmable phase relationship with the system clock is output to a corresponding group of the memory devices and not to any other group of the memory devices. For example: The local clocks generated by the RCD with a programmable phase relationship with the system clock (<i>see</i> limitations 1.c.3, 1.c.3.i, and 1.c.3.ii above) are output to a corresponding group of the memory devices and not to any other group of the memory devices. <i>See, e.g.,</i>
--	---

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

[REDACTED]

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 10 [REDACTED]

[REDACTED]

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	[REDACTED]	[REDACTED]
	[REDACTED]	

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 9

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	As explained <i>supra</i>, the Samsung Practicing '407 Products are JEDEC compliant. The JEDEC DDR5 RDIMM standard further demonstrates the local clocks generated by the Samsung Practicing '407 Products' RCD with a programmable phase relationship with the system clock (<i>see</i> limitations 1.c.3, 1.c.3.i, and 1.c.3.ii above) are output to a corresponding group of the memory devices and not to any other group of the memory devices.
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6.9.1 DIMM Connectivity Wiring

2Rx4										
DIMM Side	Row	Clock	CS	CA	Rank					
Top	Lower	QACK	QACS0	QACA	0					
Top	Upper	QBCK	QBCS0	QBCA	0					
Bottom	Lower	QCCK	QACS1	QACA	1					
Bottom	Upper	QDCK	QBCS1	QBCA	1					
2Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Top		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]
Bottom	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]
2Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Top	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]
Bottom		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]
1Rx4										
DIMM Side	Row	Clock	CS	CA	Rank					
Top	Lower	QACK	QACS0	QACA	0					
Bottom	Lower	QCCK	QBCS0	QACA	0					
1Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]
1Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]
2Rx8										
DIMM Side	Row	Clock	CS	CA	Rank					
Top	Lower	QACK	QACS0	QACA	0					
Bottom	Lower	QCCK	QACS1	QACA	1					
2Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]	
Bottom	[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]	
2Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom	[7:0]		[15:8]		[23:16]		[31:24]			CB[7:0]

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6.9.1 DIMM Connectivity Wiring (cont'd)

1Rx8 (less than 8000 Mbps)										
DIMM Side	Row	Clock	CS	CA	Rank					
Top	Lower	QACK	QACS0	QACA	0					
Bottom	Lower	QACK	QACS0	QACA	0					
1Rx8 (8000 Mbps and Faster)										
DIMM Side	Row	Clock	CS	CA	Rank					
Top	Lower	QCCK	QACS0	QACA	0					
Bottom	Lower	QCCK	QACS0	QACA	0					
1Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top			[15:8]				[31:24]			
Bottom	[7:0]				[23:16]				CB[7:0]	
1Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[7:0]				[23:16]					
Bottom			[15:8]				[31:24]		CB[7:0]	

Figure 12 — DIMM Connectivity Wiring

Ex. CC-88 (JEDEC 305B) at 31-32 (showing local clocks, e.g., QACK, QBCK, QCCK, QDCK, being output to certain groups of memory devices but not others).

[1.c.5.ii] wherein the corresponding group of the memory devices is configurable to perform memory read or write operations by communicating data signals with the memory controller via a corresponding subset of data signal lines in accordance with the respective local clock.

In the Practicing '407 Products, the corresponding group of the memory devices is configurable to perform memory read or write operations by communicating data signals with the memory controller via a corresponding subset of data signal lines in accordance with the respective local clock.

For example, groups of memory devices in the Practicing '407 Products associated with Channel A perform memory read and write operations by communicating data signals with the memory controller via a subset of data signal lines ([REDACTED] See, e.g.:

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 4

[REDACTED]

[REDACTED]

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	[REDACTED]	
	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 5 [REDACTED]	

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

A large, solid black rectangular redaction box covers the central portion of the page, obscuring all text and graphics that might have been present. The box is perfectly rectangular and extends across most of the page's width and height.

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 7 [REDACTED] [REDACTED]
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[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

[REDACTED]

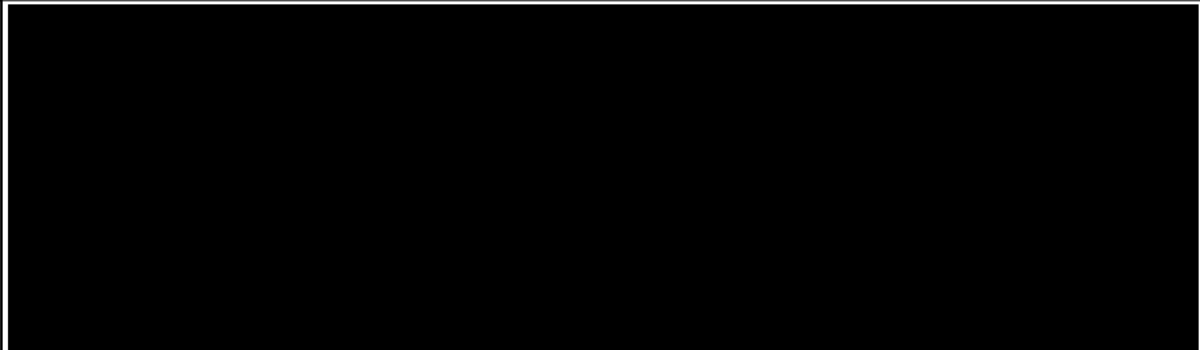
Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 8 [REDACTED]
[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 11.

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

		
	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 4 [REDACTED])	

A large, solid black rectangular redaction box covers the central portion of the page, obscuring all text and graphics that might have been present. The box is perfectly uniform in color and extends across most of the page's width and height.

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 5 ([REDACTED] [REDACTED]).
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

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[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 6 ([REDACTED])

[REDACTED].

[REDACTED]

Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 7 ([REDACTED])

[REDACTED]).

[REDACTED]

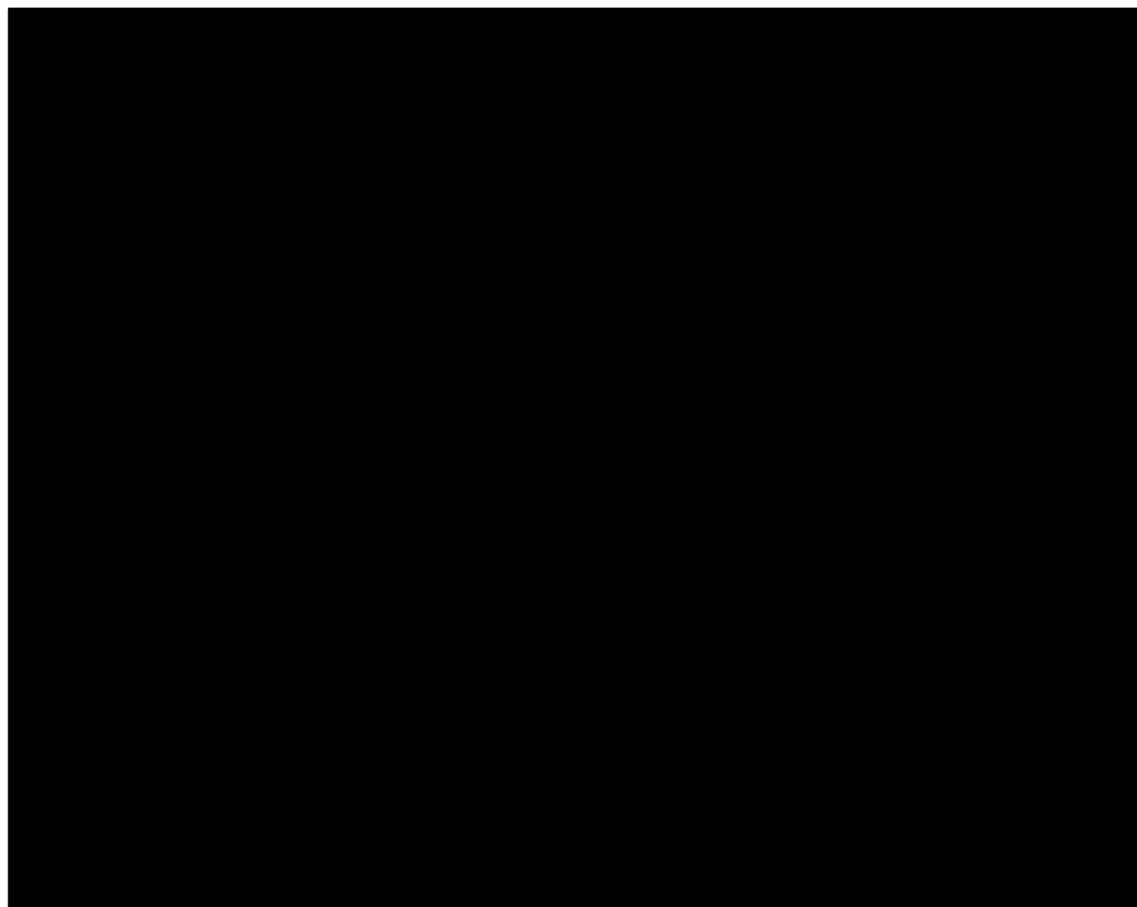
Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 10 ([REDACTED])

[REDACTED].

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Furthermore, a group of memory devices associated with Channel B perform memory read and write operations by communicating data signals with the memory controller via a subset of data signal lines (<i>e.g.</i> , [REDACTED] <i>See, e.g.,</i>
--	--



[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 4	[REDACTED]
	[REDACTED]	[REDACTED]
	[REDACTED]	[REDACTED]
	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 5	[REDACTED]
	[REDACTED]	[REDACTED]

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

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[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 7 [REDACTED] [REDACTED]
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A large, solid black rectangular redaction box covers the majority of the page, obscuring all text and graphics. The box is positioned centrally and extends nearly to the top and bottom edges of the document. Only the left and right margins of the page are visible, showing white space and thin black lines that likely represent the page's borders or margins.

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 8	[REDACTED]
	[REDACTED]	
	[REDACTED]	
	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 9	[REDACTED]
	[REDACTED]	

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

[REDACTED]

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 12 [REDACTED]

[REDACTED]

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	[REDACTED] Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 4 [REDACTED]
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

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[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 5 ([REDACTED] [REDACTED]).
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A large, solid black rectangular redaction box covers the central portion of the page, obscuring all text and graphics that might have been present. The box is perfectly uniform in color and extends across most of the page's width and height.

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 6 [REDACTED]
[REDACTED]

[REDACTED]

Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 7 [REDACTED]
[REDACTED]

[REDACTED]

[REDACTED]

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet 1.1) at 10

[REDACTED]

[REDACTED]

Furthermore, the memory devices in the Practicing '407 Products are configurable to perform memory read or write operations by communicating data signals in accordance with their respective local clock signals

[REDACTED]

[REDACTED]

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 5.

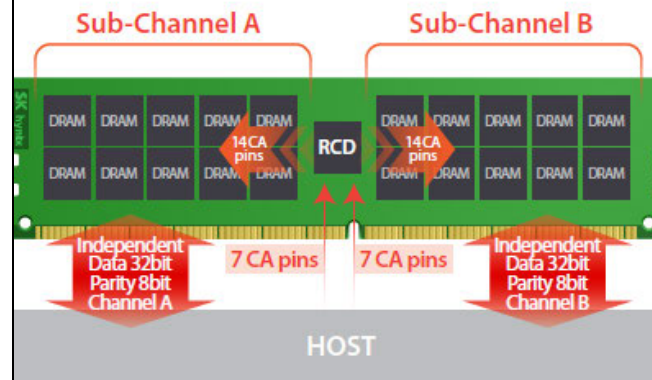
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet 1.0) at 18.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

The latest DDR5 DRAM standard turbo-boosts RDIMM module performance and technology for the era of AI and Big Data. SK hynix expects its DDR5 RDIMMs to reach over 8,000Mbps in data rates with research into continued speed enhancements.

Features



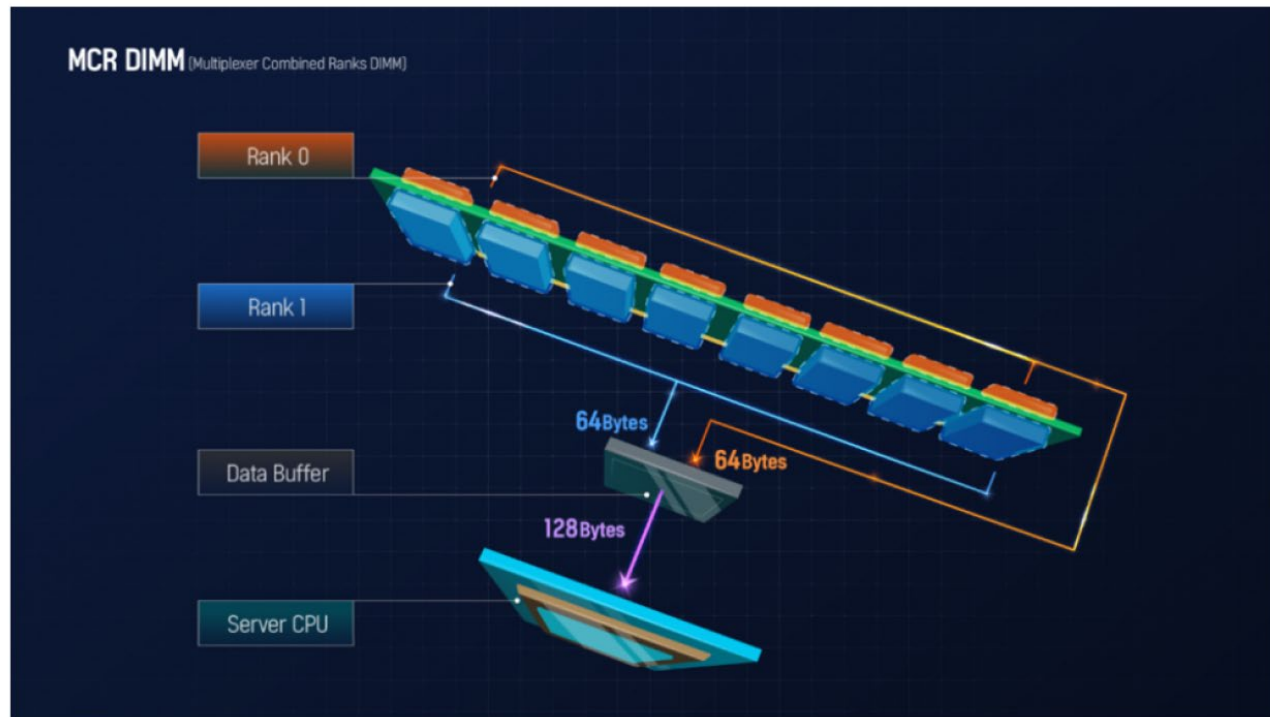
New Sub-channel DRAM Architecture

The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

Ex. CC-40 (SK Hynix A New Standard) at 5 (demonstrating each group of DRAM memory devices is configurable to perform memory read or write operations by communicating data signals with the memory controller via subsets of signal lines within two “Independent Data 32bit” channels).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

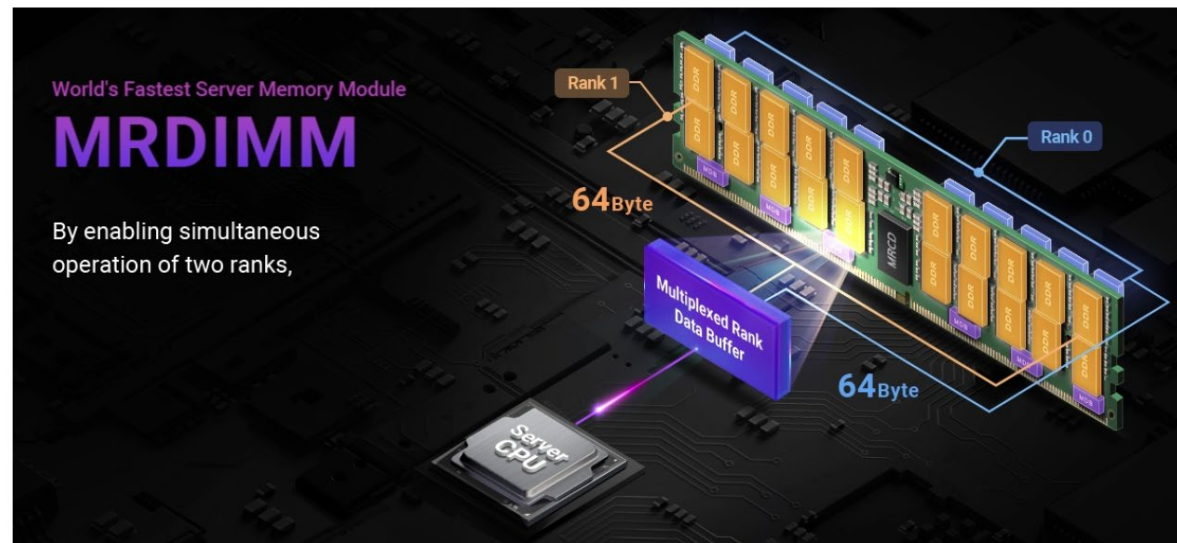
By enabling simultaneous operation of two ranks, MCR DIMM allows transmission of 128 bytes of data to CPU at once, compared with 64 bytes fetched generally in conventional DRAM module. An increase in the amount of data sent to the CPU each time supports the data transfer rate of minimum 8Gbps, twice as fast as a single DRAM.



Ex. CC-65 (SK Hynix Develops MCR DIMM) at 1 (demonstrating each group of DRAM memory devices is configurable to perform memory read or write operations by communicating data signals with the memory controller ("Server CPU") via subsets of data signal lines within two separate 64 Byte channels in accordance with their respective local clock).

How MRDIMM Works

By facilitating simultaneous operation of two ranks, MRDIMM enables the transmission of 128 bytes of data to the CPU at once, compared to the 64 bytes typically fetched by conventional DRAM modules. This increase in the amount of data sent to the CPU each time supports twice the speed of a single DRAM.



Ex. CC-42 (SK Hynix MRDIMM) at 2 (demonstrating each group of DRAM memory devices is configurable to perform memory read or write operations by communicating data signals with the memory controller ("Server CPU") via subsets of data signal lines within two separate 64 Byte channels in accordance with their respective local clock).

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

	Kim Yeong Jun provides a behind-the-scenes explanation of the newly attempted techniques that went into the product development. "In the process of developing the MCR DIMM, we introduced a new concept to increase operating speeds," he explained. "While not changing the way the existing DDR5 DRAM units are used, we designed two ranks—the operation unit of the module—to operate at the same time. While 64 bytes of data are transmitted in one rank, 128 bytes of data can be transmitted at once by operating two ranks simultaneously. A data buffer* is used for this operation as it combines the 64 bytes of data transmitted in two ranks and creates the collective transmit of 128 bytes." He added: "In particular, collaboration with global companies—Intel (U.S.) and Renesas (Japan)—played an important role in this development." Intel, which develops server CPUs, and Renesas, which develops data buffers for the MCR DIMM, collaborated with SK hynix to create synergy." *Buffer: A component that is mounted on top of the DRAM module to optimize the performance of the signal transmission between the DRAM and the CPU. It is mainly installed in server DRAM modules that require high performance and stability. Ex. CC-64 (SK hynix Newsroom Interview) at 6. As explained <i>supra</i>, the Practicing '407 Products are JEDEC compliant. The JEDEC DDR5 RDIMM standard further demonstrates groups of memory devices perform memory read and write operations by communicating data signals with the memory controller via corresponding subsets of data signal lines (e.g., DQS[]).
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U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

6.9.1 DIMM Connectivity Wiring

2Rx4										
DIMM Side	Row	Clock	CS	CA	Rank					
Top	Lower	QACK	QACS0	QACA	0					
Top	Upper	QBCK	QBCS0	QBCA	0					
Bottom	Lower	QCCK	QACS1	QACA	1					
Bottom	Upper	QDCK	QBCS1	QBCA	1					
2Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Top		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]
Bottom	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]
2Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Top	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]
Bottom		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

1Rx4										
DIMM Side	Row	Clock	CS	CA	Rank					
Top	Lower	QACK	QACS0	QACA	0					
Bottom	Lower	QCCK	QBCS0	QACA	0					
1Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]
1Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

2Rx8										
DIMM Side	Row	Clock	CS	CA	Rank					
Top	Lower	QACK	QACS0	QACA	0					
Bottom	Lower	QCCK	QACS1	QACA	1					
2Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]	
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
2Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

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6.9.1 DIMM Connectivity Wiring (cont'd)

1Rx8 (less than 8000 Mbps)										
DIMM Side	Row	Clock	CS	CA	Rank					
Top	Lower	QACK	QACS0	QACA	0					
Bottom	Lower	QACK	QACS0	QACA	0					
1Rx8 (8000 Mbps and Faster)										
DIMM Side	Row	Clock	CS	CA	Rank					
Top	Lower	QCCK	QACS0	QACA	0					
Bottom	Lower	QCCK	QACS0	QACA	0					
1Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top			[15:8]				[31:24]			
Bottom	[7:0]				[23:16]				CB[7:0]	
1Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[7:0]				[23:16]					
Bottom			[15:8]				[31:24]		CB[7:0]	

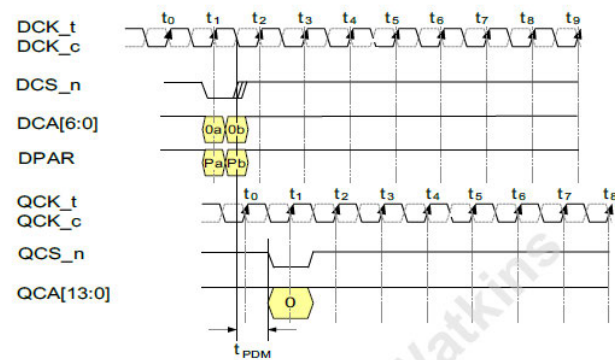
Figure 12 — DIMM Connectivity Wiring

Ex. CC-88 (JEDEC 305B) at 31-32 (showing memory devices receiving data from the memory controller via subsets of data signal lines, e.g., [3:0], [11:8], [15:12]).

The JEDEC DDR5 RCD definition further demonstrates the chip select signal (QCS) and command and address signal (QCA) received by each DDR memory device is synchronized with the local clock (QCK):

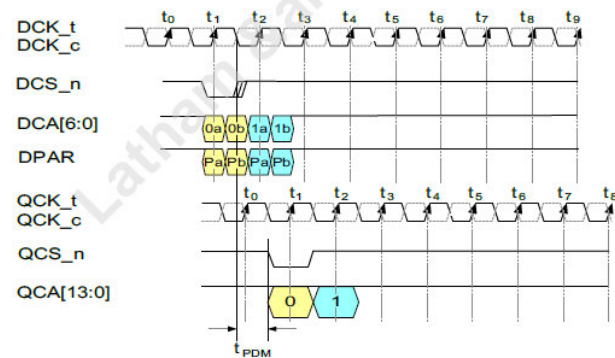
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

3.4.1 Double Data Rate DCA (cont'd)



NOTE Timing Diagram for 0 nCK command latency adder.

Figure 7 — One UI DRAM Command Timing Diagram



NOTE Timing Diagram for 0 nCK command latency adder.

Figure 8 — Two UI DRAM Command Timing Diagram

Ex. CC-32 (JEDEC 82-512) at 16. *See also* Ex. CC-30 (JEDEC 82-513) at 16.

The JEDEC DDR5 RCD definition explains the Practicing '407 Products are configurable to perform READ and WRITE operations synchronized with the local clock QCK. *See, e.g.,*:

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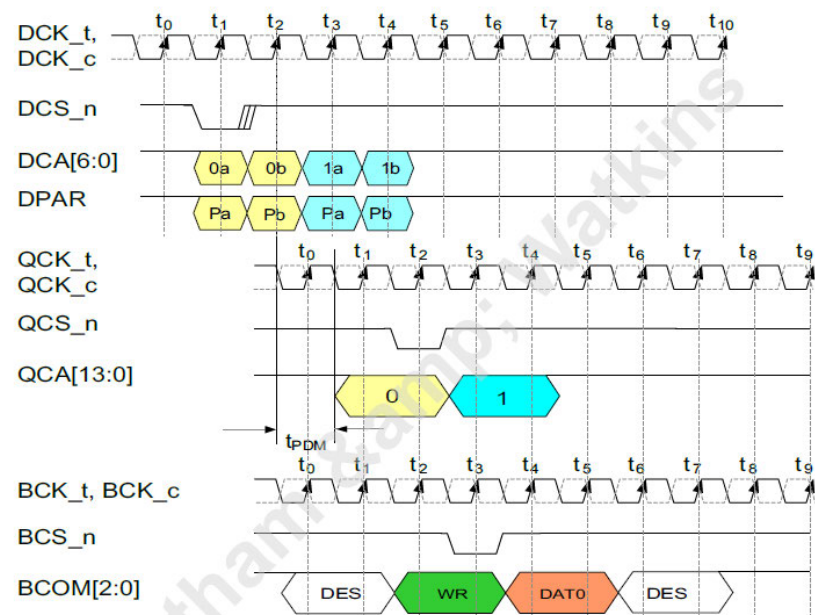


Figure 11 — SDR1 Mode Example of DRAM WR Command with $RW00[1] = 0$

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

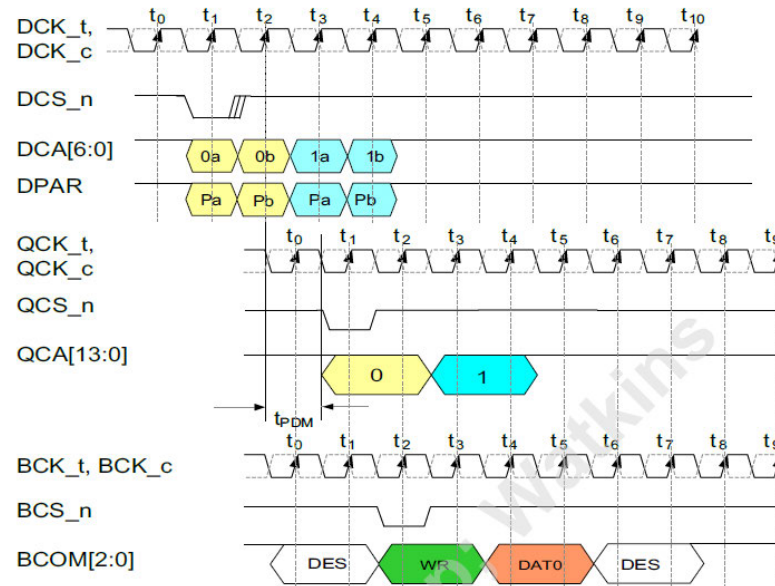


Figure 12 — SDR2 Mode Example of DRAM WR Command with $RW00[1] = 1$

Ex. CC-32 (JEDEC 82-512) at 18-19. *See also* Ex. CC-30 (JEDEC 82-513) at 18-19.

Further, the timing diagrams for read and write operations show that DQ signals appear after a set latency in terms of clock cycles after the module C/A signals are received. As another example, DQ is output/received in relation to DQS, and DQS is aligned with clock signals in predetermined manner. So the DQ output/input is in accordance with local clock signals.

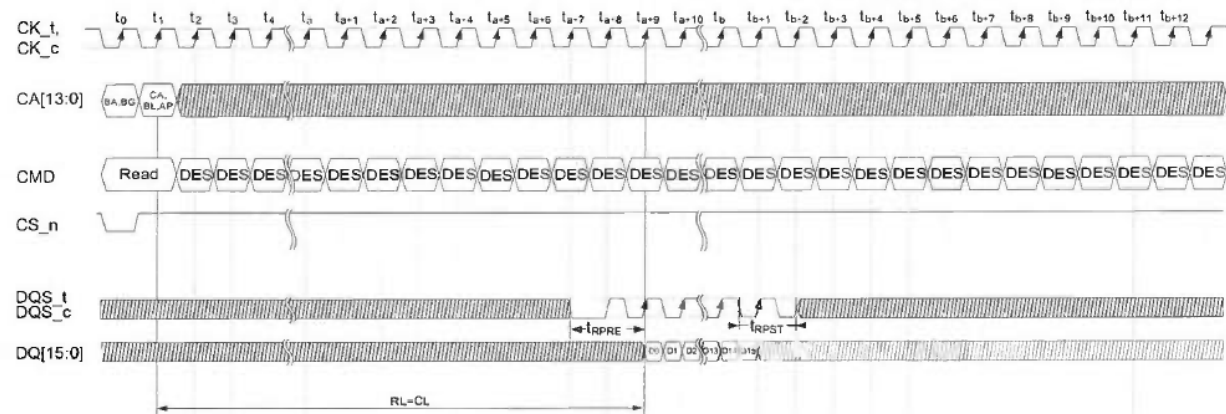
U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

4.7 Read Operation

The Read Operation causes the DRAM to retrieve and output data stored in its array. The Read Operation is initiated by the Read Command during which the beginning column address and bank/group address for the data to be retrieved from the array is provided. The data is driven by the DRAM on its DQ pins RL (CL) cycles after the Read Command along with the proper waveform on the DQS inputs. Read Latency (RL or CL) is defined from the Read command to data and is not affected by the Read DQS offset timing (MR40 OP[2:0]).

4.7.1 READ Burst Operation

During a READ or WRITE command, DDR5 shall support BC8, BL16, BL32 (optional) and BL32 OTF (optional) during the READ or WRITE. MR0[1:0] is used to select burst operation mode.



NOTES:

1. BL=16, Preamble = 2tCK - 0010 Pattern Preamble, 1.5tCK Postamble
2. DES commands are shown for ease of illustration; other commands may be valid at these times.
3. In this example, Read DQS Offset Timing is set to 0 Clocks.

Figure 35 — READ Burst Operation (BL16)

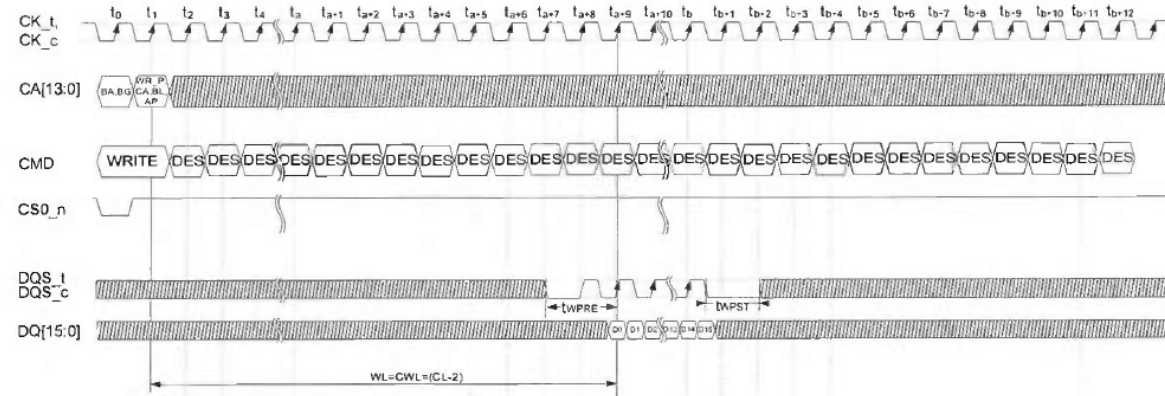
Ex. CC-05 (JESD79-5) at 135.

U.S. Patent No. 12,675,407 Domestic Industry Claim Chart

4.8.2 Write Burst Operation

The following write timing diagrams are to help understand the meaning of each write parameter; the diagrams are just examples. The details of each parameter are defined separately.

In these write timing diagrams, for clarity of illustration, CK and DQS are shown aligned. As well, DQS and DQ are shown center-aligned. Offset between CK and DQS, and between DQS and DQ may be appropriate.



NOTES:

1. BL=16, 2tCK Preamble, 1.5tCK Postamble
2. DES commands are shown for ease of illustration; other commands may be valid at these times.
3. The DQ signal is shown as "Don't Care" before the first Write data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be high for a minimum of 4UI prior to the first Write data bit for proper DFE synchronization.

Figure 49 — WRITE Burst Operation (BL16)

Ex. CC-05 (JESD79-5) at 147.